

The Fabius Function and Rvachev's Up-Function

Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics

Human-readable synthesis of the formal development

Vladimir Reshetnikov's ProveIt repository

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Abstract

This article is the human-readable exposition of the formally verified Fabius-function development in the ProveIt repository. It keeps three related functions separate: the bounded distribution function F , Rvachev's even compactly supported bump up, and the signed global extension $\mathcal{F}$. Their construction and uniqueness, differential and refinement laws, strict shape properties, optimal Lipschitz constants, derivative bounds, effective flatness, and precisely formalized analytic and non-elementarity results are presented together with direct pointers to their Lean proofs.

The probabilistic part identifies F and up with exact probability expressions for an infinite weighted sum of independent uniform coordinates. Finite atomic convolution laws, weak convergence, half-weighted step approximants, and centered uniform splines give proved finite models and quantitative approximation results. The analytic part develops the formal sinc and cosine products, Fourier inversion and Poisson summation, exact moment recurrences, and the Fourier–Legendre expansion. The arithmetic part gives reciprocal-power values, Thue–Morse formulas for arbitrary dyadic arguments, executable binary reduction, two-adic information, and the exact half-base q -binomial identities present in Lean.

The endpoint chapters report only the negative-Laplace decompositions, periodic correction, lower-Lambert formulas, saddle estimates, and correction terms that have named proved counterparts in the formal source. Plausible extensions, numerical observations, generic q -analogues, and other arguments not yet represented by an exact Lean theorem are kept in the companion volume under docs/non-formalized-research-frontiers/, not in this primary exposition.

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1 Historical note and scope

The probability distribution now attached to the name of Fabius is older than Fabius’s two-page note. It occurs in the 1935 work of Jessen and Wintner on infinite convolutions of independent random variables [1], was rediscovered several times, and entered a different mathematical culture through the compactly supported “finite function” of V. A. Rvachev. Fabius used it in 1966 as a particularly economical probabilistic construction of a nowhere analytic C^∞ function [2]. Independently, and for entirely different reasons, the Rvachevs introduced the same object in 1971 as the canonical compactly supported solution of a functional–differential equation [3], and V. A. Rvachev later developed it into the theory of *atomic functions* used in approximation theory and numerical analysis [4, 5, 6]. Rvachev’s normalization emphasizes the even compactly supported bump, whereas the Fabius normalization emphasizes its cumulative distribution function. Arias de Reyna’s 1982 paper gave a systematic treatment of the compactly supported function, including its Fourier product, probability interpretation, partitions of unity, derivatives, and dyadic values; an English translation appeared in 2017 [7]. His later arithmetic paper organized the exact rational sequences and denominator questions [8]. Haugland’s note [9] gives a practical algorithm for evaluating F at dyadic rationals, closely related to the bit-recursion of Section 8.

The present article is not merely a paraphrase of those papers. It follows the proof dependency graph of the Lean development in the `ProveIt` repository [22]. That development first constructs the bounded distribution function, then derives the Rvachev and signed-global forms, and finally builds the arithmetic and asymptotic layers on top. This order has two advantages. It prevents identities valid for the signed extension from being silently applied to the bounded distribution, and it makes the exact dyadic algorithms independent of any numerical approximation.

The formalization also settles several questions that are only implicit, conjectural, or incorrectly normalized in the source material. Most importantly, the sharp endpoint asymptotic contains a nonconstant periodic function of an exact lower-Lambert phase. The periodic term is not a cosmetic improvement: every nonzero Fourier mode is explicit and nonzero, and an elementary formula without it fails at the error scale usually claimed. The development further proves a full inverse-Lambert expansion, establishes an exact bit-recursive evaluator for every dyadic rational, and supplies an absolutely and uniformly convergent Fourier–Legendre expansion of up .

Arias de Reyna’s arithmetic paper [8] enumerates the independent occurrences of this function in chronological order, and two of them deserve mention beside Jessen–Wintner, Fabius, and the Rvachevs. In 1981 G. Kh. Kirov and G. A. Totkov arrived at it in a study of the distribution of the zeros of the derivatives of the function they wrote $\lambda(x)$ (*Godishnik Vissh. Uchebn. Zaved. Prilozhna Mat.* **17** (1981), no. 4, 157–165, in Bulgarian with English and Russian summaries); related dyadic derivative phenomena are represented in the formal Taylor and exact-evaluation APIs used below. In 1985 R. Schnabl gave a further independent construction, “Über eine C^∞ -Funktion,” in *Zahlentheoretische Analysis*, Lecture Notes in Mathematics 1114, Springer, Berlin (1985), 134–142. Neither paper is used in the proofs below, which is why their data are given here rather than in the bibliography; they are recorded because a reader tracing the function through the literature will meet both names.

Our convention is fixed throughout:

- F is the bounded cumulative distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported bump on $[-1, 1]$;
- $\mathcal{F}$ is the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$.

All logarithms are natural logarithms. We write $L = \log 2$ in the asymptotic sections, $\text{wt}_2(n)$ for the sum of the binary digits of n , and $\text{sinc } z = \sin z/z$ with the removable value 1 at the origin.

1.1 A guide to the main results

The exposition follows the dependency order of the formal development. The opening chapters separate the bounded function F , the bump up, and the signed extension $\mathcal{F}$; construct the canonical function; and record the exact uniqueness and shape theorems. The probability chapter then gives the weighted-uniform representation and its three finite approximations: atomic convolution measures, half-weighted step functions, and centered uniform splines.

The Fourier and global-extension chapters collect the proved product, inversion, Poisson, derivative, bound, and flatness APIs. They lead to the exact analytic locus of F , the proved nonanalyticity regions for up and $\mathcal{F}$, the totalized inverse on $[0, 1]$, and the formal non-elementarity results. The moment and arithmetic chapters then derive the rational recurrences, reciprocal-power identities, path and composition formulas, parity and two-adic statements, exact dyadic evaluation, binary reduction, and the specialized half-base q -binomial formulas.

The later analytic chapters summarize the formal endpoint machinery: the exact negative-Laplace product and logarithmic decomposition, the periodic Fourier correction, the lower-Lambert coordinate, the proved saddle asymptotics and correction coefficients, and the rigorous obstructions to deleting the periodic term. The Fourier–Legendre, summation, and computability chapters finish the proved story. The final module guide names the Lean declarations supporting each retained theorem.

A statement appears here only when its complete mathematical content has an exact proved Lean counterpart. Material removed from earlier drafts because that test is not yet met is preserved in the research-frontier volume. Thus the division between this document and that volume is a proof-status boundary, not a judgment about mathematical plausibility or importance.

One formal object, three global functions. The bounded F is the cumulative distribution function used for probabilities and dyadic values. The compactly supported up is obtained by folding F , and its Fourier transform has the proved sinc and cosine product descriptions. The signed $\mathcal{F}$ is the global Thue–Morse continuation and satisfies the pure dilation equation on all of $\mathbb{R}$. Domains and normalizations are stated explicitly whenever a theorem passes from one presentation to another.

2 Three functions that must not be conflated

The literature often uses the same letter F for three related, but globally different, functions. That economy of notation is harmless in a local calculation and dangerous in a long proof. We therefore begin by separating them.

2.1 The bounded Fabius distribution function

Definition 2.1 (Bounded Fabius function). A function $F : \mathbb{R} \rightarrow [0, 1]$ is called a *bounded Fabius function* if

$$F(x) = 0 \quad (x \leq 0), \quad (1)$$

$$F(x) = 1 \quad (x \geq 1), \quad (2)$$

$$F(1 - x) = 1 - F(x) \quad (0 \leq x \leq 1), \quad (3)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (4)$$

and F is infinitely differentiable on $\mathbb{R}$.

The derivative equation on the right half follows from symmetry:

$$F'(x) = 2F(2 - 2x), \quad \tfrac{1}{2} \leq x \leq 1. \quad (5)$$

In particular $F'(x) > 0$ for $0 < x < 1$, once positivity in the interior is known. Thus F is a genuine cumulative distribution function, not merely a smooth interpolation between 0 and 1.

Because F is constant on both exterior half-lines and is C^∞ , every derivative of positive order vanishes at both endpoints:

$$F^{(m)}(0) = F^{(m)}(1) = 0 \quad (m \geq 1). \quad (6)$$

This flatness is the source of both its extraordinary endpoint decay and its failure to be analytic there.

2.2 Rvachev's compactly supported bump

Definition 2.2 (Rvachev up-function). Given F , define

$$\text{up}(x) = \begin{cases} F(x+1), & x \leq 0, \\ F(1-x), & x > 0. \end{cases} \quad (7)$$

Equivalently,

$$\text{up}(x) = F(1 - |x|). \quad (8)$$

Here the right side is interpreted using the globally bounded function F .

It follows immediately that up is even, nonnegative, supported on $[-1, 1]$, and $\text{up}(0) = 1$. On the unit interval the two functions are simply translates:

$$F(x) = \text{up}(x-1), \quad 0 \leq x \leq 1. \quad (9)$$

The graph of F is an infinitely flat sigmoid; the graph of up is its symmetric fold.

The primary exposition omits sampled plots: their decimal coordinates are not themselves Lean declarations. Exact point evaluation is developed below through rational dyadic formulas and executable evaluators.

2.3 The signed global extension

Let $\text{wt}_2(n)$ denote the sum of the binary digits of n , and put

$$\varepsilon_n = (-1)^{\text{wt}_2(n)}. \quad (10)$$

The sequence (ε_n) is the $\{\pm 1\}$ form of the Thue–Morse sequence [14].

Definition 2.3 (Signed extension). Define

$$\mathcal{F}(x) = \sum_{n \geq 0} \varepsilon_n \text{up}(x - 2n - 1). \quad (11)$$

At a fixed x only finitely many summands can be nonzero, so this is a locally finite sum. On every block $[2b, 2b+2]$ only the b -th translate can be nonzero (and at the endpoints it too vanishes):

$$\mathcal{F}(x) = \varepsilon_b \text{up}(x - 2b - 1), \quad 2b \leq x \leq 2b+2. \quad (12)$$

It vanishes for $x \leq 0$, and it agrees with the bounded F for every $x \leq 1$ (both functions vanish on $(-\infty, 0]$). It does *not* remain in $[0, 1]$ to the right of that interval. This is the global object that satisfies the cleanest differential equation:

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (13)$$

Arias de Reyna's arithmetic paper uses this extension. Statements about its values outside $[0, 1]$ must therefore not be transferred blindly to the bounded cumulative distribution.

Notation used in this article. F always means the bounded distribution function; up always means the compactly supported even bump; and $\mathcal{F}$ always means the signed global extension. We have $F = \mathcal{F}$ throughout $(-\infty, 1]$ (hence on $[0, 1]$), but globally they are different functions.

2.4 Exact shape: positivity, bijectivity, and unimodality

The three defining equations (1)–(4) already determine the qualitative picture, but the statements that are actually used later are the sharp ones: not $F \geq 0$ but $F(x) > 0$ exactly for $x > 0$; not $\text{up} \leq 1$ but $\text{up}(x) = 1$ exactly at the origin. We record them here, in the form of equivalences, because the strict versions are what make the convexity and Lipschitz theorems below sharp.

Everything rests on a single formula that merges (4) and (5) into one identity valid on the whole line.

Lemma 2.4 (Unified derivative formula). *For every real x ,*

$$F'(x) = 2 \text{up}(2x - 1). \quad (14)$$

Proof. By (8), $\text{up}(2x - 1) = F(1 - |2x - 1|)$. For $0 \leq x \leq \frac{1}{2}$ one has $|2x - 1| = 1 - 2x$, so $\text{up}(2x - 1) = F(2x)$ and (14) is (4). For $\frac{1}{2} \leq x \leq 1$ one has $|2x - 1| = 2x - 1$, so $\text{up}(2x - 1) = F(2 - 2x)$ and (14) is (5). For $x < 0$ or $x > 1$ one has $|2x - 1| > 1$, hence $1 - |2x - 1| < 0$ and $\text{up}(2x - 1) = 0$ by (1), while $F'(x) = 0$ because F is constant on each exterior half-line by (1) and (2). $\square$

Since $\text{up} \geq 0$, formula (14) shows at once that F is nondecreasing on all of $\mathbb{R}$. It is also convenient to note that the reflection symmetry (3), stated on the unit interval, in fact holds globally:

$$F(1 - x) = 1 - F(x) \quad (x \in \mathbb{R}). \quad (15)$$

Indeed, for $x < 0$ both sides equal 1 by (1) and (2), and for $x > 1$ both sides equal 0.

Theorem 2.5 (Positivity, strict monotonicity, bijectivity). *The bounded Fabius function satisfies the following.*

1. $F(x) > 0$ if and only if $x > 0$.
2. $F(x) < 1$ if and only if $x < 1$.
3. $F'(x) > 0$ for $0 < x < 1$, and F is strictly increasing on $[0, 1]$.
4. F maps $[0, 1]$ bijectively onto $[0, 1]$.

Proof. The heart of the matter is the following doubling step: if $0 \leq x \leq \frac{1}{2}$ and $F(x) = 0$, then $F(2x) = 0$. To see it, observe that $F \geq 0$ everywhere, so a zero of F is a global minimum of a function differentiable on all of $\mathbb{R}$; Fermat's rule gives $F'(x) = 0$, and (4) turns this into $2F(2x) = 0$.

Suppose now that $F(x) = 0$ for some $x > 0$. Since $2^n x \rightarrow \infty$, there is a smallest integer $n \geq 0$ with $2^n x \geq \frac{1}{2}$. For $0 \leq m < n$ we have $2^m x < \frac{1}{2}$, so the doubling step applies at $2^m x$ and propagates the vanishing: from $F(x) = 0$ we obtain $F(2^m x) = 0$ for every $m \leq n$, and in particular $F(2^n x) = 0$. But F is nondecreasing and $2^n x \geq \frac{1}{2}$, so $F(2^n x) \geq F(\frac{1}{2}) = \frac{1}{2}$, where the midpoint value comes from (3) at $x = \frac{1}{2}$. This contradiction proves that F has no zero on the positive half-line; combined with (1) it proves (i).

Assertion (ii) follows from (i) and (15): $F(x) < 1$ if and only if $F(1 - x) > 0$, that is, if and only if $1 - x > 0$.

For (iii), let $0 < x < 1$. Then $2x - 1 \in (-1, 1)$, so $1 - |2x - 1| \in (0, 1]$, and (i) gives $\text{up}(2x - 1) = F(1 - |2x - 1|) > 0$. Now (14) yields $F'(x) > 0$. A continuous function on $[0, 1]$ whose derivative is positive throughout the interior is strictly increasing on $[0, 1]$.

For (iv), F is continuous with $F(0) = 0$ and $F(1) = 1$, so the intermediate value theorem makes $F([0, 1])$ contain $[0, 1]$; since F takes values in $[0, 1]$, the image is exactly $[0, 1]$. Strict monotonicity from (iii) gives injectivity. $\square$

The fold (8) now transfers the whole picture to Rvachev's bump, and it does so in strict form.

Theorem 2.6 (Strict unimodality of up). *Rvachev’s function satisfies the following.*

1. up is strictly increasing on $[-1, 0]$ and strictly decreasing on $[0, 1]$.
2. $\text{up}(x) = 1$ if and only if $x = 0$; the maximum value 1 is attained at the origin and nowhere else.
3. $\text{up}(x) = 0$ if and only if $x \notin (-1, 1)$. Thus up is nonzero exactly on $(-1, 1)$, and $\text{supp up} = [-1, 1]$.
4. $\text{up}'(x) > 0$ for $-1 < x < 0$ and $\text{up}'(x) < 0$ for $0 < x < 1$.

Proof. By (8), $\text{up}(x) = F(1 - x)$ for $x \geq 0$. As x increases through $[0, 1]$ the argument $1 - x$ decreases through $[0, 1]$, on which F is strictly increasing by [theorem 2.5\(iii\)](#); hence up is strictly decreasing on $[0, 1]$, and evenness gives (i) on $[-1, 0]$.

For (ii), $\text{up}(x) = F(1 - |x|)$ and [theorem 2.5\(ii\)](#) say that $\text{up}(x) < 1$ unless $1 - |x| \geq 1$, that is, unless $x = 0$; and $\text{up}(0) = F(1) = 1$. For (iii), $\text{up}(x) > 0$ if and only if $1 - |x| > 0$ by [theorem 2.5\(i\)](#), and $\text{up} \geq 0$, so the zero set is exactly the complement of $(-1, 1)$; the closed support is the closure $[-1, 1]$.

For (iv), the identity $\text{up}(x) = F(1 - x)$ holds on the whole open half-line $x > 0$, hence may be differentiated there: $\text{up}'(x) = -F'(1 - x)$, which is negative for $0 < x < 1$ by [theorem 2.5\(iii\)](#). Evenness gives $\text{up}'(x) = -\text{up}'(-x)$, so $\text{up}' > 0$ on $(-1, 0)$. $\square$

Parts (i) and (iii) are the classical description of the bump [6, 7]; the strict forms, and the equivalences in [theorem 2.5](#), are exactly what the formal development records.

2.5 Convexity and the maximal slope

The derivative formula transfers the monotonicity of up to the derivative of F .

Theorem 2.7 (Convexity, concavity, and the maximal slope). *The derivative F' is nondecreasing on $(-\infty, \frac{1}{2}]$ and nonincreasing on $[\frac{1}{2}, \infty)$. It is strictly increasing on $[0, \frac{1}{2}]$ and strictly decreasing on $[\frac{1}{2}, 1]$. Consequently F is convex on $(-\infty, \frac{1}{2}]$, concave on $[\frac{1}{2}, \infty)$, strictly convex on $[0, \frac{1}{2}]$, and strictly concave on $[\frac{1}{2}, 1]$. Moreover*

$$F'(\tfrac{1}{2}) = 2 = \max_{x \in \mathbb{R}} F'(x), \quad (16)$$

and equality $F'(x) = 2$ holds exactly at $x = \frac{1}{2}$.

Formal proof. The derivative monotonicity statements are `monotoneOn_deriv_fabiusReal`, `antitoneOn_deriv_fabiusReal`, `strictMonoOn_deriv_fabiusReal_Icc`, and `strictAntitoneOn_deriv_fabiusReal_Icc`. The four convexity statements are `convexOn_fabiusReal_Iic_half`, `concaveOn_fabiusReal_Ici_half`, `strictConvexOn_fabiusReal_firstHalf`, and `strictConcaveOn_fabiusReal_secondHalf`. Finally `deriv_fabiusReal_half`, `deriv_fabiusReal_eq_two_iff`, and `isGreatest_deriv_fabiusReal` prove the exact maximal-slope statement. All are in `Convexity.lean`. $\square$

The same module records the complete pointwise curvature profile. For every $x \in \mathbb{R}$,

$$F''(x) = 8(\text{up}(4x - 1) - \text{up}(4x - 3)), \quad (17)$$

and therefore

$$\begin{aligned} F''(x) > 0 &\iff 0 < x < \tfrac{1}{2}, \\ F''(x) < 0 &\iff \tfrac{1}{2} < x < 1, \\ F''(x) = 0 &\iff x \notin (0, 1) \text{ or } x = \tfrac{1}{2}. \end{aligned} \quad (18)$$

The derivative-level statement and its pointwise corollary are `deriv_fabiusReal_hasDerivAt` and `deriv_deriv_fabiusReal`; the three equivalences are `deriv_deriv_fabiusReal_pos_iff`, `deriv_deriv_fabiusReal_neg_iff`, and `deriv_deriv_fabiusReal_eq_zero_iff`, with `deriv_deriv_fabiusReal_half` naming the midpoint specialization.

2.6 The optimal Lipschitz constant

The bound $F' \leq 2$ of (16) is a Lipschitz estimate, and because the bound is attained the estimate cannot be improved.

Theorem 2.8 (Two is the least Lipschitz constant). *Both F and up are 2-Lipschitz on $\mathbb{R}$:*

$$|F(x) - F(y)| \leq 2|x - y|, \quad |\text{up}(x) - \text{up}(y)| \leq 2|x - y| \quad (x, y \in \mathbb{R}). \quad (19)$$

Moreover 2 is the least admissible constant for each: if $K \geq 0$ is such that $|F(x) - F(y)| \leq K|x - y|$ for all x, y , or such that $|\text{up}(x) - \text{up}(y)| \leq K|x - y|$ for all x, y , then $K \geq 2$.

Proof. By (14) and $0 \leq \text{up} \leq 1$ we have $0 \leq F' \leq 2$ on all of $\mathbb{R}$, so the mean value theorem gives the first estimate in (19). The map $x \mapsto 1 - |x|$ is 1-Lipschitz, so the second estimate follows from the first through the fold (8).

For optimality, let K be admissible for F . Dividing $|F(x) - F(\frac{1}{2})| \leq K|x - \frac{1}{2}|$ by $|x - \frac{1}{2}|$ and letting $x \rightarrow \frac{1}{2}$ gives $K \geq |F'(\frac{1}{2})| = 2$ by (16). For up the same limit at the same point works: the proof of theorem 2.6(iv) gives $\text{up}'(\frac{1}{2}) = -F'(\frac{1}{2}) = -2$, so $K \geq 2$. $\square$

The two extremal points are the same point in the two pictures: $\frac{1}{2}$ is the centre of the sigmoid and the midpoint of the decreasing half of the bump, and the fold sends one to the other.

Corollary 2.9 (Linear majorants at the endpoints). *For all admissible arguments,*

$$F(x) \leq 2x \quad (x \geq 0), \quad (20)$$

$$1 - F(x) \leq 2(1 - x) \quad (x \leq 1), \quad (21)$$

$$\text{up}(x) \leq 2(1 - |x|) \quad (|x| \leq 1). \quad (22)$$

Proof. Applying (19) with $y = 0$ and using $F(0) = 0$ gives (20). Replacing x by $1 - x$ there and using (15) gives (21). Substituting $1 - |x| \geq 0$ into (20) and using (8) gives (22). $\square$

The content of (20) lies on $[0, \frac{1}{2}]$; for $x \geq \frac{1}{2}$ it is weaker than $F \leq 1$. Its value is that it holds with no hypothesis at all on the size of x , so it can be used as a linear majorant in any estimate near the left endpoint, and (22) says that the bump vanishes at least linearly at both ends of its support. These majorants and the optimality of the constant 2 are contributions of the formal development; the plain 2-Lipschitz bound is used implicitly throughout [7, 6].

3 Existence and uniqueness by a contraction

The most economical construction of F is a Banach fixed-point argument. It has an additional advantage: the contraction estimate later gives a clean uniqueness theorem, without presupposing Fourier analysis or probability.

3.1 The admissible function space

Let $I = [0, 1]$ and let $C(I)$ carry the supremum norm. Define $\mathcal{A} \subset C(I)$ by

$$\mathcal{A} = \{f \in C(I) : 0 \leq f \leq 1, f(1 - x) = 1 - f(x)\}. \quad (23)$$

The set $\mathcal{A}$ is closed in the Banach space $C(I)$, hence complete. It is nonempty; for example $f(x) = x$ belongs to it.

Extend $f \in \mathcal{A}$ to a continuous function $\tilde{f}$ on $\mathbb{R}$ by clamping the argument to I :

$$\tilde{f}(t) = \begin{cases} f(0), & t \leq 0, \\ f(t), & 0 \leq t \leq 1, \\ f(1), & t \geq 1. \end{cases} \quad (24)$$

Symmetry gives

$$\int_0^1 f(t) dt = \frac{1}{2}. \quad (25)$$

Indeed, replacing t by $1 - t$ and adding the two integrals gives $2 \int_0^1 f = 1$.

3.2 The integral transform

For $f \in \mathcal{A}$, define Tf by

$$(Tf)(x) = \begin{cases} \int_0^{2x} \tilde{f}(t) dt, & 0 \leq x \leq \frac{1}{2}, \\ 1 - \int_0^{2-2x} \tilde{f}(t) dt, & \frac{1}{2} < x \leq 1. \end{cases} \quad (26)$$

The two definitions agree at $x = 1/2$ by (25). Positivity and the upper bound $f \leq 1$ show that $0 \leq Tf \leq 1$, and a direct substitution shows $(Tf)(1 - x) = 1 - (Tf)(x)$. Hence $T : \mathcal{A} \rightarrow \mathcal{A}$.

The central estimate is stronger than the naive integral bound.

Lemma 3.1 (Contraction estimate). *For $f, g \in \mathcal{A}$,*

$$\|Tf - Tg\|_\infty \leq \frac{1}{2} \|f - g\|_\infty. \quad (27)$$

Proof. Write $h = f - g$. Since both f and g have integral $1/2$, $\int_0^1 h = 0$. For $0 \leq y \leq 1$ we may therefore estimate the primitive in two ways:

$$\begin{aligned} \left| \int_0^y h(t) dt \right| &\leq y \|h\|_\infty, \\ \left| \int_0^y h(t) dt \right| &= \left| \int_y^1 h(t) dt \right| \leq (1 - y) \|h\|_\infty. \end{aligned}$$

Consequently

$$\left| \int_0^y h(t) dt \right| \leq \min(y, 1 - y) \|h\|_\infty \leq \frac{1}{2} \|h\|_\infty.$$

Every value of $Tf - Tg$ is one of these primitives, possibly with a minus sign. Taking the supremum proves (27). $\square$

Theorem 3.2 (Existence of the bounded Fabius function). *The transform T has a unique fixed point $f_* \in \mathcal{A}$. Extending f_* by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$ produces the bounded Fabius function F .*

Proof. The space $\mathcal{A}$ is complete and T is a strict contraction with constant $1/2$, so Banach's fixed-point theorem gives a unique fixed point. On $[0, 1/2]$ its fixed-point equation is

$$F(x) = \int_0^{2x} F(t) dt. \quad (28)$$

Differentiating gives $F'(x) = 2F(2x)$. The second half of (26) supplies symmetry and the matching right-hand formula. The endpoint values are immediate. $\square$

3.3 The differential equation for up and smoothness

Folding (4) and (5) yields the refinement-type differential equation

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)) \quad (x \in \mathbb{R}). \quad (29)$$

At $x = 0$ the two one-sided derivatives glue because $\text{up}(1) = \text{up}(-1) = 0$. Once up is continuously differentiable, (29) bootstraps regularity: if $\text{up} \in C^r$, then the right side is C^r , so $\text{up} \in C^{r+1}$. Induction gives $\text{up} \in C^\infty$ and then $F \in C^\infty$.

Proposition 3.3 (Basic shape). *The functions F and up have the following properties.*

1. F is strictly increasing on $(0, 1)$ and satisfies $0 < F(x) < 1$ there.
2. up is even, strictly positive on $(-1, 1)$, and zero outside $[-1, 1]$.
3. $\int_{-1}^1 \text{up}(x) dx = 1$.
4. Every derivative of up vanishes at $x = \pm 1$.

Proof. For $0 < x \leq 1/2$, (4) gives $F'(x) = 2F(2x)$. Positivity follows first near $1/2$ from $F(1/2) = 1/2$, and then propagates toward zero by repeatedly doubling the argument; symmetry handles the right half. This proves strict monotonicity and, by (8), positivity of up in the interior of its support. Finally,

$$\int_{-1}^1 \text{up}(x) dx = 2 \int_0^1 F(t) dt = 1,$$

where the middle identity uses the fold and the last uses symmetry. Flatness at the support endpoints follows from smoothness and the fact that up is identically zero just outside the support. $\square$

3.4 The original Rvachev characterization

The formal predicate `IsOriginalFabius` packages the source characterization exactly. For a function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ and a scale k , it requires C^∞ -smoothness, topological support $[-1, 1]$, strict positivity on $(-1, 1)$, the normalization $\varphi(0) = 1$, positivity of k , and

$$\varphi'(x) = k(\varphi(2x + 1) - \varphi(2x - 1)) \quad (x \in \mathbb{R}). \quad (30)$$

The positivity of the scale is redundant: the smart constructor `IsOriginalFabius.mk_of_derivative_law` derives it from the other hypotheses.

Theorem 3.4 (Original characterization). *If $\text{IsOriginalFabius } \varphi \ k$, then*

$$k = 2, \quad \varphi = \text{up}. \quad (31)$$

Conversely, the canonical up satisfies this predicate at scale 2; in particular there is exactly one original solution.

Formal proof. The scale is fixed by `originalFabius_scale_eq_two`; the canonical witness is `canonical_isOriginalFabius`. Uniqueness and the displayed characterization are `originalFabius_eq_canonical`, `isOriginalFabius_iff_eq_canonical`, and `existsUnique_originalFabius` in `OriginalUniqueness.lean`. $\square$

The fold loses only the unused right tail of a bounded candidate. Precisely,

$$\text{up}_G = \text{up}_H \iff G(t) = H(t) \text{ for every } t \leq 1. \quad (32)$$

Thus a bounded candidate is canonical exactly when its fold satisfies the original characterization at scale 2 and its right tail is constantly 1. These are the exact declarations `rvachevUp_eq_iff_eqOn_Iic_one`, `isFabius_iff_isOriginalFabius_rvachevUp_and_rightTail`, and `isOriginalFabius_iff_existsUnique_isFabius` in `OriginalUniqueness.lean`.

4 The probability law behind the functions

The fixed-point construction is analytic. A probabilistic construction explains the same self-similarity almost without calculation and makes monotonicity and positivity transparent.

4.1 An infinite weighted sum of uniforms

Let $U_0, U_1, \dots$ be independent random variables, each uniformly distributed on $[0, 1]$, and define

$$X = \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \quad (33)$$

The series converges absolutely for every outcome and satisfies $0 \leq X \leq 1$. Removing the first coordinate gives an independent copy X' of X , and

$$X = \frac{U_0 + X'}{2}. \quad (34)$$

Reflecting every coordinate, $U_n \mapsto 1 - U_n$, preserves the product measure and sends X to $1 - X$.

Let $H(x) = \mathbb{P}(X \leq x)$ be the cumulative distribution function. Conditioning on U_0 in (34) gives

$$H(y) = \int_0^1 H(2y - u) \, du. \quad (35)$$

For $0 \leq y \leq 1/2$, the integrand vanishes when $u > 2y$, so

$$H(y) = \int_0^{2y} H(t) \, dt. \quad (36)$$

Reflection gives $H(1 - y) = 1 - H(y)$. Thus the restriction of H to $[0, 1]$ is a fixed point of the contraction T from [section 3](#).

Theorem 4.1 (Global probabilistic representation). *For every $x, y \in \mathbb{R}$,*

$$F(x) = \mathbb{P} \left(\sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \leq x \right), \quad (37)$$

$$\text{up}(y) = \mathbb{P} \left(\sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \leq 1 - |y| \right). \quad (38)$$

The second identity includes both tails: if $|y| \geq 1$, both sides vanish.

Formal proof. The product-space construction proves continuity and measurability of the uniformly convergent coordinate sum, identifies its pushforward distribution, establishes the head–tail self-similarity and the coordinate-reflection law, and invokes uniqueness of the bounded Fabius fixed point. The exact canonical specializations are `fabiusReal_eq_weightedSum_probability` and `rvachevUp_eq_weightedSum_probability_global` in `ProbabilityRepresentation.le an`. $\square$

4.2 Finite atomic laws and weak convergence

There is also a completely discrete approximation theorem. For $k \geq 0$, let β_k be the probability measure assigning mass $1/2$ to each of $\pm 2^{-k-1}$. Starting from $\nu_0 = \delta_0$, define

$$\nu_{n+1} = \nu_n * \beta_{n+1}^{*(n+1)}. \quad (39)$$

Thus ν_n is a finite atomic probability measure. Its characteristic function is the finite weighted cosine product

$$\widehat{\nu}_n(t) = \prod_{k=0}^{n-1} \left(\cos \frac{t}{2^{k+2}} \right)^{k+1}. \quad (40)$$

The indexing in (39) is worth noticing: the new scale at stage $n + 1$ is 2^{-n-2} , and it occurs with multiplicity $n + 1$.

Let μ_{up} denote the formal measure `rvachevMeasure`; by definition it is Lebesgue measure with density of `Real` `up`. It is absolutely continuous with respect to Lebesgue measure; in particular, every Lebesgue-null set is also μ_{up} -null. These are `rvachevMeasure_absolutelyContinuous` and `rvachevMeasure_eq_zero_of_volume_eq_zero` in `StepMeasureBridge.lean`. Its characteristic function uses the positive-frequency normalization

$$\text{charFun}(\mu_{\text{up}}, t) = \widehat{\text{up}} \left(\frac{t}{2\pi} \right) \quad (t \in \mathbb{R}), \quad (41)$$

the exact identity `rvachevMeasure_charFun_pos` in `WeakConvergence.lean`. The finite laws converge weakly to μ_{up} :

$$\nu_n \Longrightarrow \mu_{\text{up}}. \quad (42)$$

Equivalently, for every bounded continuous real-valued map g ,

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} g(x) \, d\nu_n(x) = \int_{\mathbb{R}} g(x) \, d\mu_{\text{up}}(x). \quad (43)$$

The same test-function theorem is formalized for complex-valued, and more generally real-or-complex-like, bounded continuous functions.

Formal proof. The definitions are `centeredBernoulliMeasure` and `finiteConvolutionMeasure`. Their probability-measure instances and (40) are proved in `EarlyApproximants.lean`. Pointwise convergence of the characteristic functions, weak convergence (42), and the test-function formulations (43) are respectively `finiteConvolutionMeasure_charFun_tendsto`, `finiteConvolutionProbability_tendsto_fabius`, and `integral_finiteConvolutionMeasure_tendsto_fabius` in `WeakConvergence.lean`. $\square$

4.3 Combinatorial step functions and the half-weight convention

A second finite model replaces each atom by a narrow interval. Define polynomials with nonnegative integer coefficients by

$$\mathcal{P}_0(X) = 1, \quad \mathcal{P}_m(X) = \mathcal{P}_{m-1}(X^2)(1 + X)^m \quad (m \geq 1). \quad (44)$$

They also have the finite product form

$$\mathcal{P}_m(X) = \prod_{j=1}^m (1 + X + \cdots + X^{2^j-1}). \quad (45)$$

Writing $\mathcal{P}_m(X) = \sum_r a_{m,r} X^r$, the coefficient $a_{m,r}$ counts the tuples

$$(s_1, \dots, s_m), \quad 0 \leq s_j \leq 2^j - 1, \quad s_1 + \cdots + s_m = r. \quad (46)$$

Moreover,

$$g_m := \deg \mathcal{P}_m = 2^{m+1} - m - 2, \quad \mathcal{P}_m(1) = 2^{\binom{m+1}{2}}. \quad (47)$$

For $\alpha < \beta$, let $\mathbf{1}_{[\alpha, \beta]}^*$ be 1 in the interior, $1/2$ at either endpoint, and 0 elsewhere. Put

$$I_{m,r} = \left[\frac{2r-1-g_m}{2^{m+1}}, \frac{2r+1-g_m}{2^{m+1}} \right] \quad (48)$$

and

$$\mathcal{W}_m(x) = \frac{2^m}{2^{\binom{m+1}{2}}} \sum_{r=0}^{g_m} a_{m,r} \mathbf{1}_{I_{m,r}}^*(x). \quad (49)$$

Theorem 4.2 (Corrected step approximation). *For every $m \geq 0$ and $x \in \mathbb{R}$,*

$$0 \leq \mathcal{W}_m(x) \leq 1, \quad \mathcal{W}_m(0) = 1, \quad \int_{\mathbb{R}} \mathcal{W}_m(x) dx = 1.$$

The function $\mathcal{W}_m$ is nondecreasing on $(-\infty, 0)$ and nonincreasing on $(0, \infty)$. For every real x ,

$$\mathcal{W}_m(x) \longrightarrow \text{up}(x). \quad (50)$$

The proof of pointwise convergence passes through a useful exact comparison. Set $h_m = 2^{-m-1}$, and use the finite atomic law ν_m from (39). Whenever $a \leq b$,

$$\nu_m((a + h_m, b - h_m]) \leq \int_a^b \mathcal{W}_m(x) dx \leq \nu_m((a - h_m, b + h_m]). \quad (51)$$

Consequently,

$$\lim_{m \rightarrow \infty} \int_a^b \mathcal{W}_m(x) dx = \int_a^b \text{up}(x) dx. \quad (52)$$

Monotonicity on the two open half-lines, the exact value at the origin, and continuity of up then upgrade these interval limits to (50).

Formal proof. The polynomial recursion, product, degree, total coefficient mass, and bounded-partition interpretation are respectively proved by `approximationPolynomial_succ`, the definition `approximationPolynomial`, `approximationPolynomial_natDegree`, `approximationDegree_eq_sub`, `approximationPolynomial_eval_one`, and `approximationPolynomial_coeff_eq_card` in `EarlyApproximants.lean`. The finite atomic description is identified with `finiteConvolutionMeasure` by `polynomialMeasure_eq_finiteConvolutionMeasure` in `EarlyMeasureBridge.lean`.

For nonnegativity, see `stepApproximant_nonneg` in `EarlyApproximants.lean`. The remaining analytic assertions are `stepApproximant_le_one`, `stepApproximant_monotoneOn_Iio`, `stepApproximant_antitoneOn_Ioi`, `stepApproximant_apply_zero`, and `stepApproximant_tendsto_fabius` in `StepApproximationLimit.lean`. The normalization, the two sides of (51), and (52) are `integral_stepApproximant`, `finiteConvolutionMeasure_inner_le_intervalIntegral_stepApproximant`, `intervalIntegral_stepApproximant_le_finiteConvolutionMeasure_outer`, and `intervalIntegral_stepApproximant_tendsto_of_le` in `StepMeasureBridge.lean`. $\square$

Warning 4.3 (Why the endpoint convention is explicit). The half-weighted cell representative agrees almost everywhere with the ordinary interval indicator, so it has the same integral. Its point values are nevertheless part of the formal statement: adjacent cells share endpoints, and each contributes exactly half of its coefficient there. In particular, the corrected representative satisfies $\mathcal{W}_m(0) = 1$ at every level. These facts are formalized by `halfEndpointIntervalIndicator_ae_eq_indicator`, `intervalIntegral_halfEndpointIntervalIndicator_of_le`, and `stepApproximant_apply_zero`.

4.4 Centered finite splines and an explicit error bound

The preceding finite random sums also give a completely explicit polynomial approximation to F . Put

$$T_p = \sum_{i=0}^{p-1} \frac{U_i}{2^{i+1}}, \quad C_p(x) = \mathbb{P}(T_p \leq x - 2^{-p-1}), \quad (53)$$

where $p \geq 1$. The shift by half of the omitted tail is important: it centers the one-sided approximation and improves its error by a factor of two.

Let

$$N_p(x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right), \quad \varepsilon_r = (-1)^{\text{wt}_2(r)}, \quad (54)$$

and interpret a sum with $N_p(x) = 0$ as empty. Define

$$S_p(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left(r - 2^p x + \frac{1}{2}\right)_+^p. \quad (55)$$

Equivalently,

$$S_p(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r \geq 0} \varepsilon_r \left(2^p x - \frac{1}{2} - r\right)_+^p, \quad (56)$$

where only the terms $r < N_p(x)$ can be nonzero and $t_+ = \max(t, 0)$. The cutoff in (54) is a safe natural-number form of the inclusive upper limit

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor.$$

It handles the empty case without assigning a negative upper endpoint to a finite sum.

Theorem 4.4 (Centered spline as a finite distribution function). *For $p \geq 1$ and $0 \leq x \leq 1$,*

$$S_p(x) = C_p(x). \quad (57)$$

Consequently $0 \leq S_p(x) \leq 1$ on the unit interval.

Proof. For $p = 1$, $T_1 = U_0/2$, and direct evaluation gives

$$C_1(x) = \max\left(0, \min\left(1, 2x - \frac{1}{2}\right)\right).$$

Formula (56) gives exactly the same broken-linear function: the term $r = 0$ starts at $x = 1/4$, and the term $r = 1$ subtracts the excess after $x = 3/4$.

For the induction step, split off the first uniform coordinate. The finite sums satisfy

$$T_{p+1} \stackrel{d}{=} \frac{U_0 + T'_p}{2},$$

where T'_p is an independent copy of T_p . Conditioning on U_0 and accounting for the midpoint shifts gives the exact smoothing recurrence

$$C_{p+1}(x) = \int_0^1 C_p(2x - u) du. \quad (58)$$

On the spline side, integrate each truncated power in (56). The two binary children $2r$ and $2r + 1$ have opposite Thue–Morse signs, because $\varepsilon_{2r} = \varepsilon_r$ and $\varepsilon_{2r+1} = -\varepsilon_r$. After the change of scale this yields

$$S_{p+1}(x) = \int_0^1 S_p(2x - u) du. \quad (59)$$

The normalization is exactly the one required by

$$\sum_{r=0}^{2^p-1} \varepsilon_r r^p = (-1)^p 2^{\binom{p}{2}} p!,$$

while every lower power sum vanishes. Thus the two recurrences and their common base case prove (57). $\square$

Theorem 4.5 (Quantitative centered approximation). *For $p \geq 1$ and $0 \leq x \leq 1$,*

$$F(x - 2^{-p-1}) \leq S_p(x) \leq F(x + 2^{-p-1}), \quad (60)$$

and hence

$$|S_p(x) - F(x)| \leq 2^{-p}. \quad (61)$$

Proof. Write $X = T_p + R_p$, where the omitted tail satisfies

$$0 \leq R_p = \sum_{i=p}^{\infty} \frac{U_i}{2^{i+1}} \leq 2^{-p}.$$

If $X \leq x - 2^{-p-1}$, then certainly $T_p \leq x - 2^{-p-1}$. Conversely, if $T_p \leq x - 2^{-p-1}$, then $X \leq x + 2^{-p-1}$. Taking probabilities and using (37) and (57) proves (60).

The bounded Fabius function is globally 2-Lipschitz. Indeed, on the open unit interval its derivative is a rescaled value of up (or of the signed extension), whose absolute value is at most 1, and outside the unit interval F is constant. Thus

$$|F(x \pm 2^{-p-1}) - F(x)| \leq 2 \cdot 2^{-p-1} = 2^{-p}.$$

Combining these two inequalities with the sandwich gives (61). $\square$

The bound is uniform on the closed interval and includes both endpoints. It is stronger than mere weak convergence of the finite distributions, and it is the quantitative input for the computability proof in section 15.

The same finite formula also has the correct signed global behavior. For $x \leq 0$ its safe cutoff is empty, so $S_p(x) = \mathcal{F}(x) = 0$. On the nonnegative half-line, binary block translation and midpoint reflection extend the local identification. Consequently, for every $p \geq 1$ and every $x \in \mathbb{R}$,

$$|S_p(x) - \mathcal{F}(x)| \leq 2^{-p}. \quad (62)$$

Indeed, every nonnegative two-unit block is a common signed copy of the first block, and on the second unit cell both functions use the same complement. Neither operation changes the absolute error in (61); the nonpositive half-line is exact. This pointwise all-real bound is `abs_fabiusUniformSpline_sub_extendedFabius_le`; it implies the separately proved uniform convergence `fabiusUniformSpline_tendstoUniformly_extendedFabius`. This global form strengthens and packages the analytic input used by the complex-shift and discrete-limit results below.

5 Fourier products, inversion, and Poisson summation

Use the convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i x z} dx. \quad (63)$$

The formal Fourier transform of up is entire and has two exact infinite-product descriptions.

Theorem 5.1 (Sinc and cosine products). *For every $z \in \mathbb{C}$,*

$$\widehat{\text{up}}(z) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^n}\right), \quad (64)$$

$$= \prod_{m=1}^{\infty} \left(\cos \frac{\pi z}{2^m}\right)^m. \quad (65)$$

The transform also satisfies

$$\widehat{\text{up}}(z) = \text{sinc}(\pi z) \widehat{\text{up}}(z/2). \quad (66)$$

Formal proof. The finite-product refinement and its infinite limit are `rvachevFourier_eq_finite_product` and `rvachevFourier_eq_product` in `FourierProduct.lean`. The refinement equation is `rvachevFourier_scaling` in the same module. Writing Φ for the first product in (64), the standalone one-step and n -fold renormalizations

$$\Phi(z) = \text{sinc}(\pi z) \Phi(z/2), \quad \Phi(z) = \left(\prod_{j < n} \text{sinc}(\pi z/2^j) \right) \Phi(z/2^n)$$

are `rvachevFourierProduct_scaling` and `rvachevFourierProduct_shell` in `RenormalizationIdentity.lean`. The whole-plane analyticity theorem is `rvachevFourier_analyticOnNhd` in `FourierAnalytic.lean`. The weighted cosine product is `rvachevFourierProduct_eq_cosineProduct` in `OriginalPaperSupplement.lean`; its finite telescoping identity is proved internally in the same module. $\square$

Theorem 5.2 (Dyadic shell factorization). *For every $k \in \mathbb{N}$ and $z \in \mathbb{C}$, dilation by 2^k extracts exactly the first k factors of the sinc product:*

$$\widehat{\text{up}}(2^k z) = \left(\prod_{j=0}^{k-1} \text{sinc}(\pi 2^{j+1} z) \right) \widehat{\text{up}}(z). \quad (67)$$

Formal proof. This is `rvachevFourier_two_pow_mul` in `RvachevFourierDecay.lean`. Its standalone-product form is `rvachevFourierProduct_two_pow_mul` in `SincProductShells.lean`. On the real axis the removable singularity can be cleared without excluding the zero ray: for every $k \in \mathbb{N}$ and $y \in \mathbb{R}$,

$$\begin{aligned} 2^{k(k+1)/2} (\pi|y|)^k \left| \widehat{\text{up}}(2^k y) \right| \\ = \left(\prod_{j=0}^{k-1} |\sin(2^{j+1} \pi y)| \right) |\widehat{\text{up}}(y)|. \end{aligned} \quad (68)$$

This total form includes $y = 0$, and at $k = 0$ both sides are the same Fourier modulus. It is `norm_rvachevFourier_two_pow_mul_cross` in `RvachevFourierDecay.lean`; its product-level source is `norm_rvachevFourierProduct_two_pow_mul_cross` in `SincProductShells.lean`. $\square$

The mean-value input behind shell estimates is more general than the dyadic case. Let $m_0, m_1, \dots$ be positive integer frequencies satisfying $2m_j \leq m_{j+1}$. Then, for every $n \in \mathbb{N}$,

$$\int_0^1 \prod_{j=0}^{n-1} \sin^2(\pi m_j t) \, dt = \left(\frac{1}{2}\right)^n, \quad (69)$$

$$\int_0^1 \prod_{j=0}^{n-1} |\sin(\pi m_j t)| \, dt \leq \left(\frac{\sqrt{2}}{2}\right)^n. \quad (70)$$

The exact square mean is `integral_prod_sin_sq` in `LacunaryRieszIntegral.lean`. Applying Cauchy–Schwarz to it gives `integral_prod_abs_sin_le_sqrt_two` in `LacunaryMeanSharp.lean`. The geometric-base and dyadic specializations are `integral_prod_abs_sin_pow_le_sqrt_two` and `integral_prod_abs_sin_two_pow_le_sqrt_two`.

The logarithm of the normalized dyadic sine shell has an exact square mean as well. For every $n \in \mathbb{N}$, including the empty product $n = 0$,

$$\int_0^1 \log^2 \left| \prod_{j=0}^{n-1} 2 \sin(\pi 2^j t) \right| dt = \frac{\pi^2}{4} n - \frac{\pi^2}{3} (1 - 2^{-n}). \quad (71)$$

The product statement is `integral_sq_log_prod_two_sin_all` in `BirkhoffProductBridge.lean`. Its doubling-cocycle form is `integral_sq_birkhoff_sum_all` in `BirkhoffVariance.lean`, and the finite covariance summation is `variance_closed_form_all` in `VarianceBookkeeping.lean`. The all-index declarations make the zero-length boundary an identity rather than a side condition.

For a quantitative global form, set

$$\kappa_\infty = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2}, \quad E_\kappa(x) = \exp\left(-\frac{(\log x)^2}{2 \log 2}\right) x^{-\kappa} \quad (x > 0). \quad (72)$$

There is a constant $C > 0$ such that the single two-sided estimate

$$|\widehat{\text{up}}(x)| \leq C E_{\kappa_\infty}(|x|) \quad (|x| \geq 1) \quad (73)$$

holds on both real tails. This is `norm_rvachevFourier_le_decayGauge_abs` in `RvachevFourierDecay.lean`. On the distinguished dyadic ray the same gauge occurs in an exact identity:

$$|\widehat{\text{up}}(2^k 2/3)| = \frac{|\widehat{\text{up}}(2/3)|}{E_{\kappa_\infty}(2/3)} E_{\kappa_\infty}(2^k 2/3), \quad (74)$$

namely `norm_rvachevFourier_peak_ray_envelope` in the same module.

The exponent κ_∞ is sharp within this family of gauges. Write Φ for the standalone sinc product, which equals $\widehat{\text{up}}$ by `rvachevFourier_eq_product`. For every $\kappa > \kappa_\infty$,

$$\frac{|\Phi(2^k 2/3)|}{E_\kappa(2^k 2/3)} \rightarrow +\infty \quad (k \rightarrow \infty). \quad (75)$$

More explicitly, for every real C there is a $k \in \mathbb{N}$ such that $x_k = 2^k 2/3 \geq 1$ and

$$C E_\kappa(x_k) < |\Phi(x_k)|. \quad (76)$$

Consequently no $C > 0$ satisfies

$$|\Phi(x)| \leq C E_\kappa(x) \quad \text{for every real } x \geq 1.$$

The limit, explicit peak-ray obstruction, and positive-ray nonexistence theorem are `norm_rvachevFourierProduct_peak_ray_ratio_tendsto_atTop`, `exists_peak_ray_norm_gt_mul_decayGauge`, and `not_exists_norm_rvachevFourierProduct_le_decayGauge_of_kappaInf_lt` in `GlobalDecayEnvelope.lean`.

The transform also has the exact conjugation, parity, and real-axis identities

$$\overline{\widehat{\text{up}}(z)} = \widehat{\text{up}}(-\bar{z}), \quad \widehat{\text{up}}(-z) = \widehat{\text{up}}(z), \quad \text{Im } \widehat{\text{up}}(t) = 0 \quad (t \in \mathbb{R}). \quad (77)$$

These are `rvachevFourier_conj` in `FourierAnalytic.lean` and `rvachevFourier_neg`, `rvachevFourier_ofReal_im_eq_zero` in `FourierProduct.lean`.

Theorem 5.3 (Exact Fourier zero set and positive-integer zero orders). *The complex zeros of the Fourier transform are precisely the nonzero integers:*

$$\widehat{\text{up}}(z) = 0 \iff \exists m \in \mathbb{Z} \setminus \{0\} \text{ such that } z = m, \quad z \in \mathbb{C}. \quad (78)$$

At every positive integer m , its real-axis zero order is exactly $v_2(m) + 1$, in the following quantitative sense: there is a constant $C_m > 0$ such that

$$\lim_{\substack{x \rightarrow m \\ x \neq m}} \frac{|\widehat{\text{up}}(x)|}{|x - m|^{v_2(m)+1}} = C_m. \quad (79)$$

The corresponding canonical product records all of these orders at once:

$$\widehat{\text{up}}(z) = \prod_{r=0}^{\infty} \left(1 - \frac{z^2}{(r+1)^2}\right)^{1+v_2(r+1)}. \quad (80)$$

Formal proof. The complete complex zero-set equivalence is `rvachevFourier_eq_zero_iff` in `FourierProduct.lean`. The positive-integer limit is `tendsto_norm_rvachevFourier_div_pow_zero_order` in `RvachevFourierDecay.lean`; its limiting constant is exhibited there as an exponential, and is therefore strictly positive. The canonical product is `rvachevFourier_eq_canonical` in the same facade module, built from `rvachevFourierProduct_eq_canonical` in `SincCanonicalProduct.lean`. $\square$

On the real axis the origin is the unique global point of unit modulus:

$$|\widehat{\text{up}}(x)| = 1 \iff x = 0, \quad x \neq 0 \implies |\widehat{\text{up}}(x)| < 1. \quad (81)$$

These are `norm_rvachevFourier_eq_one_iff` and `norm_rvachevFourier_lt_one_of_ne_zero`. On each positive real lobe the sign and peak are also exact. If $m \in \mathbb{N}$ and $m < x < m + 1$, then

$$\widehat{\text{up}}(x) = \varepsilon_m |\widehat{\text{up}}(x)|, \quad (82)$$

by `rvachevFourier_eq_thueMorse_sign_mul_norm`. If moreover $m \geq 1$, there is exactly one point of $(m, m + 1)$ maximizing the modulus over that open lobe and exactly one point of the reflected lobe $(-(m + 1), -m)$ maximizing it there. These are `existsUnique_isMaxOn_rvachevFourier_lobe` and `existsUnique_isMaxOn_rvachevFourier_neg_lobe`. All five facade declarations are in `RvachevFourierDecay.lean`.

Globally, the origin is the unique real frequency of unit modulus:

$$|\widehat{\text{up}}(x)| \leq 1, \quad |\widehat{\text{up}}(x)| = 1 \iff x = 0. \quad (83)$$

Equivalently, the first inequality is strict whenever $x \neq 0$. These statements are `norm_rvachevFourier_le_one`, `norm_rvachevFourier_lt_one_of_ne_zero`, and `norm_rvachevFourier_eq_one_iff` in `RvachevFourierDecay.lean`.

5.1 Rapid decay and inversion

For every $N \in \mathbb{N}$ there is a constant $C_N > 0$ such that

$$|\xi|^N |\widehat{\text{up}}(\xi)| \leq C_N \quad (\xi \in \mathbb{R}), \quad (84)$$

and the companion formal theorem gives the corresponding polynomially weighted bound for every iterated real-frequency derivative. Fourier inversion holds in the exact form

$$\text{up}(x) = \int_{\mathbb{R}} \widehat{\text{up}}(\xi) e^{2\pi i x \xi} d\xi. \quad (85)$$

These are `rvachevFourier_real_rapidDecay`, `rvachevFourier_real_iteratedDeriv_rapidDecay`, and `rvachev_fourier_inversion_analytic` in `PoissonSummation.lean` and `FourierAnalytic.lean`.

5.2 Poisson summation and partitions of unity

For $u > 0$ and $t \in \mathbb{R}$, the formal Poisson identity is

$$\sum_{k \in \mathbb{Z}} \text{up}(t + uk) = \frac{1}{u} \sum_{k \in \mathbb{Z}} \widehat{\text{up}}(k/u) e^{2\pi i kt/u}. \quad (86)$$

It is `rvachev_poisson_summation`. Combining it with (78) gives, for every positive integer m ,

$$\sum_{k \in \mathbb{Z}} \text{up}\left(t + \frac{k}{m}\right) = m, \quad (87)$$

and in particular

$$\sum_{k \in \mathbb{Z}} \text{up}(t + k) = 1. \quad (88)$$

These are `rvachev_partition_one_over_nat` and `rvachev_partition_unity` in `PoissonSummation.lean`.

5.3 Moment expansion of the Fourier transform

With

$$c_n = \int_{-1}^1 x^{2n} \text{up}(x) dx. \quad (89)$$

the exact entire series is

$$\widehat{\text{up}}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n c_n}{(2n)!} (2\pi z)^{2n}. \quad (90)$$

This is `rvachevFourier_eq_momentSeries` in `AnalyticMoments.lean`.

6 The global extension, derivatives, flatness, and inverse

6.1 The signed extension and all iterated derivatives

The translate series (11) is locally finite. It agrees with the bounded F on $[0, 1]$, vanishes on $(-\infty, 0]$, and satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (91)$$

These facts are `extendedFabius_eq_fabiusReal`, `extendedFabius_eq_zero_of_nonpos`, and `extendedFabius_hasDerivAt` in `GlobalExtension.lean`.

Iterating the dilation law gives the exact identities

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x), \quad (92)$$

$$\text{up}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m(x+1)) \quad (-1 \leq x \leq 1). \quad (93)$$

They are `iteratedDeriv_extendedFabius` and `iteratedDeriv_rvachev`. The global extension obeys

$$|\mathcal{F}(x)| \leq 1, \quad (94)$$

by `abs_extendedFabius_le_one`. Hence, for all $m \in \mathbb{N}$ and $x \in \mathbb{R}$,

$$|F^{(m)}(x)| \leq 2^{\binom{m+1}{2}}, \quad (95)$$

and this bound is attained at $x = 2^{-m}$:

$$F^{(m)}(2^{-m}) = 2^{\binom{m+1}{2}}. \quad (96)$$

The exact bound, attainment, and supremum statements are `abs_iteratedDeriv_fabiusReal_le`, `iteratedDeriv_fabiusReal_inv_two_pow_all`, and `isGreatest_abs_iteratedDeriv_fabiusReal_all` in `BoundedDerivatives.lean`.

6.2 Effective flatness

For every $n \in \mathbb{N}$, $x \geq 0$, and $2^n x \leq 1$, the formal development proves both

$$F(x) \leq 2^{\binom{n+1}{2}} x^n, \quad (97)$$

$$F(x) \leq \frac{2^{\binom{n+1}{2}}}{n!} x^n. \quad (98)$$

These are `fabiusReal_le_two_pow_mul_pow` in `EffectiveFlatness.lean` and `fabiusReal_le_two_pow_div_factorial_mul_pow` in `SharpFlatness.lean`. Their reflected forms at 1 and their folded forms for up are also formalized in those modules. In particular, for each natural power,

$$F(x) = o(x^n) \quad (x \rightarrow 0),$$

as stated by `fabiusReal_isLittleO_pow_at_zero` in `FabiusFlatness.lean`.

6.3 The precise analytic statements

The exact real-analytic locus of the bounded function is

$$F \text{ is analytic at } x \iff x \notin [0, 1]. \quad (99)$$

This is `fabius_analyticAt_iff`, with canonical specialization `canonical_fabius_analyticAt_iff`. For the other global forms the current proved statements are deliberately one-sided:

$$\text{up is not analytic at any } x \in [-1, 1], \quad (100)$$

$$\mathcal{F} \text{ is not analytic at any } x \in [0, 2). \quad (101)$$

They are `canonical_rvachev_not_analyticAt` and `extendedFabius_not_analyticAt` in `NowhereAnalytic.lean`. No stronger global locus is inferred here.

6.4 The totalized inverse

Let $I : \mathbb{R} \rightarrow [0, 1]$ be the inverse order isomorphism on $[0, 1]$, extended constantly by 0 on the left and 1 on the right. This is the formal `fabiusInv` constructed from `fabiusOrderIso`. It is continuous and nondecreasing; on $[0, 1]$ it is a strictly increasing bijection and satisfies

$$F(I(y)) = y, \quad I(F(x)) = x \quad (x, y \in [0, 1]). \quad (102)$$

For $0 < y < 1$, it is smooth and

$$I'(y) = \frac{1}{F'(I(y))} = \frac{1}{2 \text{up}(2I(y) - 1)} > 0. \quad (103)$$

Formal proof. The range and inverse identities are `fabiusInv_mem_Icc`, `fabiusReal_fabiusInv`, and `fabiusInv_fabiusReal`. The order, continuity, and bijectivity statements are `strictMonoOn_fabiusInv`, `monotone_fabiusInv`, `continuous_fabiusInv`, and `bijOn_fabiusInv`. Interior smoothness and the derivative chain are `fabiusInv_contDiffOn_Ioo`, `deriv_fabiusInv`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, and `deriv_fabiusInv_pos`, all in `FabiusInverse.lean`. $\square$

The same module controls the curvature of the inverse without asserting differentiability at the clamped endpoints. For $0 < y < 1$,

$$I''(y) = -\frac{F''(I(y))}{F'(I(y))^3}. \quad (104)$$

This is `deriv_deriv_fabiusInv`, with the corresponding derivative witness `deriv_fabiusInv_hasDerivAt`. On the closed half intervals, continuity at the endpoints and the interior derivative criterion give

$$I \text{ is strictly concave on } [0, \tfrac{1}{2}] \text{ and strictly convex on } [\tfrac{1}{2}, 1]. \quad (105)$$

The exact declarations are `strictConcaveOn_fabiusInv_firstHalf` and `strictConvexOn_fabiusInv_secondHalf`.

The pointwise statement is sharper. The midpoint jet is

$$I(1/2) = 1/2, \quad I'(1/2) = 1/2, \quad I''(1/2) = 0, \quad (106)$$

and, for $0 < y < 1$,

$$\begin{aligned} I''(y) < 0 &\iff 0 < y < \tfrac{1}{2}, \\ I''(y) > 0 &\iff \tfrac{1}{2} < y < 1, \\ I''(y) = 0 &\iff y = \tfrac{1}{2}. \end{aligned} \quad (107)$$

The exact declarations are `fabiusInv_half`, `deriv_fabiusInv_half`, `deriv_deriv_fabiusInv_half`, and the `deriv_deriv_fabiusInv_neg_iff`, `deriv_deriv_fabiusInv_pos_iff`, and `deriv_deriv_fabiusInv_eq_zero_iff` family. The hypothesis $0 < y < 1$ is essential in the last three statements: the totalized inverse is not differentiable at the clamping endpoints.

The inverse inherits the exact symmetry and midpoint

$$I(1 - y) = 1 - I(y), \quad I(1/2) = 1/2, \quad (108)$$

and for $0 \leq a \leq 2^n$,

$$I\left(F\left(\frac{a}{2^n}\right)\right) = \frac{a}{2^n}. \quad (109)$$

These are `fabiusInv_one_sub`, `fabiusInv_half`, and `fabiusInv_fabiusDyadicUnit_of_1e`. Its exact analytic locus is likewise

$$I \text{ is analytic at } y \iff y \notin [0, 1], \quad (110)$$

by `fabiusInv_analyticAt_iff`; in the open interval its derivative never vanishes by `deriv_fabiusInv_ne_zero`.

Flatness of F becomes infinite endpoint steepness of I . For every $n \in \mathbb{N}$,

$$y = o(I(y)^n) \quad (y \downarrow 0), \quad (111)$$

$$1 - y = o((1 - I(y))^n) \quad (y \uparrow 1). \quad (112)$$

These are `id_isLittleO_fabiusInv_pow_at_zero_right` and `one_sub_isLittleO_one_sub_fabiusInv_pow_at_one_left`. In particular, the two one-sided difference quotients diverge:

$$\frac{I(y)}{y} \longrightarrow +\infty \quad (y \downarrow 0), \quad \frac{1 - I(y)}{1 - y} \longrightarrow +\infty \quad (y \uparrow 1). \quad (113)$$

The formally proved counterparts are `tendsto_fabiusInv_div_atTop` and `tendsto_one_sub_fabiusInv_div_one_sub_atTop`. Thus the totalized inverse is not differentiable at either clamping point, exactly as stated by `fabiusInv_not_differentiableAt_zero` and `fabiusInv_not_differentiableAt_one`; no finite endpoint derivative is being smuggled into the curvature statement above. The stronger limits of the positive interior derivative itself are also formalized:

$$I'(y) \longrightarrow +\infty \quad (y \downarrow 0), \quad I'(y) \longrightarrow +\infty \quad (y \uparrow 1). \quad (114)$$

They are `tendsto_deriv_fabiusInv_atTop_at_zero_right` and `tendsto_deriv_fabiusInv_atTop_at_one_left`; these are limits through the open interval, not finite derivatives at 0 or 1.

6.5 Precise non-elementarity

The inductive predicate `IsElementary` contains constants and the identity and is closed under arithmetic operations, composition, exponential, logarithm, trigonometric operations, and variable real powers whose elementary base is globally nonvanishing. The last closure statement is `IsElementary.rpow_of_ne_zero`; the positive-base and positive-region interfaces are `IsElementary.rpow_of_pos` and `eqOn_rpow_of_pos`. Every expression in this class has a dense real-analytic locus:

$$\text{IsElementary } g \implies \overline{\{x : g \text{ is analytic at } x\}} = \mathbb{R}.$$

This is `IsElementary.dense_analyticLocus`. It yields the proved global obstructions

$$F, \quad \text{up}, \quad \mathcal{F}, \quad I \quad \text{are not elementary.} \quad (115)$$

The exact declarations are `canonical_fabijs_not_isElementary`, `rvachev_not_isElementary`, `extendedFabijs_not_isElementary`, and `fabijsInv_not_isElementary`; local versions for F and I on $(0, 1)$ are `canonical_fabijs_not_isElementary_on_Ioo` and `canonical_fabijsInv_not_isElementary_on_Ioo`.

The larger formal class `IsElementaryOrInverse` admits an inverse-branch constructor only with an open branch domain, continuity and a right-inverse identity there, and analyticity of the chosen total extension on the interior of the complement. The canonical global functions themselves are not members of that larger class:

$$\neg \text{IsElementaryOrInverse } F, \quad \neg \text{IsElementaryOrInverse } I. \quad (116)$$

These are `fabijs_not_isElementaryOrInverse` and `fabijsInv_not_isElementaryOrInverse`. More strongly in a different direction, no member of the class can represent F or I even merely on $(0, 1)$, by `canonical_fabijs_not_isElementaryOrInverse_on_Ioo` and `canonical_fabijsInv_not_isElementaryOrInverse_on_Ioo` in `InverseNotElementary.lean`.

7 Moments, generating functions, and reciprocal powers

7.1 Even and half moments

Retain the even moments c_n from (89); in particular, $c_0 = 1$. Define

$$d_0 = 1, \quad d_n = n \int_0^1 t^{n-1} \text{up}(t) \, dt \quad (n \geq 1). \quad (117)$$

Both are positive rational sequences. Their exact triangular recurrences, for $n \geq 1$, are

$$(2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k, \quad (118)$$

$$(n+1)(2^n-1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k. \quad (119)$$

These are the rational declarations `moment_succ` and `halfMoment_succ` in `Arithmetic.lean`; positivity is `moment_pos` in `Paper06487Supplement.lean` and `halfMoment_pos` in `Arithmetic.lean`. The displayed integral definitions are `integral_even_pow_mul_rvachev_eq_moment`, `moment_eq_integral_formula`, and `halfMoment_eq_integral_formula_all` in `AnalyticMoments.lean`.

The entire generating function

$$G(z) = 1 + z \int_0^1 \text{up}(t) e^{zt} dt = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!} \quad (120)$$

satisfies

$$G(2z) = \frac{e^z - 1}{z} G(z), \quad (121)$$

with the removable value at $z = 0$. The exact formal statements are `proposition_two_formula` and `complexGeneratingFunction_eq_series_formula` in `AnalyticMoments.lean`; entire differentiability is `differentiable_complexGeneratingFunction` in `NegativeLaplaceVerticalLog.lean`.

The two sequences are linked by

$$d_n = 2^{-n} \sum_{0 \leq k \leq n/2} \binom{n}{2k} c_k, \quad (122)$$

and, at odd indices,

$$d_{2k+1} = \frac{2k+1}{2} c_k. \quad (123)$$

These are `halfMoment_eq_evenMomentSum` and `halfMoment_odd_eq_moment` in `MomentPowerSeries.lean`. Division-free natural numerators for c_n and d_n , together with their exact recurrences and odd-index bridge, are formalized by `momentNumerator`, `halfMomentNumerator`, `halfMomentNumerator_succ`, `halfMomentNumerator_odd_index_eq`, and `halfMomentNumerator_odd_index_formula`. No unproved numerical table is needed.

More explicitly, if $C_n = \text{momentNumerator}(n)$, then C_n is a positive integer and

$$c_n = \frac{C_n}{n \prod_{j=1}^n (2^{2j} - 1)}. \quad (124)$$

The exact rational identity is `moment_eq_momentNumerator_div`; the product reindexing is `evenMersenneProduct_eq_prod_Ico`, and positivity is `momentNumerator_pos`. Thus (124) records the proved natural normalization, not a claim that the displayed denominator is already reduced.

7.2 Universal Bell and cumulant transforms

The algebra converting moments to cumulants is independent of the particular Fabius law. Let R be any commutative $\mathbb{Q}$ -algebra and let $a : \mathbb{N} \rightarrow R$. Write

$$(\mathcal{N}a)_n = \frac{a_n}{n!}, \quad (\mathcal{D}a)_n = n! a_n. \quad (125)$$

These changes of coefficient convention are mutually inverse. For cumulants $\kappa : \mathbb{N} \rightarrow R$ and moments $\mu : \mathbb{N} \rightarrow R$, define the complete Bell and moment–cumulant transforms by the formal identities

$$\mathcal{B}(\kappa)_n = n! [X^n] \exp \left(\sum_{j \geq 1} \kappa_j \frac{X^j}{j!} \right), \quad (126)$$

$$\mathcal{K}(\mu)_n = n! [X^n] \log \left(1 + \sum_{j \geq 1} \mu_j \frac{X^j}{j!} \right). \quad (127)$$

Here the exponential and logarithm are formal power series implemented by finite coefficient recurrences; no topology or field structure on R is being assumed.

Proposition 7.1 (Bell recurrence and exact inversion). *For every $n \in \mathbb{N}$,*

$$\mathcal{B}(\kappa)_0 = 1, \quad \mathcal{K}(\mu)_0 = 0, \quad (128)$$

$$\mathcal{B}(\kappa)_{n+1} = \sum_{j=0}^n \binom{n}{j} \kappa_{j+1} \mathcal{B}(\kappa)_{n-j}. \quad (129)$$

At every index, including zero, the two composites are exactly

$$\mathcal{B}(\mathcal{K}(\mu))_n = \begin{cases} 1, & n = 0, \\ \mu_n, & n > 0, \end{cases} \quad \mathcal{K}(\mathcal{B}(\kappa))_n = \begin{cases} 0, & n = 0, \\ \kappa_n, & n > 0. \end{cases} \quad (130)$$

Consequently the transforms are inverse as functions when $\mu_0 = 1$ and $\kappa_0 = 0$, respectively. More precisely, if $f : R \rightarrow S$ is a morphism of commutative $\mathbb{Q}$ -algebras, then

$$f(\mathcal{B}_R(\kappa)_n) = \mathcal{B}_S(f \circ \kappa)_n, \quad f(\mathcal{K}_R(\mu)_n) = \mathcal{K}_S(f \circ \mu)_n. \quad (131)$$

More generally, if E and A are ordinary coefficient sequences for the formal exponent and mass series, then the coefficient-convention bridge is

$$\mathcal{B}(\mathcal{D}E) = \mathcal{D}(\text{expCoeff}(E)), \quad \mathcal{K}(\mathcal{D}A) = \mathcal{D}(\text{logCoeff}(A)). \quad (132)$$

Thus the same theorem applies without modification to polynomial rings, function rings, and scalar extensions, not only to real- or rational-valued moment sequences.

Formal proof. The operations $\mathcal{N}, \mathcal{D}, \mathcal{B}, \mathcal{K}$ are respectively `factorialNormalize`, `factorialDenormalize`, `completeBellPolynomial`, and `momentCumulant`. The two identities following (125) are `factorialNormalize_factorialDenormalize` and `factorialDenormalize_factorialNormalize`. The zero values and (129) are `completeBellPolynomial_zero`, `momentCumulant_zero`, and `completeBellPolynomial_succ`.

Equation (130) is `completeBellPolynomial_momentCumulant` together with `momentCumulant_completeBellPolynomial`; its normalized function-level forms are `completeBellPolynomial_momentCumulant_of_zero_eq_one` and `momentCumulant_completeBellPolynomial_of_zero_eq_zero`. The bridge (132) is `completeBellPolynomial_factorialDenormalize` and `momentCumulant_factorialDenormalize`. Finally, functoriality is (131), formalized by `map_completeBellPolynomial` and `map_momentCumulant`. All declarations are in `MomentCumulantAlgebra.lean` and assume only a commutative $\mathbb{Q}$ -algebra. $\square$

7.3 The probability–Laplace bridge

Let μ_X be the distribution of the weighted sum X from (33). Its survival function on the nonnegative half-line is exactly

$$\mu_X((t, \infty)) = \text{up}(t) \quad (t \geq 0). \quad (133)$$

If $g, g' : \mathbb{R} \rightarrow \mathbb{R}$ are continuous and g has derivative $g'(t)$ at every $t \in [0, 1]$, then

$$\int_{[0,1]} g(x) \, d\mu_X(x) = g(0) + \int_0^1 g'(t) \, \text{up}(t) \, dt. \quad (134)$$

For $s \in \mathbb{R}$ and $k \in \mathbb{N}$, put

$$L_k(s) = \int_{[0,1]} e^{-sx} x^k \, d\mu_X(x). \quad (135)$$

These raw tilted moments agree with the formal analytic Fabius Laplace moments, and at zero tilt

$$L_n(0) = \int_{[0,1]} (1-x)^n \, d\mu_X(x) = d_n. \quad (136)$$

On the real axis, every derivative of the generating function is a tilted moment:

$$G^{(n)}(s) = L_n(-s), \quad G^{(n)}(0) = d_n. \quad (137)$$

Reflection of the law about $1/2$ gives

$$L_0(s) = e^{-s} L_0(-s), \quad e^{s/2} L_0(s) = e^{-s/2} L_0(-s), \quad (138)$$

so $s \mapsto e^{s/2} L_0(s)$ is even. Two useful quantitative forms, with all variables real, are

$$L_0\left(\frac{a+b}{2}\right)^2 \leq L_0(a)L_0(b) \quad (a, b \in \mathbb{R}), \quad (139)$$

$$L_k(s) \leq (1/a)^k k! L_0(s-a) \quad (a > 0). \quad (140)$$

Formal proof. The survival identity and integration-by-parts formula are `weightedSumDistribution_real_Ioi_eq_rvachevUp_of_nonneg` and `integral_unit_eq_zero_add_in_tegral_deriv_mul_rvachevUp`. The tilted and zero-tilt identities are `unitLaplaceMoment_weightedSumDistribution_eq_fabiusLaplaceMoment`, `unitLaplaceMoment_weightedSumDistribution_zero_eq_halfMoment`, and `unitEndpointMoment_weightedSumDistribution_eq_halfMoment` in `ProbabilityLaplaceMoments.lean`. The derivative identities are `iteratedDeriv_generatingFunction` and `iteratedDeriv_generatingFunction_zero` in `LaplaceMoments.lean`; (138) is `fabiusLaplaceMoment_zero_reflection` and `fabiusLaplaceMoment_zero_centered_even`. Finally, (139)–(140) are `fabiusLaplaceMoment_midpoint_sq_le_all` and `fabiusLaplaceMoment_le_shift` in `UnitLaplaceMomentBounds.lean`. $\square$

7.4 Exact values at reciprocal powers of two

For every $n \in \mathbb{N}$,

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}), \quad F(2^{-n}) = \frac{d_n}{n! 2^{\binom{n}{2}}}. \quad (141)$$

This is `halfMoment_eq_fabius_formula` in `AnalyticMoments.lean`. The numerator-cleared version is `fabiusAtInverseTwoPow_eq_halfMomentNumerator_formula` in `ExactInversePower.lean`.

Writing $V_n = F(2^{-n})$, one has $V_0 = 1$, and for $n \geq 1$,

$$V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} V_k. \quad (142)$$

This is `fabiusAtInverseTwoPow_recurrence`. The same sequence has exact finite path and ordered-composition descriptions, proved by `fabiusAtInverseTwoPow_eq_pathSum`, `fabiusAtInverseTwoPow_eq_composition_formula`, `pathWeight_eq_boundary_product`, and `fabiusRecurrenceSequence_eq_sum_compositions` in `FabiusInverseDyadicClosedForm.lean`. Their positivity and recurrence uniqueness are `fabiusCompositionSum_pos` and `triangularRecurrence_unique`.

The totalized inverse sends both exact rational presentations straight back to their arguments. In particular, for every $n \in \mathbb{N}$, including $n = 0$,

$$I(\text{fabiusAtInverseTwoPow } n) = 2^{-n}, \quad (143)$$

$$I\left(2^{-\binom{n}{2}} \text{fabiusCompositionSum } n\right) = 2^{-n}. \quad (144)$$

These are `fabiusInv_fabiusAtInverseTwoPow` in `FabiusInverse.lean` and `fabiusInv_two_pow_choose_mul_fabiusCompositionSum` in `FabiusInverseDyadicClosedForm.lean`.

7.5 Parity and the exact two-adic valuation

Lucas parity is packaged as `odd_binomial_coefficient_counts` and `card_odd_inner_binomial_coefficients`. It implies the exact numerator theorem `momentNumerator_odd`. For every $n \geq 1$, the reduced numerator and denominator of $2d_n$ are both odd, equivalently

$$\nu_2(d_n) = -1. \quad (145)$$

This is `two_mul_halfMoment_num_den_odd` together with `halfMoment_padicVal_two`.

Consequently, for $n \geq 1$,

$$\nu_2(F(2^{-n})) = -\binom{n}{2} - 1 - \nu_2(n!), \quad (146)$$

by `fabijsAtInverseTwoPow_padicVal_two`. The binary form uses the proved Legendre identity

$$\nu_2(n!) = n - \text{wt}_2(n), \quad (147)$$

namely `factorial_padicVal_two`; the combined declaration is `fabijsAtInverseTwoPow_padicVal_two_closed` in `TwoAdic.lean`.

7.6 Bernoulli recurrences

The formal development also proves Bernoulli convolution and recurrence forms for both sequences. For $n \geq 1$,

$$c_n = \frac{1}{2^{2n} - 1} \sum_{k=1}^n 2^{2n-2k} (2^{2k} - 2) \binom{2n}{2k} B_{2k} c_{n-k}, \quad (148)$$

$$d_n = \frac{n 2^n}{4(2^n - 1)} d_{n-1} - \frac{1}{2^n - 1} \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{2k} 2^{n-2k} B_{2k} d_{n-2k}. \quad (149)$$

Here $B_1 = -1/2$. These are `moment_bernoulli_recurrence` and `halfMoment_bernoulli_recurrence`; their convolution forms and combined source-paper statement are `moment_bernoulli_convolution`, `halfMoment_bernoulli_convolution`, and `proposition_twenty_two_formula` in `BernoulliRecurrences.lean`.

8 Exact evaluation at arbitrary dyadic rationals

The rationality theorem for dyadic arguments comes with two complementary exact presentations:

1. a closed finite formula involving Thue–Morse signs and the rational moments c_k ;
2. an executable bit-recursive Taylor evaluator.

8.1 The closed Thue–Morse formula

Theorem 8.1 (Exact dyadic formula). *Let $n, a \in \mathbb{N}$ with $0 \leq a \leq 2^n$. Then*

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-(\frac{n+1}{2})}}{n!} \sum_{h=0}^{a-1} (-1)^{\text{wt}_2(h)} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a - 2h - 1)^{n-2k}. \quad (150)$$

An empty outer sum has value 0.

Formal proof. The rational expression on the right is the definition `fabiusDyadic`. Its equality, after coercion to $\mathbb{R}$, with $F(a/2^n)$ under the displayed bound on a is `fabiusDyadic_cast` in `DyadicAnalytic.lean`. $\square$

Every quantity on the right of (150) is rational, so dyadic rationality is immediate. More importantly, the formula lives entirely in $\mathbb{Q}$ and can be evaluated without roundoff.

Theorem 8.2 (Global dyadic formula). *For all $n, a \in \mathbb{N}$ the signed global extension satisfies*

$$\mathcal{F}\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a - 2h - 1)^{n-2k}. \quad (151)$$

No upper bound on a is imposed. An empty outer sum has value 0.

Formal proof. The displayed rational identity is `fabiusDyadic_cast_extended_nat` in `GlobalDyadic.lean`, after unfolding the closed form `fabiusDyadic`. $\square$

This closed form depends only on the represented dyadic rational, not on the chosen numerator–exponent pair. Writing $\text{Dyad}(n, m) = \text{fabiusDyadic } n \, m$, Lean proves

$$\text{Dyad}(n + k, 2^k m) = \text{Dyad}(n, m), \quad (152)$$

$$\frac{m_1}{2^{n_1}} = \frac{m_2}{2^{n_2}} \text{ in } \mathbb{Q} \implies \text{Dyad}(n_1, m_1) = \text{Dyad}(n_2, m_2). \quad (153)$$

The exact declarations are `fabiusDyadic_refine_pow` and `fabiusDyadic_eq_of_rat_eq` in `GlobalDyadic.lean`.

Two special cases are the ones used in practice.

Corollary 8.3 (Bounded case). *If $0 \leq a \leq 2^n$ then $\mathcal{F}(a/2^n) = F(a/2^n)$, so (151) reduces to (150).*

Proof. The two functions agree on $[0, 1]$ and $a/2^n \in [0, 1]$. $\square$

Corollary 8.4 (Bump case). *Let $n \in \mathbb{N}$ and $q \in \mathbb{Z}$. If $|q| < 2^n$, then*

$$\text{up}\left(\frac{q}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{q+2^n-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2q + 2^{n+1} - 2h - 1)^{n-2k}, \quad (154)$$

that is, (151) evaluated at $a = q + 2^n$. If $|q| \geq 2^n$, then $\text{up}(q/2^n) = 0$.

Proof. Taking $b = 0$ in (12) gives $\mathcal{F}(x) = \text{up}(x - 1)$ for $0 \leq x \leq 2$. With $x = 1 + q/2^n = (q + 2^n)/2^n$ and $|q| < 2^n$ we have $0 < x < 2$, so $\text{up}(q/2^n) = \mathcal{F}((q + 2^n)/2^n)$, and (151) with $a = q + 2^n$ applies. The last statement is the support condition $\text{supp up} = [-1, 1]$. $\square$

Corollary 8.4 is the global form invoked in the proof of the uniform denominator multiplier theorem below. The point is that it genuinely requires Theorem 8.2 and not (150): for $0 < q < 2^n$ the numerator $a = q + 2^n$ exceeds 2^n , so the bounded statement does not apply. At the endpoints $|q| = 2^n$ the two clauses agree, both giving 0; for $q = -2^n$ the outer sum is empty and for $q = 2^n$ the value is $\mathcal{F}(2) = 0$.

Remark 8.5 (Why the sum is finite). Both index sets in (151) are finite: the outer range is $0 \leq h < a$, and the inner range is $0 \leq k \leq \lfloor n/2 \rfloor$. The moments c_k are rational by the proved recurrence (118). Thus the formula is an exact computation in $\mathbb{Q}$, with no analytic truncation or floating-point roundoff.

Remark 8.6 (Exact evaluation without uncertified examples). The finite sum is an exact rational algorithm, and its agreement with the analytic Fabius function is `fabiusDyadic_cast`. Concrete numerical specializations that do not yet have their own named Lean theorems are recorded only in the consolidated research-frontier volume.

8.2 Thue–Morse cancellation

The mechanism that makes the dyadic formulas simplify is the exact Prouhet cancellation carried by the Thue–Morse signs.

Lemma 8.7 (Affine power cancellation). *If $d < b$, then for all $X, Y \in \mathbb{Q}$,*

$$\sum_{h=0}^{2^b-1} (-1)^{\text{wt}_2(h)} (X + Yh)^d = 0. \quad (155)$$

Formal proof. This is the strict-degree case of `thueMorse_affine_power_sum_of_1e` in `ThueMorsePrefix.lean`; the monomial statement used internally is `thueMorsePowerSum_eq_zero_of_1t`. $\square$

The finite block polynomial packages the same signs without introducing an analytic exponential identity:

$$\sum_{h < 2^b} \varepsilon_h z^h = \prod_{j=0}^{b-1} (1 - z^{2^j}). \quad (156)$$

This is `thueMorseBlockPolynomial_eq_product` in `ThueMorseGenerating.lean`. At the level of the centered formal exponential series, the corresponding zero has exact order b , by `order_thueMorseCenteredPowerSeries` and `thueMorseCenteredPowerSeries_eq_product` in `ThueMorseExponential.lean`.

Writing $P_b(z)$ for either side of (156), its zero at $z = 1$ has exact multiplicity b over $\mathbb{Z}[z]$:

$$(1 - z)^b \mid P_b(z), \quad (1 - z)^{b+1} \nmid P_b(z). \quad (157)$$

The divisibility and matching nondivisibility statements are respectively `one_sub_X_pow_dvd_thueMorseBlockPolynomial` and `one_sub_X_pow_succ_not_dvd_thueMorseBlockPolynomial` in `ThueMorseBlockAlgebra.lean`.

The ordinary derivatives of the finite block polynomial make that exact order visible without passing to formal power series.

Proposition 8.8 (Block-polynomial derivatives at one). *For every $b, r \geq 0$,*

$$\left. \frac{d^r}{dz^r} \sum_{h < 2^b} \varepsilon_h z^h \right|_{z=1} = r! \sum_{h < 2^b} \varepsilon_h \binom{h}{r}. \quad (158)$$

Consequently this derivative is zero when $r < b$, while the first surviving derivative is

$$\left. \frac{d^b}{dz^b} \prod_{j=0}^{b-1} (1 - z^{2^j}) \right|_{z=1} = (-1)^b b! 2^{\binom{b}{2}}. \quad (159)$$

Formal proof. The derivative–binomial dictionary is `iterate_derivative_thueMorseBlockPolynomial_eval_one`; its strict vanishing and first surviving cases are respectively `iterate_derivative_thueMorseBlockPolynomial_eval_one_of_1t` and `iterate_derivative_thueMorseBlockPolynomial_eval_one_self`, all in `ThueMorseBlockAlgebra.lean`. $\square$

Proposition 8.9 (The first surviving Thue–Morse moment). *For every $k \geq 0$,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r r^k = (-1)^k 2^{\binom{k}{2}} k!. \quad (160)$$

Formal proof. This is `thueMorsePowerSum_self` in `ThueMorsePrefix.lean`. Together with `thueMorsePowerSum_eq_zero_of_1t`, it is also the coefficient calculation behind the exact-order theorem above. $\square$

Corollary 8.10 (Translation invariance). *For every $k \geq 0$ and $Q \in \mathbb{Q}$,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r (r + Q)^k = (-1)^k 2^{\binom{k}{2}} k!. \quad (161)$$

Formal proof. Apply `thueMorse_affine_power_sum_self` with $X = Q$ and $Y = 1$. $\square$

Proposition 8.11 (Centered reflection and the characteristic-two boundary). *Let R be a commutative ring, let $c \in R$ satisfy $c + c = 2^m - 1$, and put*

$$S_{m,r}(c) = \sum_{n=0}^{2^m-1} \varepsilon_n (n - c)^r \in R. \quad (162)$$

Then

$$S_{m,r}(c) = (-1)^{m+r} S_{m,r}(c). \quad (163)$$

Consequently, if $m + r$ is odd, then $2S_{m,r}(c) = 0$. This is the characteristic-free conclusion: in a ring with 2-torsion one cannot in general cancel the factor 2. Over $\mathbb{Q}$, taking $c = (2^m - 1)/2$ gives the familiar parity rule

$$\sum_{n=0}^{2^m-1} \varepsilon_n \left(n - \frac{2^m - 1}{2} \right)^r = 0 \quad (m + r \text{ odd}). \quad (164)$$

Formal proof. The fixed-point identity is `sum_thueMorseSign_mul_centered_pow_reflect`, and its universal 2-torsion consequence is `two_mul_sum_thueMorseSign_mul_centered_pow_eq_zero`. The rational specialization, including the legitimate cancellation of $2 \neq 0$, is `sum_thueMorseSign_mul_midpoint_pow_eq_zero`. All three declarations are in `ThueMorseMoments.lean`. $\square$

The centered specialization $Q = -2^k$ is therefore

$$\sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k)^k = (-1)^k 2^{\binom{k}{2}} k!, \quad (165)$$

the exact declaration `thueMorseCenteredPowerSum_self`.

8.3 The exact leading-bit recurrence

Let $e, a \in \mathbb{N}$ with $0 < a \leq 2^e$, and write

$$b = \lfloor \log_2 a \rfloor, \quad r = e - b, \quad q = a - 2^b, \quad y = \frac{q}{2^e}. \quad (166)$$

Thus $0 \leq q < a$: the leading binary power has been removed. Put $V_m = F(2^{-m})$, whose exact half-moment form is (141), and define

$$T_r(y) = \sum_{j=0}^r \frac{2^{\binom{j+1}{2}} V_{r-j}}{j!} y^j. \quad (167)$$

Then the exact rational recurrence is

$$\boxed{F\left(\frac{a}{2^e}\right) = T_r\left(\frac{q}{2^e}\right) - F\left(\frac{q}{2^e}\right)}. \quad (168)$$

The formal route stays entirely inside the rational dyadic model: `fabijsTaylorHorner_eq_sum` identifies the Horner routine with (167); `fabijsInversePowTwoTable_hasBlockTaylor` proves the block identity; `blockTaylor_implies_bitRecurrence` turns it into the leading-bit recurrence; and `fabijsInversePowTwoTable_hasBitRecurrence` instantiates that recurrence for the precomputed inverse-power table. Finally, `fabijsDyadicUnit_eq_fabijsDyadic` and `fabijsDyadic_cast` identify the executable rational result with the analytic function. These declarations are in `DyadicClosedForm.lean` and `DyadicAnalytic.lean`.

8.4 The bit-recursive evaluator

Use b, r, q, y , and T_r from (166)–(167). Equation (168) reduces the numerator from a to the strictly smaller q .

Algorithm 8.12 (Exact dyadic evaluator). Precompute $V_j = F(2^{-j})$ for $0 \leq j \leq e$ from (141). Define $\text{Eval}(e, a)$ recursively by

$$\text{Eval}(e, a) = \begin{cases} 0, & a \leq 0, \\ 1, & a \geq 2^e, \\ T_r(q/2^e) - \text{Eval}(e, q), & 0 < a < 2^e, \end{cases}$$

where b, r, q are given by (166); evaluate T_r by Horner’s rule.

A literal pseudocode version is:

```

Eval(e, a, V):
  if a <= 0:      return 0
  if a >= 2^e:    return 1

  b = floor_log2(a)
  r = e - b
  q = a - 2^b
  y = q / 2^e

  p = V[0]                // highest Taylor coefficient, j = r
  for m = 1 .. r:
    j = r - m
    p = V[m] + (2^(j+1) * y / (j+1)) * p

  return p - Eval(e, q, V)

```

The displayed loop is one convenient Horner ordering; any exact evaluation of (167) is equivalent. In the formal source this evaluator is `fabijsDyadicValue`.

Theorem 8.13 (Correctness and termination). *For every $e \in \mathbb{N}$ and $a \in \mathbb{Z}$,*

$$\text{Eval}(e, a) = F(a/2^e) \tag{169}$$

under the bounded convention $F = 0$ to the left and $F = 1$ to the right.

Formal proof. The pseudocode is the rational routine `fabijsDyadicValue`, whose analytic correctness is `fabijsDyadicValue_cast` in `PaperStatements.lean`. The interior recursion is `fabijsInversePowTwoTable_hasBitRecurrence`; the corresponding total recursive correctness theorem is `fabijsDyadicUnitAux_eq_of_bitRecurrence`. $\square$

8.5 Reduction of an arbitrary dyadic argument

Algorithm 8.12 evaluates the bounded function F on $[0, 1]$. A total evaluator for the signed global extension first folds the argument back into the unit interval and records the associated sign. The fold is a reduction modulo 2 followed by a reflection.

Theorem 8.14 (Triangular reduction). *Let $n \in \mathbb{N}$ and put $s = 2^n$. Let $a \in \mathbb{N}$ and write*

$$a = 2sB + R, \quad B = \left\lfloor \frac{a}{2s} \right\rfloor, \quad 0 \leq R < 2s, \quad (170)$$

and set

$$C = \begin{cases} R, & R \leq s, \\ 2s - R, & R > s. \end{cases} \quad (171)$$

Then $0 \leq C \leq s$ and

$$\boxed{\mathcal{F}\left(\frac{a}{2^n}\right) = \varepsilon_B F\left(\frac{C}{2^n}\right)}. \quad (172)$$

For $a \in \mathbb{Z}$ with $a < 0$ one has $\mathcal{F}(a/2^n) = 0$.

Proof. Write $x = a/2^n = 2B + R/s$. Since $0 \leq R < 2s$ we have $2B \leq x < 2B + 2$, so x lies in the B th block and (12) gives

$$\mathcal{F}(x) = \varepsilon_B \text{up}(x - 2B - 1) = \varepsilon_B \text{up}\left(\frac{R}{s} - 1\right).$$

Put $t = R/s - 1$, so that $-1 \leq t < 1$. The fold relation (7) says $\text{up}(t) = F(t + 1)$ for $t \leq 0$ and $\text{up}(t) = F(1 - t)$ for $t > 0$; on $[-1, 1]$ both clauses read $\text{up}(t) = F(1 - |t|)$. If $R \leq s$ then $|t| = 1 - R/s$ and $F(1 - |t|) = F(R/s)$. If $R > s$ then $|t| = R/s - 1$ and $F(1 - |t|) = F(2 - R/s) = F((2s - R)/2^n)$. In both cases the argument is $C/2^n$ with C as in (171), and $0 \leq C \leq s$ is immediate from $0 \leq R < 2s$. The final statement is the vanishing of $\mathcal{F}$ on the negative half-line. $\square$

Combining (172) with Algorithm 8.12 gives a total exact evaluator for $\mathcal{F}$ at every dyadic rational: reduce with (170)–(171), compute the block sign $\varepsilon_B = (-1)^{\text{wt}_2(B)}$ from the binary digits of $B = \lfloor a/2^{n+1} \rfloor$, and call $\text{Eval}(n, C)$.

The reduction is compatible with unreduced presentations. Replacing (n, a) by $(n + 1, 2a)$ doubles s and R , leaves B unchanged, and doubles C ; the right side of (172) is therefore unchanged, as it must be.

Remark 8.15 (Recognizing dyadic arguments exactly). A reduced rational p/q with $q > 0$ is dyadic exactly when q is a power of two, and in that case $n = \log_2 q$ and $a = p$ are read off directly. An exact rational interface to F , up , and $\mathcal{F}$ therefore tests a single integer condition on the reduced denominator, returns “not dyadic” when it fails, and otherwise invokes the signed integer evaluator above. No floating-point recognition of powers of two is involved anywhere, so the interface is exact on every input and cannot be defeated by rounding in the recognition step.

8.6 Denominator bounds

Let

$$M_m = \prod_{j=1}^m (2^{2^j} - 1), \quad (2m + 1)!! = 1 \cdot 3 \cdots (2m + 1). \quad (173)$$

Theorem 8.16 (Uniform denominator multiplier). *Let $n \geq 1$ and $a \in \mathbb{Z}$ with $|a| < 2^n$. Then*

$$\text{up}(a/2^n) \, n! \, 2^{\binom{n+1}{2}} (2 \lfloor n/2 \rfloor + 1)!! \, M_{\lfloor n/2 \rfloor} \quad (174)$$

is an integer.

Formal proof. This is `rvachevDyadic_mul_denominatorBound_isNatural_all` in `DenominatorBound.lean`. Its proof uses `moment_mul_common_denominator` for the moments and `fabiusDyadic_mul_denominatorBound_isInteger` for the bounded dyadic values. $\square$

Let Δ_n be the least common multiple of the reduced denominators of the values $\text{up}(a/2^n)$ with odd a in the support. The theorem gives the explicit divisibility bound

$$\Delta_n \mid n! 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!! M_{\lfloor n/2 \rfloor}. \quad (175)$$

A complementary lower statement is also formalized.

Proposition 8.17 (The reciprocal power is one of the samples). *For every $n \geq 1$ the reduced denominator of $F(2^{-n})$ divides Δ_n . Consequently*

$$v_2(\Delta_n) \geq \binom{n}{2} + 1 + v_2(n!). \quad (176)$$

Formal proof. The denominator divisibility is `proposition_fifteen` in `PaperStatements.lean`, after the bridge `dyadicDenominator_eq_fabiusDyadicDenominator`. The exact two-adic exponent is `fabiusAtInverseTwoPow_padicVal_two_closed`. $\square$

Together, (175) and (176) give the formally proved denominator information used in this article. Proposed exact factorizations of the least common denominator, together with the associated numerical tables, are kept in the consolidated research-frontier volume until their individual statements are formalized.

8.7 The exact half-base q -binomial layer

The dyadic dilation forces the specialization $q = 1/2$. For $a, q \in \mathbb{Q}$, define

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (177)$$

and write $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{1/2}$ for the Gaussian coefficient at that fixed base. For $k \leq n$, its exact Mersenne normalization is

$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{1/2} = \frac{\mathfrak{m}_n}{\mathfrak{m}_k \mathfrak{m}_{n-k} 2^{k(n-k)}}, \quad \mathfrak{m}_j = \prod_{\ell=1}^j (2^\ell - 1). \quad (178)$$

In particular the row is symmetric under $k \mapsto n - k$.

Theorem 8.18 (Finite half-base q -binomial theorem). *For every $n \in \mathbb{N}$ and $z \in \mathbb{Q}$,*

$$(z; 1/2)_n = \sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{1/2} z^k. \quad (179)$$

At every dyadic node this gives the single closed evaluation

$$\sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{1/2} (2^m)^k = \begin{cases} 0, & m < n, \\ (-1)^n \mathfrak{m}_m / \mathfrak{m}_{m-n}, & n \leq m. \end{cases} \quad (180)$$

In particular, the sum vanishes if and only if $m < n$. Moreover,

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n. \quad (181)$$

Formal proof. These are `halfQBinomial_theorem`, `qBinomial_half_two_pow_sum_eq_ite`, `qBinomial_half_two_pow_sum_eq_zero_iff`, `halfQPochhammer_eq_mersenne_div`, `halfQBinomial_eq_mersenne`, and `halfQBinomial_symm` in `HalfQBinomial.lean`. $\square$

The polynomial in (179) has the complete rational root locus

$$\sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2} z^k = 0 \iff \exists j < n \text{ such that } z = 2^j, \quad z \in \mathbb{Q}. \quad (182)$$

This is `qBinomial_half_sum_eq_zero_iff`. The same statement is stable under scalar extension at embedded rational arguments. Let K be a field with a $\mathbb{Q}$ -algebra structure, and write $\iota : \mathbb{Q} \rightarrow K$ for its canonical map. Then, for every $z \in \mathbb{Q}$,

$$\sum_{k=0}^n \iota \left((-1)^k 2^{-\binom{k}{2}} \left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2} \right) (\iota z)^k = 0 \iff \exists j < n \text{ such that } \iota z = 2^j \text{ in } K. \quad (183)$$

This is `qBinomial_half_sum_algebraMap_eq_zero_iff` in `HalfQBinomial.lean`. The scalar statement classifies the embedded rational arguments ιz ; it does not assert a root classification for arbitrary elements of K .

The same Gaussian coefficients form a geometric Prouhet functional. Set

$$w_{n,k} = (-1)^k (1/2)^{\binom{k}{2}} \left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2}, \quad \xi_k = -2^k.$$

For every rational polynomial p with $\deg p \leq n$,

$$\sum_{k=0}^n w_{n,k} p(\xi_k) = [X^n]_p 2^{\binom{n+1}{2}} (1/2; 1/2)_n = [X^n]_p m_n. \quad (184)$$

Thus the functional annihilates every polynomial of degree below n . At $n = 0$, the degree-valued extractor covers all constant polynomials; the strict-degree cancellation statement admits exactly the zero polynomial. The two forms of (184) are `halfQBinomial_negativeDyadic_polynomial_1_sum_eq_coeff` and `halfQBinomial_negativeDyadic_polynomial_sum_eq_mersenne`; strict-degree cancellation is `halfQBinomial_negativeDyadic_polynomial_sum_eq_zero_of_degree_lt`, all in `HalfQBinomial.lean`.

The inner cancellation in the dyadic formula is the exact Thue–Morse block identity

$$\sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} z^r = \prod_{j=0}^{k-1} (1 - z^{2^j}), \quad (185)$$

formalized as `thueMorseBlockPolynomial_eq_product`. Its affine power-sum form annihilates every degree below k , while its first surviving moment is $(-1)^k 2^{\binom{k}{2}} k!$; see `thueMorse_affine_power_sum_of_le` and `thueMorse_affine_power_sum_self`.

For comparison with the moment recurrence, put

$$a_n = \frac{d_n}{n!}, \quad \mathcal{A}(X) = \sum_{n \geq 0} a_n X^n, \quad E(X) = \frac{e^X - 1}{X}, \quad (186)$$

with the removable value $E(0) = 1$. The refinement is

$$\mathcal{A}(2X) = E(X) \mathcal{A}(X), \quad (187)$$

and its coefficient recurrence is

$$(2^n - 1)a_n = \sum_{j=0}^{n-1} \frac{a_j}{(n-j+1)!} \quad (n \geq 1), \quad a_0 = 1. \quad (188)$$

The functional identity follows from `proposition_two_formula` and `complexGeneratingFunction_eq_series_formula` in `AnalyticMoments.lean`; the coefficient recurrence is `halfMoment_original_recurrence` in `BernoulliRecurrences.lean`. These are the formal-series bridge used by the specialized dyadic theorem.

For $m, k, d \in \mathbb{N}$ and $Q \in \mathbb{C}$, write

$$\sigma_{m;k,d}(Q) = \sum_{r=0}^{m2^k-1} (-1)^{\text{wt}_2(r)} (r - m2^k + Q)^d. \quad (189)$$

The exact result is the following finite identity.

Theorem 8.19 (Translated global dyadic formula). *For all $m, n \in \mathbb{N}$ and $Q \in \mathbb{C}$,*

$$\begin{aligned} \mathcal{F}(m/2^n) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\quad \times \sum_{r=0}^{m2^k-1} (-1)^{\text{wt}_2(r)} (r - m2^k + Q)^{n+k}. \end{aligned} \quad (190)$$

The real number on the left is embedded in $\mathbb{C}$. The entire right-hand side is independent of the common translation Q . When $m \leq 2^n$, the left side is the bounded value $F(m/2^n)$.

Formal proof. The complete normalized finite expression is packaged as a polynomial in the translation and proved equal to the constant centered formula by `qBinomialThueMorseDyadicTranslatedFormulaPolynomial_eq_const`. Its identification with the signed dyadic value over $\mathbb{C}$ is `globalFabijs_dyadic_eq_qBinomialThueMorseDyadic_translated_sum_complex`; the rational core and real, complex, and Gaussian-rational specializations are in `FabijsQBinomialFormula.lean` and `FabijsDyadicQBinomialScalar.lean`. $\square$

The rational specialization is also independent of how the dyadic argument is written. If

$$\frac{m_1}{2^{n_1}} = \frac{m_2}{2^{n_2}},$$

then the two translated rational formulas agree even when their common translations $Q_1, Q_2 \in \mathbb{Q}$ differ. This is `qBinomialThueMorseDyadicTranslatedFormula_eq_of_rat_eq`; the one-step refinement $(m, n) \mapsto (2m, n+1)$ is `qBinomialThueMorseDyadicTranslatedFormula_refine`. Over every field over $\mathbb{Q}$, both conclusions persist even when the scalar translations and the dyadic presentations are changed simultaneously; these stronger wrappers are `qBinomialThueMorseDyadicTranslatedFormulaIn_eq_of_rat_eq` and `qBinomialThueMorseDyadicTranslatedFormulaIn_refine` in `FabijsDyadicQBinomialScalar.lean`.

Binary complementation gives the raw inverse-dyadic form

$$\boxed{\mathcal{F}(2^{-n}) = \frac{1}{(-2)^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r + Q)^{n+k}.} \quad (191)$$

Its translated numerator is constant by `qBinomialThueMorseRawTranslatedNumeratorPolynomial_eq_const`. The algebraic statement is in fact valid over every field K equipped with an

$\mathbb{Q}$ -algebra structure and every scalar translation $Q \in K$: the fully expanded right-hand side equals the canonical inverse-dyadic rational value after its scalar embedding by `fabiusAtInverseTwoPow_cast_eq_qBinomialThueMorse_rawTranslated_sum`. Translation independence and the intermediate value identification are `qBinomialThueMorseRawTranslatedFormulaIn_eq_centered` and `qBinomialThueMorseRawTranslatedFormulaIn_eq_fabiusAtInverseTwoPow`. For every scalar type K carrying Mathlib’s `RCLike` structure, the corresponding theorem for an arbitrary bounded Fabius solution is `fabiusFunction_inverse_two_pow_eq_qBinomialThueMorseRawTranslatedFormulaIn`. All five declarations are in `FabiusRawQBinomialScalar.lean`; the final one uses the `RCLike` hypothesis only to embed the real Fabius value. Variable-base q -interpolation formulas, barycentric identities, numerical cancellation studies, and complexity comparisons are maintained in the consolidated research-frontier volume rather than used as theorems here.

9 The global binary-reduction telescope

The Taylor-block recursion also applies to the signed global extension $\mathcal{F}$. It can therefore be iterated along the binary expansion of an arbitrary nonnegative real number, rather than only along the finite expansion of a dyadic rational. The reductions need not terminate, but their residuals tend geometrically to zero.

Define the finite reduction polynomial

$$\mathcal{T}_m(y) = \sum_{j=0}^m 2^{\binom{j+1}{2} - \binom{m-j}{2}} \frac{d_{m-j}}{(m-j)!} \frac{y^j}{j!}. \quad (192)$$

All powers of 2 here are integer powers. In particular, negative exponents are allowed, and $\mathcal{T}_0(y) = 1$. The signed Taylor-block identity is

$$\mathcal{F}(2^{-m} + y) = -\mathcal{F}(y) + \mathcal{T}_m(y), \quad 0 \leq y < 2^{-m}. \quad (193)$$

Fix $x \geq 0$. At scale m , put

$$p_m = \lfloor 2^m x \rfloor, \quad y_m = x - \frac{p_m}{2^m}, \quad 0 \leq y_m < 2^{-m}. \quad (194)$$

The previous prefix requires a genuine half-scale value at $m = 0$:

$$p_{m-1}^* = \begin{cases} \lfloor x/2 \rfloor, & m = 0, \\ p_{m-1}, & m \geq 1. \end{cases} \quad (195)$$

Uniformly, $p_{m-1}^* = \lfloor p_m/2 \rfloor$. Thus $p_m = 2p_{m-1}^* + \delta_m$ with $\delta_m \in \{0, 1\}$. Define

$$\alpha_m(x) = (-1)^{\text{wt}_2(p_m)} (2p_{m-1}^* - p_m), \quad R_m(x) = (-1)^{\text{wt}_2(p_m)} \mathcal{F}(y_m). \quad (196)$$

The coefficient α_m is zero or a sign, and it is nonzero precisely when the newly exposed binary digit is 1.

Lemma 9.1 (One-step residual identity). *For $m \geq 1$,*

$$\alpha_m(x) \mathcal{T}_m(y_m) = R_{m-1}(x) - R_m(x). \quad (197)$$

Proof. If $p_m = 2p_{m-1}$, then $y_m = y_{m-1}$, the Thue–Morse signs agree, and $\alpha_m = 0$, so both sides vanish. If $p_m = 2p_{m-1} + 1$, then

$$y_{m-1} = 2^{-m} + y_m, \quad (-1)^{\text{wt}_2(p_m)} = -(-1)^{\text{wt}_2(p_{m-1})}, \quad \alpha_m = (-1)^{\text{wt}_2(p_{m-1})}.$$

Apply (193) to y_{m-1} and multiply by the old prefix sign; the two copies of $\mathcal{F}(y_m)$ combine exactly into $R_{m-1} - R_m$. $\square$

The scale-zero identity is separate and retains the parity of the integer block:

$$\mathcal{F}(x) = \alpha_0(x)\mathcal{T}_0(y_0) + R_0(x). \quad (198)$$

Indeed, write $p_0 = 2p_{-1}^* + \delta_0$ and use the two-unit translation law (12). When $\delta_0 = 0$, the polynomial term vanishes. When $\delta_0 = 1$, the identity $\mathcal{F}(1 + y_0) = -\mathcal{F}(y_0) + 1$ supplies it.

Theorem 9.2 (Exact finite binary telescope). *For $x \geq 0$ and $N \in \mathbb{N}$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^N \alpha_m(x)\mathcal{T}_m(y_m(x)) + R_N(x).} \quad (199)$$

Proof. Start with (198). Summing (197) for $m = 1, \dots, N$ telescopes all intermediate residuals. The case $N = 0$ is exactly the scale-zero identity. $\square$

For $N \geq 1$, $0 \leq y_N < 2^{-N} \leq 1/2$. On this interval $0 \leq \mathcal{F}(y) = F(y) \leq 2y$, so

$$|R_N(x)| \leq 2^{1-N}. \quad (200)$$

The estimate is independent of x . Therefore, if $B_N(x)$ denotes the finite sum in (199), then for every $x \in \mathbb{R}$ and $N \geq 1$,

$$|B_N(x) - \mathcal{F}(x)| \leq 2^{1-N}. \quad (201)$$

This is `norm_globalBinaryReductionSum_sub_extendedFabius_le`. Lean’s natural-floor convention extends every B_N by zero on $x \leq 0$, where $\mathcal{F}$ also vanishes, and `globalBinaryReductionSum_tendstoUniformly_extendedFabius` proves that the finite telescopes converge uniformly on the whole line. The one-step identity then gives, for $m \geq 2$,

$$|\alpha_m(x)\mathcal{T}_m(y_m)| \leq |R_{m-1}| + |R_m| \leq 4 \cdot 2^{-(m-1)}. \quad (202)$$

Thus the limiting series is absolutely, not merely conditionally, convergent.

Theorem 9.3 (Absolutely convergent global binary series). *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x)\mathcal{T}_m(y_m(x)).} \quad (203)$$

On $0 \leq x \leq 1$, the sum is equivalently $F(x)$.

Lean’s natural-floor definitions make the telescope genuinely all-real: when $x \leq 0$, every binary prefix, summand, residual, and $\mathcal{F}(x)$ vanishes termwise. Thus the finite identity and the $m = 0$ infinite series hold for every $x \in \mathbb{R}$, as `extendedFabius_eq_globalBinaryReductionSum_add_remainder_all` and `extendedFabius_eq_tsum_globalBinaryReductionSummand_all` in `FabiusBinaryReductionSeries.lean`. The fully expanded parity-power version has the same all-real scope by `globalFabius_eq_tsum_fabiusParityPower_literal_all` in `FabiusParityPowerSeries.lean`. The displayed floor formulas above remain ordinary nonnegative-floor formulas on their stated domain.

The index $m = 0$ is indispensable globally. For $0 \leq x < 1$ it vanishes, so a series starting at $m = 1$ happens to be valid there. At $x = 1$, however, the zeroth term is 1 and all later coefficients vanish. More generally it is the zeroth term that distinguishes even and odd integer blocks of the signed extension.

9.1 The corrected parity-power presentation

The same summands have a more literal presentation in terms of prefix parity and inverse dyadic values. Define, using integer exponents,

$$\kappa_j = \frac{((j : \mathbb{Z}) + 2)((j : \mathbb{Z}) - 1)}{2} = \binom{j+1}{2} - 1. \quad (204)$$

The integer interpretation matters: $\kappa_0 = -1$, hence $2^{\kappa_0} = 1/2$; natural subtraction would silently give the wrong term. Put

$$\mathcal{I}_m(y) = \sum_{j=0}^m 2^{\kappa_j} \mathcal{F}(2^{(j:\mathbb{Z})-(m:\mathbb{Z})}) \frac{y^j}{j!}. \quad (205)$$

Lemma 9.4. *For every m and y ,*

$$\mathcal{I}_m(y) = \frac{1}{2} \mathcal{T}_m(y). \quad (206)$$

Proof. For $j \leq m$, set $r = m - j$. The inverse-dyadic identity (141) gives

$$\mathcal{F}(2^{j-m}) = \frac{d_{m-j}}{(m-j)! 2^{\binom{m-j}{2}}}.$$

Together with $\kappa_j = \binom{j+1}{2} - 1$, this identifies each summand with one half of the corresponding summand in (192). $\square$

Let $\tau(a) = \text{wt}_2(a) \bmod 2$. Since

$$\frac{(-1)^a - 1}{2} = 2 \left\lfloor \frac{a}{2} \right\rfloor - a,$$

the literal summand

$$\begin{aligned} \Lambda_m(x) &:= ((-1)^{\lfloor 2^m x \rfloor} - 1) (-1)^{\tau(\lfloor 2^m x \rfloor)} \\ &\quad \times \sum_{j=0}^m 2^{((j+2)(j-1))/2} \mathcal{F}(2^{j-m}) \frac{(x - \lfloor 2^m x \rfloor / 2^m)^j}{j!} \end{aligned} \quad (207)$$

is exactly $\alpha_m(x) \mathcal{T}_m(y_m)$. Both exponents $((j+2)(j-1))/2$ and $j-m$ in this display are integer exponents, not truncated natural subtractions.

Warning 9.5. In (189) and (190), the summation indices are cast into the ambient characteristic-zero field before the subtraction in $r - m2^k + Q$ is performed. This is the typing used by the formal theorem; the displayed subtraction is therefore field subtraction, never truncated subtraction on natural numbers.

Theorem 9.6 (Corrected parity-power series). *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \Lambda_m(x)}, \quad (208)$$

and the series converges absolutely. On the half-open unit interval $0 \leq x < 1$, the zeroth summand vanishes and one may start at $m = 1$.

Proof. The elementary parity identity and (206) prove $\Lambda_m = \alpha_m \mathcal{T}_m(y_m)$ term by term. The claim follows from (203). The half-open corollary follows from $\lfloor x \rfloor = 0$; the closed endpoint $x = 1$ requires the restored zeroth term. $\square$

9.2 The global complex q -binomial series

The finite constant-polynomial identity from [section 8.7](#) can be inserted at every scale of the binary telescope. For $Q \in \mathbb{C}$, define

$$\mathcal{N}_n^{(Q)} = \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r - 2^k + Q)^{n+k}. \quad (209)$$

Translation invariance and the inverse-dyadic normalization give

$$\frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n} = \frac{2^n d_n}{n!}. \quad (210)$$

The normalization differs from the 2^{n^2} in (190): here it produces a Taylor coefficient rather than a function value.

For $m \in \mathbb{N}$, put

$$\mathcal{Q}_m^{(Q)}(y) = 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n}. \quad (211)$$

Theorem 9.7 (q -binomial Taylor identity). *For every m , every $y \in \mathbb{R}$, and every $Q \in \mathbb{C}$,*

$$\mathcal{Q}_m^{(Q)}(y) = \mathcal{T}_m(y), \quad (212)$$

where the real polynomial on the right is embedded in $\mathbb{C}$. In particular, each finite summand is pointwise independent of Q before any infinite sum is taken.

Proof. Insert (210) into (211) and set $j = m - n$. Collecting the powers of 2 gives

$$2^{-\binom{m+1}{2}} 2^{(m+1)j} 2^{m-j} = 2^{\binom{j+1}{2} - \binom{m-j}{2}}.$$

The remaining factorials and moments are exactly those in (192). $\square$

Theorem 9.8 (Global complex q -binomial series). *For fixed $Q \in \mathbb{C}$ and every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{Q}_m^{(Q)}(y_m(x))}, \quad (213)$$

with the left side embedded in $\mathbb{C}$. The series converges absolutely.

The displayed floor presentation is stated on the nonnegative half-line. The literal Lean summand has a stronger all-real scope: for every $x \in \mathbb{R}$, its series is absolutely summable and sums to the signed global extension, while for $x \leq 0$ every term and the target vanish. These are `summable_norm_qBinomialFabiusGlobalSummand_all`, `hasSum_qBinomialFabiusGlobalSummand_all`, `extendedFabius_eq_tsum_qBinomialFabiusGlobalSummand_all`, `globalFabius_eq_tsum_qBinomialFabiusGlobalSummand_all`, and `qBinomialFabiusGlobalSummand_eq_zero_of_nonpos` in `FabiusGlobalQBinomialSeries.lean`. The older nonnegative signatures are retained as compatibility wrappers.

This is not a new infinite basic-hypergeometric identity. The outer sum is the analytic binary telescope, while at each scale a finite q -binomial identity presents the same Taylor polynomial. Expanding one summand makes the architecture explicit:

$$\begin{aligned} & \alpha_m(x) 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1}x - 2p_m(x))^{m-n}}{(m-n)!} \frac{1}{2^{\binom{n}{2}} (1/2; 1/2)_n} \\ & \times \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r - 2^k + Q)^{n+k}. \end{aligned} \quad (214)$$

Convergence comes entirely from the binary-tail majorant (202); the special-function manipulations are finite and algebraic.

10 A complex q -binomial discrete limit

There is a second way in which the same Gaussian row produces $\mathcal{F}$. It is a genuine limit of finite shifted splines rather than a sequence of partial sums of (213). Reuse the safe midpoint cutoff $N_p(x)$ from (54). For $x \geq 0$ it is equivalent to the inclusive source range

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor, \quad (215)$$

including the empty case where the upper endpoint is -1 .

For fixed $Q \in \mathbb{C}$ and $x \in \mathbb{R}$, define

$$\begin{aligned} \mathcal{D}_n^{(Q)}(x) &= \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ &\quad \times \sum_{r=0}^{N_{n+k}(x)-1} (-1)^{\text{wt}_2(r)} (r - 2^{n+k}x + Q)^{n+k}. \end{aligned} \quad (216)$$

Theorem 10.1 (Complex discrete-limit formula). *For every fixed $Q \in \mathbb{C}$ and every $x \in \mathbb{R}$,*

$$\boxed{\mathcal{D}_n^{(Q)}(x) \longrightarrow \mathcal{F}(x)} \quad (217)$$

in $\mathbb{C}$. If $0 \leq x \leq 1$, the limit is equivalently $F(x)$.

Proof. If $x \leq 0$, every safe cutoff $N_p(x)$ is zero. Hence every shifted spline and every approximant (216) is zero, as is $\mathcal{F}(x)$. It remains to treat $x \geq 0$.

For a degree p and complex shift Q , introduce the finite shifted spline

$$S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} (-1)^{\text{wt}_2(r)} (r - 2^p x + Q)^p. \quad (218)$$

At $Q = 1/2$ this is exactly the centered spline (55), so by (62),

$$S_p^{(1/2)}(x) \longrightarrow \mathcal{F}(x).$$

Lemma 10.2 (Fixed-cutoff Taylor expansion of a complex shift). *Read (55) at $p = 0$ as*

$$S_0(x) = \sum_{r=0}^{N_0(x)-1} \varepsilon_r, \quad (219)$$

so that a centered branch S_d is defined for every integer $d \geq 0$. Fix $p \geq 1$ and $x \geq 0$, and abbreviate $M = N_p(x)$. For every d with $0 \leq d \leq p$, the degree- d branch carried by that one fixed cutoff satisfies

$$\frac{1}{2^{\binom{d}{2}} d!} \sum_{r=0}^{M-1} \varepsilon_r \left(2^p x - \frac{1}{2} - r \right)^d = S_d(2^{p-d}x), \quad (220)$$

and in particular $|S_d(2^{p-d}x)| \leq 1$. Moreover, for every $Q \in \mathbb{C}$,

$$S_p^{(Q)}(x) - S_p^{(1/2)}(x) = \sum_{d=0}^{p-1} \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}} (p-d)!} \left(\frac{1}{2} - Q \right)^{p-d} S_d(2^{p-d}x). \quad (221)$$

Proof. Write the shifted spline in its descending orientation. Since $(-1)^p(r - 2^p x + Q)^p = (2^p x - Q - r)^p$, definition (218) reads

$$S_p^{(Q)}(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r=0}^{M-1} \varepsilon_r (2^p x - Q - r)^p, \quad M = N_p(x), \quad (222)$$

and at $Q = 1/2$ this is (55). The cutoff M depends on p and x only. It is unaffected by Q , and it is held fixed in what follows while the degree of the summand varies; that is the whole point of (220).

We prove (220). The dilation $x \mapsto 2^{p-d}x$ compensates exactly for the drop in degree, in the sense that

$$N_d(2^{p-d}x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right) = N_p(x) = M \quad (223)$$

by (54). If $M = 0$, both sides of (220) are empty sums and there is nothing to prove. Assume $M \geq 1$. Every index in the range satisfies

$$r \leq M - 1 = \left\lfloor 2^p x + \frac{1}{2} \right\rfloor - 1 \leq 2^p x - \frac{1}{2},$$

so every base $2^p x - \frac{1}{2} - r$ occurring in (220) is nonnegative and the truncation in (56) is inactive. For $d \geq 1$, formula (56) evaluated at the point $2^{p-d}x$, whose own cutoff is M by (223), is therefore literally the left side of (220). For $d = 0$ both sides equal $\sum_{r=0}^{M-1} \varepsilon_r$, by (219) and (223).

For the uniform bound, the rescaled argument $2^{p-d}x$ is nonnegative. The all-degree declaration `abs_fabiusUniformSpline_le_one` therefore gives the claim directly, including $d = 0$.

Now expand. Put $w = 2^p x - \frac{1}{2}$ for the descending argument at $Q = 1/2$ and $t = \frac{1}{2} - Q$ for its increment, so that $2^p x - Q = w + t$. The binomial theorem gives

$$(w + t - r)^p = \sum_{d=0}^p \binom{p}{d} t^{p-d} (w - r)^d.$$

Multiply by $\varepsilon_r / (2^{\binom{p}{2}} p!)$, sum over $0 \leq r \leq M - 1$, and interchange the two finite sums. Using (222) on the left and (220) on each inner sum,

$$\begin{aligned} S_p^{(Q)}(x) &= \sum_{d=0}^p \frac{t^{p-d}}{2^{\binom{p}{2}} p!} \binom{p}{d} \sum_{r=0}^{M-1} \varepsilon_r (w - r)^d \\ &= \sum_{d=0}^p \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}} (p-d)!} t^{p-d} S_d(2^{p-d}x), \end{aligned} \quad (224)$$

because $\binom{p}{d}/p! = 1/(d!(p-d)!)$ and the inner sum is $2^{\binom{d}{2}} d! S_d(2^{p-d}x)$. The term $d = p$ is $S_p(x) = S_p^{(1/2)}(x)$; subtracting it leaves (221). $\square$

Remark 10.3 (Which convention makes the sign disappear). The expansion (221) is free of alternating signs only because the increment is measured in the *descending* variable $2^p x - Q$ of (222), that is, because the coefficient carries $(\frac{1}{2} - Q)^{p-d}$ and not $(Q - \frac{1}{2})^{p-d}$. If one instead measures the shift by the ascending increment $q = Q - \frac{1}{2}$, which is the natural reading when the branch is written as a function of the ascending argument $r - 2^p x + Q$ of (218), then the identity acquires exactly the factor $(-1)^{p+d}$:

$$S_p^{(Q)}(x) - S_p^{(1/2)}(x) = \sum_{d=0}^{p-1} (-1)^{p+d} \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}} (p-d)!} q^{p-d} S_d(2^{p-d}x), \quad q = Q - \frac{1}{2}. \quad (225)$$

The two displays are the same statement, since $(-1)^{p+d} = (-1)^{p-d}$ and therefore $(-1)^{p+d}q^{p-d} = (\frac{1}{2} - Q)^{p-d}$. It is worth being explicit about where the sign does *not* live: it is not a matter of renormalizing the lower branches. The alternating prefactor $(-1)^d$ of (55) is cancelled exactly by the $(-1)^d$ produced when the ascending argument is reversed, so the branch values $S_d(2^{p-d}x)$ appearing in (221) and in (225) are the same numbers. The sign belongs to the increment alone, and the convention that removes it is the descending one.

Remark 10.4 (Formal bound). Only the exponential bound (226) is asserted here; it is `norm_fabiusComplexShiftSpline_sub_center_le_half_pow_mul_exp` in `FabiusComplexShiftSpline.lean`. A proposed sharper constant is recorded in the research-frontier gap register pending a corrected named theorem.

Expanding a fixed cutoff branch about $Q = 1/2$ gives an exact finite Taylor combination of lower-degree centered branches. The normalized Thue–Morse bounds then give, for $p \geq 1$ and $x \geq 0$,

$$\left| S_p^{(Q)}(x) - S_p^{(1/2)}(x) \right| \leq 2^{-(p-1)} \exp(|Q - \frac{1}{2}|). \quad (226)$$

The nonpositive half-line is handled separately by exact vanishing, and `fabiusComplexShiftSpline_tendsto_globalFabius_all` proves that every fixed complex shift has the same pointwise spline limit for every real x .

Now reindex the outer sum in (216) by $j = n - k$. Direct collection of signs, powers, and factorials gives the exact finite identity

$$\mathcal{D}_n^{(Q)}(x) = \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x), \quad (227)$$

where

$$\omega_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_n}. \quad (228)$$

The finite q -binomial theorem gives the row normalization

$$\sum_{j=0}^n \omega_{n,j} = 1. \quad (229)$$

The same theorem identifies the entire row rather than only its value at one.

Proposition 10.5 (Toeplitz row polynomial and module-valued q -Richardson cancellation). *For $z \in \mathbb{Q}$, put $W_n(z) = \sum_{j=0}^n \omega_{n,j} z^j$. Then*

$$W_n(z) = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}. \quad (230)$$

Consequently its rational zeros are exactly

$$W_n(z) = 0 \iff z \in \{2, 2^2, \dots, 2^n\}. \quad (231)$$

Moreover, let M be any additive commutative monoid carrying a $\mathbb{Q}$ -module structure. If an M -valued sequence has the finite geometric-error model $u_p = L + \sum_{r=1}^n 2^{-rp} \cdot a_r$, where $\cdot$ denotes scalar multiplication, then

$$\sum_{j=0}^n \omega_{n,j} \cdot u_{2n-j} = L. \quad (232)$$

Thus the cancellation needs no multiplication, topology, order, or norm on the target. Taking $M = \mathbb{Q}$ recovers the rational formula $u_p = L + \sum_{r=1}^n a_r 2^{-rp}$. This module-valued extension concerns cancellation only; the root classification above remains a statement for rational z . All statements include $n = 0$: the row polynomial is then 1, the root set is empty, and the error sum is empty.

Formal proof. The full generating identity and its root classification are `sum_range_discreteLimitWeight_mul_pow` and `sum_range_discreteLimitWeight_mul_pow_eq_zero_iff`. The reusable single-mode identity is `sum_range_discreteLimitWeight_mul_geometricMode_smul_eq_zero`, and the module-valued cancellation theorem is `discreteLimitWeight_qRichardson_exact_module`; its $M = \mathbb{Q}$ specialization is `discreteLimitWeight_qRichardson_exact`. All are in `FabijsDiscreteLimitToeplitz.lean`. The former proof of (229) is now its specialization at $z = 1$. $\square$

Applying it with the opposite sign also gives the uniform variation bound

$$\sum_{j=0}^n |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n} \leq 16. \quad (233)$$

Finally, every sampled degree satisfies $2n - j \geq n$. A finite-row Toeplitz lemma therefore carries the convergence of $S_p^{(Q)}(x)$ to the weighted rows (227). Explicitly, given $\epsilon > 0$, choose P so that $|S_p^{(Q)}(x) - \mathcal{F}(x)| < \epsilon/16$ for $p \geq P$. If $n \geq P$, then every $2n - j \geq n \geq P$, and the row-sum identity together with (233) gives

$$\left| \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x) - \mathcal{F}(x) \right| \leq \sum_{j=0}^n |\omega_{n,j}| |S_{2n-j}^{(Q)}(x) - \mathcal{F}(x)| < \epsilon.$$

This proves (217). $\square$

Two distinctions prevent common misreadings of this theorem. First, unlike the complete finite dyadic numerator and each summand of the global q -series, a finite value $\mathcal{D}_n^{(Q)}(x)$ can depend on Q ; it is the limit that is independent of a fixed shift. Second, the Toeplitz row (227) is not a finite binary telescope or a partial sum of (213). These finite expressions share a limit but are not equal term by term.

The two properties of the reindexed row (228) used above are both evaluations of the finite q -binomial theorem (179), at $z = 1/2$ and at $z = -1/2$ respectively. Both use the exponent identity $\binom{j}{2} + j = \binom{j+1}{2}$.

For the row sum, evaluate (179) at $z = 1/2$. The left side is $(1/2; 1/2)_n$ by definition, and the right side is

$$(1/2; 1/2)_n = \sum_{j=0}^n (-1)^j 2^{-\binom{j}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} \left(\frac{1}{2} \right)^j = \sum_{j=0}^n (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-(j+1)}. \quad (234)$$

Dividing by $(1/2; 1/2)_n$ gives exactly $\sum_{j=0}^n \omega_{n,j} = 1$, which is (229).

For the total variation, evaluate (179) at $z = -1/2$ instead. Each factor $(-1)^j (-1/2)^j = 2^{-j}$ is now positive, so the alternating signs disappear:

$$(-1/2; 1/2)_n = \sum_{j=0}^n (-1)^j 2^{-\binom{j}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} \left(-\frac{1}{2} \right)^j = \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-(j+1)}. \quad (235)$$

Since $\begin{bmatrix} n \\ j \end{bmatrix}_{1/2} > 0$, the summands of (235) are the absolute values of those of (234). Dividing by $(1/2; 1/2)_n$ gives the exact evaluation

$$\sum_{j=0}^n |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n}, \quad (236)$$

which is the first equality in (233). The formally proved finite-product estimate supplies the bound by 16 used in the Toeplitz argument above.

11 Iterated prefix sums of the signed Thue–Morse sequence

The signed Thue–Morse sequence $\varepsilon_n = (-1)^{\text{wt}_2(n)}$ has already appeared twice in this article: once as the coefficient sequence of the locally finite global expansion (11), and once as the sign pattern that makes the finite dyadic sums of section 8 collapse. There is a third, entirely discrete, way in which the same signs produce the Fabius function. Sum the sequence, sum the result, and iterate. After k summations the sequence has been convolved with an explicit binomial kernel, and after a dyadic normalization the resulting integer grid reproduces, in the limit, the Rvachev bump and hence F .

This section develops that construction: the convolution law and its explicit binomial kernel, the signed dyadic reflection preceding each terminal zero run, the zero runs themselves and their exact boundary values, the two generating-series identities, their analytic realization and natural-boundary theorem, the discrete functional equation satisfied by the normalized grid, and the precise sense in which the normalized grid converges. The convergence statement requires one shift of the prefix order and the centered reading of the grid cells; the literal normalization does not converge, and that failure is recorded separately in section 11.6. Two further pieces of foundational material are attached at the end: the lower branch of Lambert’s W -function, which the asymptotic sections of this article use but never construct, and an exact binomial–logarithm formula for the Thue–Morse bit.

The two binary recurrences used throughout this section are

$$\varepsilon_{2m} = \varepsilon_m, \quad \varepsilon_{2m+1} = -\varepsilon_m, \quad (237)$$

$$\text{wt}_2(2m) = \text{wt}_2(m), \quad \text{wt}_2(2m+1) = \text{wt}_2(m) + 1. \quad (238)$$

They are exactly `thueMorseSign_two_mul`, `thueMorseSign_two_mul_add_one`, `binaryWeight_two_mul`, and `binaryWeight_two_mul_add_one` in `DyadicClosedForm.lean`.

11.1 Iterated prefix sums and the binomial convolution law

Definition 11.1 (Iterated prefix sums). Set $\sigma_0(n) = \varepsilon_n$ and, for $k \geq 0$,

$$\sigma_{k+1}(n) = \sum_{j=0}^n \sigma_k(j). \quad (239)$$

All sums are *inclusive*: the index n itself is summed.

Every $\sigma_k(n)$ is an integer, and $\sigma_k(0) = 1$ for all k . The inclusive convention has one consequence that must be kept in view throughout, because it is the source of an index shift later:

$$\sigma_{k+1}(n+1) - \sigma_{k+1}(n) = \sigma_k(n+1). \quad (240)$$

The increment of a prefix sum is the lower-order value at the *new* index, not at the old one.

The first prefix sum nevertheless vanishes at every odd index:

$$\sigma_1(2m+1) = 0 \quad (m \in \mathbb{N}).$$

This is exactly `iteratedPrefix_one_two_mul_add_one` in `ThueMorsePrefix.lean`.

The general structure is a discrete convolution. Write

$$\mathcal{B}_k(i) = \binom{i+k-1}{k-1} \quad (k \geq 1, i \geq 0) \quad (241)$$

which is `iteratedPrefixKernel`. Because the argument $i = n - m$ is nonnegative in the sums below, no separate indicator is needed.

Theorem 11.2 (Binomial convolution law). *For every $k \geq 1$ and every $n \geq 0$,*

$$\sigma_k(n) = \sum_{m=0}^n \mathcal{B}_k(n-m) \varepsilon_m = \sum_{m=0}^n \binom{n-m+k-1}{k-1} \varepsilon_m. \quad (242)$$

Equivalently, in a form valid for every $k \geq 0$,

$$\sigma_{k+1}(n) = \sum_{m=0}^n \binom{n-m+k}{k} \varepsilon_m. \quad (243)$$

Proof. Induct on k in the form (243). For $k = 0$ the binomial coefficient is 1 and the assertion is the definition of σ_1 . Assume (243) for k . Then

$$\begin{aligned} \sigma_{k+2}(n) &= \sum_{j=0}^n \sigma_{k+1}(j) = \sum_{j=0}^n \sum_{m=0}^j \binom{j-m+k}{k} \varepsilon_m \\ &= \sum_{m=0}^n \varepsilon_m \sum_{i=0}^{n-m} \binom{i+k}{k} = \sum_{m=0}^n \binom{n-m+k+1}{k+1} \varepsilon_m, \end{aligned}$$

where the interchange is the triangular reindexing $i = j - m$ and the inner evaluation is the hockey-stick identity $\sum_{i=0}^N \binom{i+k}{k} = \binom{N+k+1}{k+1}$. $\square$

Remark 11.3. The kernel in (242) is exactly the coefficient sequence of $(1-z)^{-k}$. [Theorem 11.2](#) is therefore the coefficient form of the series identity proved in [theorem 11.11](#) below, and either statement may be taken as the definition of the other. Both are recorded here because the coefficient form is what one actually evaluates, while the series form is what one manipulates.

11.2 Prouhet annihilation, signed dyadic reciprocity, and exact zero runs

[Equation \(155\)](#) already records that a complete dyadic block of Thue–Morse signs annihilates every polynomial of degree below the block exponent. For the prefix sums we also need the sharp boundary case, which identifies the first power that survives.

Lemma 11.4 (Sharp Prouhet extraction). *Write $S_{r,d} = \sum_{h < 2^r} \varepsilon_h h^d$. Then $S_{r,d} = 0$ for $0 \leq d < r$, and*

$$S_{r,r} = (-1)^r 2^{\binom{r}{2}} r!. \quad (244)$$

Consequently, for all rational numbers $a_0, \dots, a_r$,

$$\sum_{h < 2^r} \varepsilon_h \sum_{d=0}^r a_d h^d = a_r (-1)^r 2^{\binom{r}{2}} r!, \quad (245)$$

so summation against a complete dyadic block of Thue–Morse signs annihilates every polynomial of degree below r and, on polynomials of degree at most r , extracts the leading coefficient up to the explicit nonzero factor $(-1)^r 2^{\binom{r}{2}} r!$.

Proof. The vanishing statement is (155) with $X = 0, Y = 1$. For the boundary case, split the block by parity rather than by halves. Since $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$ by (237),

$$S_{r+1,d} = \sum_{j < 2^r} \varepsilon_j [(2j)^d - (2j+1)^d] = - \sum_{c=0}^{d-1} \binom{d}{c} 2^c S_{r,c},$$

because $(2j+1)^d - (2j)^d = \sum_{c<d} \binom{d}{c} (2j)^c$. Take $d = r+1$. All terms with $c < r$ vanish by the first assertion, leaving only $c = r$:

$$S_{r+1,r+1} = -\binom{r+1}{r} 2^r S_{r,r} = -(r+1) 2^r S_{r,r}.$$

Since $S_{0,0} = \varepsilon_0 = 1$, induction gives $S_{r,r} = (-1)^r r! 2^{0+1+\dots+(r-1)} = (-1)^r r! 2^{\binom{r}{2}}$, which is (244). Formula (245) follows by exchanging the two sums: every index $d < r$ contributes $a_d S_{r,d} = 0$, and the index $d = r$ contributes $a_r S_{r,r}$. $\square$

Corollary 11.5 (Affine sharp case). *For all $X, Y \in \mathbb{Q}$,*

$$\sum_{h < 2^r} \varepsilon_h (X + Yh)^r = (-1)^r Y^r 2^{\binom{r}{2}} r!.$$

In particular the sum does not depend on the translation X .

Proof. Expand $(X + Yh)^r$ by the binomial theorem and apply (245): the coefficient of h^r is Y^r . $\square$

The public Boolean-cube theorem uses the genuine degree $\deg p \in \mathbb{N} \cup \{-\infty\}$, rather than the natural degree that assigns the zero polynomial degree zero. Over any commutative ring, for a finite set S , weights w_j , and a polynomial satisfying $\deg p < |S|$, it states

$$\sum_{T \subseteq S} (-1)^{|T|} p\left(x + \sum_{j \in T} w_j\right) = 0. \quad (246)$$

This is `sum_powerset_neg_one_pow_eval_of_degree_1t` in `ThueMorseSparseProuhet.lean`. The degree convention matters at the boundary $S = \emptyset$: the hypothesis then forces $p = 0$, so the theorem correctly includes the empty-support case instead of declaring it vacuous. With S the bit support of n and $w_j = 2^j h$, the corresponding sparse submask parametrization is

$$\sum_{T \subseteq \text{bits}(n)} (-1)^{|T|} p\left(x + h \sum_{j \in T} 2^j\right) = 0 \quad \text{when } \deg p < \text{wt}_2(n), \quad (247)$$

formalized as `sum_submask_neg_one_pow_eval_of_degree_1t`. It is valid also at $n = 0$, where again the only polynomial satisfying the strict degree bound is zero.

At the level of the centered formal exponential series, the corresponding zero has exact order r ; this is `order_thueMorseCenteredPowerSeries`. For the binomial kernel (241), the directly applicable formally proved form is instead the coefficient-extraction theorem `thueMorse_polynomial_sum_eq_coeff`, displayed in (245).

Theorem 11.6 (Signed dyadic reciprocity). *Let $k, r, d \in \mathbb{N}$ satisfy $k \leq r$ and $k + d < 2^r$. Then the part of the order- k prefix row immediately preceding its final k positions is the initial part of that row, read backwards and multiplied by one fixed sign:*

$$\sigma_k(2^r - k - 1 - d) = (-1)^{r-k} \sigma_k(d). \quad (248)$$

Consequently reflection preserves, rather than excludes, its zero locus:

$$\sigma_k(2^r - k - 1 - d) = 0 \iff \sigma_k(d) = 0. \quad (249)$$

The identity (248) and equivalence (249) are formalized respectively as `iteratedPrefix_dyadic_reverse_window` and `iteratedPrefix_dyadic_reverse_window_eq_zero_iff`. At $k = 0$ the first identity is precisely the dyadic bit-complement law `thueMorseSign_dyadic_complement`. Its proof below uses the independently established left boundary in [theorem 11.7](#) as the seed for the prefix recurrence.

Theorem 11.7 (Exact dyadic zero runs). *Let $1 \leq k \leq r$. Then*

$$\sigma_k(n) = 0 \quad \text{for } 2^r - k \leq n \leq 2^r - 1, \quad (250)$$

$$\sigma_k(2^r - k - 1) = (-1)^{r-k}, \quad (251)$$

$$\sigma_k(2^r) = -1. \quad (252)$$

Formula (251) also holds for $k = 0$, where it reduces to $\varepsilon_{2^r-1} = (-1)^r$, and (252) also holds for $k = 0$. Hence the zero run terminating at $2^r - 1$ has length exactly k : it is bounded on the left by a value of modulus one and on the right by the value -1 .

Formal proof and Lean correspondence. The terminal block is formalized by `iteratedPrefix_dyadic_zero_run`; its endpoint and reverse-indexed forms are `iteratedPrefix_dyadic_endpoint` and `iteratedPrefix_dyadic_zero_reverse`. Its left and right boundaries are `iteratedPrefix_before_dyadic_run` and `iteratedPrefix_at_dyadic`. All five declarations are in `ThueMorsePrefix.lean`. The mathematical argument behind them is as follows.

Endpoint. Put $q = k - 1$, so $0 \leq q < r$. By (242),

$$\sigma_k(2^r - 1) = \sum_{m=0}^{2^r-1} \binom{2^r-1-m+q}{q} \varepsilon_m = \frac{1}{q!} \sum_{m < 2^r} \varepsilon_m \prod_{i=1}^q (2^r - 1 - m + i).$$

The product is a polynomial in m of degree $q < r$, so the sum vanishes by (245).

Rest of the run. We prove, by induction on d , that $\sigma_k(2^r - 1 - d) = 0$ whenever $1 \leq k \leq r$ and $0 \leq d < k$. The case $d = 0$ is the endpoint. Suppose the claim holds for d and that $d + 1 < k$, so $k \geq 2$ and $1 \leq k - 1 \leq r$. The induction hypothesis applied at order k gives $\sigma_k(2^r - 1 - d) = 0$, and applied at order $k - 1$, which is legitimate because $d < k - 1$, gives $\sigma_{k-1}(2^r - 1 - d) = 0$. Now use (240) with $n = 2^r - 1 - (d + 1)$, so that $n + 1 = 2^r - 1 - d$:

$$0 - \sigma_k(n) = \sigma_k(n + 1) - \sigma_k(n) = \sigma_{k-1}(n + 1) = 0.$$

This proves (250), since the indices $2^r - k, \dots, 2^r - 1$ are exactly the $2^r - 1 - d$ with $0 \leq d < k$.

Left boundary. Induct on k . For $k = 0$ the number $2^r - 1$ has r binary digits, all equal to one, so $\varepsilon_{2^r-1} = (-1)^r$. Assume (251) at order k and let $k + 1 \leq r$. Put $n = 2^r - (k + 1) - 1$, so $n + 1 = 2^r - k - 1$. By the induction hypothesis $\sigma_k(n + 1) = (-1)^{r-k}$, and by (250) at order $k + 1$ we have $\sigma_{k+1}(n + 1) = \sigma_{k+1}(2^r - (k + 1)) = 0$. Then (240) gives $-\sigma_{k+1}(n) = (-1)^{r-k}$, that is, $\sigma_{k+1}(n) = (-1)^{r-(k+1)}$.

Right boundary. Induct on k . For $k = 0$, $\text{wt}_2(2^r) = 1$, so $\varepsilon_{2^r} = -1$. If (252) holds at order k and $k + 1 \leq r$, then by (240) and (250), $\sigma_{k+1}(2^r) = \sigma_{k+1}(2^r - 1) + \sigma_k(2^r) = 0 - 1 = -1$. $\square$

Proof of theorem 11.6. Induct first on the prefix order k . At order zero, $\sigma_0 = \varepsilon$, and reflecting d in the block $0, \dots, 2^r - 1$ complements its first r binary digits. Thus $\varepsilon_{2^r-1-d} = (-1)^r \varepsilon_d$, which is (248) for $k = 0$.

Now suppose the identity is known at order k and prove it at order $k + 1$ by induction on d . For $d = 0$, equation (251) and $\sigma_{k+1}(0) = 1$ give exactly the required identity. For the induction step, put

$$n = 2^r - (k + 1) - 1 - (d + 1), \quad s = (-1)^{r-(k+1)}.$$

The same-order induction hypothesis at offset d and the lower-order hypothesis at offset $d + 1$ have the common reflected index $n + 1$ and give

$$\sigma_{k+1}(n + 1) = s \sigma_{k+1}(d), \quad \sigma_k(n + 1) = (-1)^{r-k} \sigma_k(d + 1) = -s \sigma_k(d + 1).$$

Applying the forward-difference law (240) first at n and then at d therefore yields

$$\begin{aligned} \sigma_{k+1}(n) &= \sigma_{k+1}(n + 1) - \sigma_k(n + 1) \\ &= s(\sigma_{k+1}(d) + \sigma_k(d + 1)) = s \sigma_{k+1}(d + 1), \end{aligned}$$

which closes both inductions. Finally, the multiplier $(-1)^{r-k}$ is nonzero, so (248) is equivalent to (249). $\square$

Remark 11.8. At the level of statements, setting $d = 0$ in (248) and using `iteratedPrefix_at_zero`, namely $\sigma_k(0) = 1$, recovers the left boundary (251). Reciprocity pairs any zeros that occur in the pre-run window; it does not claim that window is zero-free. Nor does it imply the terminal zero run (250) or the separate right boundary (252), both of which were proved independently above.

Remark 11.9. The right boundary value is not $+1$. Every iterated prefix sum, at every order $k \leq r$, takes the value -1 at the index 2^r . This single sign is what makes the literal normalization of section 11.5 fail at the right endpoint; see section 11.6.

11.3 Two generating-series identities

All series in this subsection are formal power series over $\mathbb{Z}$; no convergence hypothesis on a complex variable is used or needed.

Let

$$\mathcal{K}_r(z) = \sum_{n < 2^r} \varepsilon_n z^n, \quad \mathcal{K}(z) = \sum_{n \geq 0} \varepsilon_n z^n, \quad \mathcal{S}_k(z) = \sum_{n \geq 0} \sigma_k(n) z^n. \quad (253)$$

Proposition 11.10 (Block product and coefficient stabilization). *For every $r \geq 0$,*

$$\mathcal{K}_r(z) = \prod_{j=0}^{r-1} (1 - z^{2^j}), \quad (254)$$

and, whenever $n < 2^r$, the coefficient of z^n in this finite product is ε_n . Hence each coefficient stabilizes once the finite product reaches a large enough dyadic block.

Formal proof. The finite product is `thueMorseBlockPolynomial_eq_product`; the stable coefficient identity is `coeff_finite_thueMorse_product`, both in `ThueMorseGenerating.lean`. $\square$

Theorem 11.11 (Prefix generating series). *For every $k \geq 0$,*

$$\mathcal{S}_k(z) = \frac{\mathcal{K}(z)}{(1 - z)^k}, \quad (255)$$

or equivalently

$$(1 - z)^k \mathcal{S}_k(z) = \mathcal{K}(z). \quad (256)$$

For each fixed coefficient index, either side can be computed from a sufficiently long finite product (254).

Formal proof. The two identities are `iteratedPrefixSeries_eq` and `one_sub_X_pow_mul_iteratedPrefixSeries`; their finite coefficient bridge is `coeff_eq6_finite` in `ThueMorseGenerating.lean`. $\square$

We use $\prod_{j \geq 0} (1 - z^{2^j})$ only as shorthand for the coefficientwise stabilization of the finite products (254); no separate infinite-product object is asserted.

The two identities of theorem 11.11 connect the prefix sums to the polynomial approximants used in section 11.7. These are the combinatorial step polynomials already introduced in (45); their coefficient interpretation and exact degree were proved there. In the index convention of this section,

$$\mathcal{P}_n(z) = \prod_{i=1}^n (1 + z + \cdots + z^{2^i-1}), \quad g_n = \deg \mathcal{P}_n = \sum_{i=1}^n (2^i - 1) = 2^{n+1} - n - 2. \quad (257)$$

Since $1 - z^{2^j} = (1 - z)(1 + z + \cdots + z^{2^j-1})$ and the factor with $j = 0$ is trivial, (254) factors as

$$\mathcal{K}_{n+1}(z) = (1 - z)^{n+1} \mathcal{P}_n(z). \quad (258)$$

Comparing (258) with (256) at order $k = n + 1$ and using stabilization below 2^{n+1} gives the exact coefficient identification

$$\sigma_{n+1}(m) = [z^m] \mathcal{P}_n(z) \quad (0 \leq m < 2^{n+1}). \quad (259)$$

Note that (259) must fail at $m = 2^{n+1}$, because $g_n < 2^{n+1}$ makes the right side zero while the left side is -1 by (252). That single index is the exceptional right endpoint of [theorem 11.19](#).

11.4 The analytic Thue–Morse series and its natural boundary

The series $\mathcal{K}$ in (253) was interpreted there as a formal power series. On the open unit disc the same coefficients define an analytic function, and we retain the notation $\mathcal{K}$:

$$\mathcal{K}(z) = \sum_{n=0}^{\infty} \varepsilon_n z^n, \quad |z| < 1. \quad (260)$$

Proposition 11.12 (Analytic product and Mahler equation). *For every $z \in \mathbb{C}$ with $|z| < 1$, the series (260) converges and*

$$\mathcal{K}(z) = \prod_{j=0}^{\infty} (1 - z^{2^j}). \quad (261)$$

The function $\mathcal{K}$ is holomorphic on $|z| < 1$ and satisfies

$$\mathcal{K}(z) = (1 - z) \mathcal{K}(z^2), \quad (262)$$

$$\mathcal{K}(z) = \left(\prod_{j=0}^{k-1} (1 - z^{2^j}) \right) \mathcal{K}(z^{2^k}) \quad (k \in \mathbb{N}). \quad (263)$$

On the real radius,

$$|\mathcal{K}(x)| \leq 1 - x \quad (0 \leq x < 1), \quad (264)$$

and $\mathcal{K}(0) = 1$.

Formal proof. The analytic function is `thueMorseDiscSeries`. Convergence, the product identity, and holomorphy are respectively `summable_thueMorseDiscSeries`, `thueMorseDiscSeries_eq_tprod`, and `thueMorseDiscSeries_differentiableOn`. Convergence of the product itself is `multipliable_one_sub_pow_two_pow_complex` in `ThueMorseInfiniteProduct.lean`. Equations (262) and (263) are `thueMorseDiscSeries_mahler` and `thueMorseDiscSeries_iterate`; (264) and the value at zero are `thueMorseDiscSeries_real_le` and `thueMorseDiscSeries_zero`. Except for the product-convergence declaration just cited, all these declarations are in `ThueMorseDiscSeries.lean`. $\square$

Theorem 11.13 (Dyadic radial zeros and the natural boundary). *If $k \in \mathbb{N}$ and $\zeta \in \mathbb{C}$ satisfy $\zeta^{2^k} = 1$, then*

$$\lim_{\rho \rightarrow 1^-} \mathcal{K}(\rho \zeta) = 0. \quad (265)$$

Moreover, for every $z_0 \in \mathbb{C}$ with $|z_0| = 1$ and every $R > 0$, there is no function $g : \mathbb{C} \rightarrow \mathbb{C}$ which is holomorphic on $\{z : |z - z_0| < R\}$ and satisfies

$$g(z) = \mathcal{K}(z) \quad \text{whenever} \quad |z - z_0| < R \quad \text{and} \quad |z| < 1. \quad (266)$$

Thus, in precisely this local-disc sense, every point of the unit circle is singular and the unit circle is a natural boundary for $\mathcal{K}$.

Formal proof. The radial limit is `tendsto_thueMorseDiscSeries_radial_zero`. More generally, if $n \in \mathbb{N}$ is nonzero and $\zeta^n = 1$, then $|\zeta| = 1$; this is `norm_eq_one_of_pow_eq_one`. The exact quantified nonextension statement (266), and hence the natural-boundary conclusion in the stated local-disc sense, is `thueMorse_natural_boundary`. All three declarations are in `ThueMorseNaturalBoundary.lean`. $\square$

11.5 The functional equation of the normalized prefix grid

The natural dyadic normalization of the prefix sums is

$$\mathcal{V}_k(j) = \frac{\sigma_k(j)}{2^{\binom{k}{2}}}, \quad x_{k,j} = \frac{j}{2^k}, \quad (267)$$

so that $\mathcal{V}_k$ is a rational-valued function on the level- k dyadic grid. The exponent $\binom{k}{2}$ is forced: it is the exponent that appears in the sharp Prouhet value (244), in the centered spline (55), and in the derivative scaling (92).

Theorem 11.14 (Discrete functional equation). *Let $q \geq 0$ and $0 \leq j < 2^q$, and put $h = 2^{-(q+1)}$ and $x = x_{q+1,j}$. Then*

$$\frac{\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j)}{h} = 2 \mathcal{V}_q(j+1), \quad (268)$$

and the abscissa attached to the right-hand value is

$$x_{q,j+1} = 2(x+h) = 2x_{q+1,j+1} \in [0, 1]. \quad (269)$$

The same identity holds verbatim for the shifted grid $\tilde{\mathcal{V}}_k(j) = \sigma_{k+1}(j)/2^{\binom{k}{2}}$ of section 11.7.

Proof. Because $\binom{q+1}{2} = \binom{q}{2} + q$, the forward difference (240) gives

$$\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j) = \frac{\sigma_{q+1}(j+1) - \sigma_{q+1}(j)}{2^{\binom{q}{2}+q}} = \frac{\sigma_q(j+1)}{2^{\binom{q}{2}}} \cdot \frac{1}{2^q} = \frac{\mathcal{V}_q(j+1)}{2^q}.$$

Dividing by $h = 2^{-(q+1)}$ multiplies by 2^{q+1} and gives (268). For (269), $2(x+h) = 2(j+1)/2^{q+1} = (j+1)/2^q = x_{q,j+1}$, and the hypothesis $j < 2^q$ is exactly the condition $j+1 \leq 2^q$ that keeps this abscissa in the unit interval. The shifted grid satisfies the same computation with σ replaced by $\sigma_{\bullet+1}$ throughout. $\square$

Equation (268) is the discrete counterpart of the global differential equation (91): a difference quotient at level $q+1$ equals twice a level- q value at the doubled abscissa. Two features of the discrete statement have no continuous analogue and are easy to lose. First, the right-hand side is evaluated at the *forward* node $2(x+h)$ rather than at $2x$; this is the inclusive-prefix convention (240). Second, the identity is an assertion about a level- q value, so it is meaningful as a statement about F or up only when the doubled abscissa lies in the relevant interval, that is, only when $j < 2^q$. Both features are used in section 11.7.

11.6 The literal right endpoint does not converge

The unshifted normalization printed in the source formula cannot converge to a function whose right-endpoint value is 1: the sequence of literal values at the index 2^k does not tend to 1. This exact negative statement is `paperPrefixGridValue_endpoint_not_tendsto_one` in `ThueMorseGenerating.lean`. The corrected normalization below is therefore a mathematical requirement, not a cosmetic reindexing.

11.7 Corrected normalization and pointwise convergence

We now identify the normalized prefix grid with the histogram approximants of the probabilistic construction, and deduce convergence. Two adjustments to the naive reading are required, and both are forced by identities already proved: the prefix order must be shifted by one, because (259) matches σ_{n+1} with $\mathcal{P}_n$; and the grid index m must be read as labelling the *center* of a cell of width 2^{-n} , not the left endpoint of a cell of width $2^{-(n+1)}$.

For the polynomial $\mathcal{P}_n$ of (257), place the coefficient of the monomial z^m at

$$\xi_{n,m} = \frac{2m - g_n}{2^{n+1}}, \quad (270)$$

and spread it uniformly over the cell $I_{n,m}$ of width 2^{-n} centered at $\xi_{n,m}$. The consecutive centers are spaced 2^{-n} apart, so these finitely many cells tile their support interval

$$\left[-1 + \frac{n+1}{2^{n+1}}, 1 - \frac{n+1}{2^{n+1}} \right].$$

The resulting step approximant is

$$\mathcal{W}_n(x) = \frac{2^n}{2^{\binom{n+1}{2}}} \sum_{m=0}^{g_n} [z^m] \mathcal{P}_n(z) \cdot \mathbf{1}_{I_{n,m}}(x), \quad (271)$$

where a point shared by two adjacent cells is given the weight $\frac{1}{2}$ in each, so that the two contributions add to the single value the formula is intended to have. This endpoint convention is not cosmetic: with ordinary closed-interval indicators the value at $x = 0$ would be counted twice already for $n = 1$.

Theorem 11.15 (Histogram approximants). *For every $n \geq 0$ the function $\mathcal{W}_n$ satisfies $0 \leq \mathcal{W}_n \leq 1$, is nondecreasing on $(-\infty, 0)$, is nonincreasing on $(0, \infty)$, and satisfies $\mathcal{W}_n(0) = 1$. Moreover $\mathcal{W}_n(x) \rightarrow \text{up}(x)$ for every real x .*

Proof. For $n \geq 1$, this is exactly theorem 4.2, since (257) and (45) define the same polynomial, while (270) and (271) give the same cells and normalization as (48) and (49). At $n = 0$ the empty product is $\mathcal{P}_0 = 1$, $g_0 = 0$, and $\mathcal{W}_0$ is the half-weighted indicator of $[-1/2, 1/2]$, so the stated order and normalization properties are immediate. The pointwise convergence of the sequence is already the final assertion of theorem 4.2. $\square$

Theorem 11.16 (Exact identification of the corrected grid). *Put $\tilde{\mathcal{V}}_n(m) = \sigma_{n+1}(m)/2^{\binom{n}{2}}$. For every $n \geq 0$ and every m with $0 \leq m < 2^{n+1}$,*

$$\tilde{\mathcal{V}}_n(m) = \mathcal{W}_n(\xi_{n,m}). \quad (272)$$

Proof. If $m \leq g_n$, the point $\xi_{n,m}$ is the center of $I_{n,m}$ and lies in no other cell, so (271) evaluates there to

$$\frac{2^n}{2^{\binom{n+1}{2}}} [z^m] \mathcal{P}_n(z).$$

Since $\binom{n+1}{2} = \binom{n}{2} + n$, this is $[z^m] \mathcal{P}_n / 2^{\binom{n}{2}} = \tilde{\mathcal{V}}_n(m)$ by (259). If $g_n < m < 2^{n+1}$, then $\xi_{n,m}$ lies strictly to the right of the finite support interval above, so $\mathcal{W}_n(\xi_{n,m}) = 0$; meanwhile $[z^m] \mathcal{P}_n = \sigma_{n+1}(m) = 0$ because $\deg \mathcal{P}_n = g_n$. Thus both sides again agree. $\square$

Sampling at the dyadic floor now gives an approximation scheme. For $x \in \mathbb{R}$ set

$$j_n(x) = \max(0, \lfloor 2^{n+1}x \rfloor), \quad \mathcal{J}_n(x) = \tilde{\mathcal{V}}_n(j_n(x)) = \frac{\sigma_{n+1}(j_n(x))}{2^{\binom{n}{2}}}. \quad (273)$$

Lemma 11.17 (Moving cell centers). *For every $x \in [0, 1)$, $\xi_{n,j_n(x)} \rightarrow 2x - 1$ as $n \rightarrow \infty$.*

Proof. By (270), $\xi_{n,j_n(x)} = 2 \cdot j_n(x)2^{-(n+1)} - g_n2^{-(n+1)}$. The first term tends to $2x$ because $0 \leq 2^{n+1}x - j_n(x) < 1$, and $g_n2^{-(n+1)} = 1 - (n+2)2^{-(n+1)} \rightarrow 1$. $\square$

Lemma 11.18 (Stability under moving arguments). *If $u_n \rightarrow y$ in $\mathbb{R}$, then $\mathcal{W}_n(u_n) \rightarrow \text{up}(y)$.*

Proof. Let $\varepsilon > 0$. Suppose first $y > 0$. By continuity of up choose $d \in (0, y/2]$ with $|\text{up}(y \pm d) - \text{up}(y)| < \varepsilon/2$. For all large n we have $u_n \in [y - d, y + d] \subset (0, \infty)$, so the monotonicity of theorem 11.15 on $(0, \infty)$ gives

$$\mathcal{W}_n(y + d) \leq \mathcal{W}_n(u_n) \leq \mathcal{W}_n(y - d).$$

By pointwise convergence at the two fixed points $y \pm d$, for all large n both outer terms lie within $\varepsilon/2$ of $\text{up}(y \pm d)$, hence within ε of $\text{up}(y)$. The case $y < 0$ is identical using monotonicity on $(-\infty, 0)$. For $y = 0$ the upper bound is $\mathcal{W}_n(u_n) \leq 1 = \text{up}(0)$, and the lower bound is $\mathcal{W}_n(-d)$, $\mathcal{W}_n(0) = 1$, or $\mathcal{W}_n(d)$ according to the sign of u_n ; each of the three converges to a value within $\varepsilon/2$ of $\text{up}(0)$. $\square$

Theorem 11.19 (Corrected prefix approximation). *For every $x \leq 1$,*

$$\mathcal{J}_n(x) \rightarrow \text{up}(2x - 1), \quad (274)$$

and on the whole nonpositive half-line the finite-stage values are exactly

$$\mathcal{J}_n(x) = \mathcal{J}_n(0) = 2^{-\binom{n}{2}} \quad (x \leq 0). \quad (275)$$

and consequently, for every $x \in [0, 1]$,

$$\mathcal{J}_n\left(\frac{x}{2}\right) \rightarrow F(x). \quad (276)$$

Proof. For $x \leq 0$, the natural-floor clamp is exact: `correctedPrefixGridSample_eq_at_zero_of_nonpos` and `correctedPrefixGridSample_zero` give (275), while `correctedPrefixGridSample_tendsto_of_nonpos` gives the asserted limit. Now let $0 < x < 1$. Then $2^{n+1}x < 2^{n+1}$, so $j_n(x) < 2^{n+1}$ and theorem 11.16 applies: $\mathcal{J}_n(x) = \mathcal{W}_n(\xi_{n,j_n(x)})$. Combining lemma 11.17 with lemma 11.18 gives $\mathcal{J}_n(x) \rightarrow \text{up}(2x - 1)$.

At $x = 1$ the identification fails, because $j_n(1) = 2^{n+1}$ is exactly the first index not covered by (259). Here the value is computed instead from (252) with $k = r = n + 1$:

$$\mathcal{J}_n(1) = \frac{\sigma_{n+1}(2^{n+1})}{2^{\binom{n}{2}}} = -\frac{1}{2^{\binom{n}{2}}} \rightarrow 0 = \text{up}(1),$$

since $\binom{n}{2} \rightarrow \infty$ and $\text{up}(1) = F(0) = 0$. Altogether this is the public theorem `correctedPrefixGridSample_tendsto_rvachevUp_of_le_one` in `ThueMorseApproximation.lean`. For (276), apply it at $x/2 \in [0, 1]$ and use $2(x/2) - 1 = x - 1 \leq 0$, so that $\text{up}(x - 1) = F(x)$ by (7). $\square$

Warning 11.20. Theorem 11.19 is a pointwise statement and nothing more. No uniform rate is asserted, and none is proved. The proof passes through lemma 11.18, whose only quantitative inputs are the shared monotonicity of the $\mathcal{W}_n$ and the modulus of continuity of the limit up at the single point $2x - 1$; the sandwich it uses degrades as that modulus degrades. A uniform bound of the kind supplied for the centered splines by (61) is available from section 4.4, but it is a different approximation scheme, and the two should not be conflated.

Remark 11.21. The two corrections are independent of one another and both are necessary. Shifting the prefix order from k to $k + 1$ is what makes (259) available; reading the index m as the center (270) of a cell of width 2^{-n} rather than as an abscissa $m2^{-k}$ is what supplies the affine change of variable $x \mapsto 2x - 1$ between the unit interval and the support of up . In particular the corrected grid is a sample of the compactly supported bump, not of the distribution function; this is why its right-endpoint limit is 0 and not 1.

11.8 The lower Lambert branch

The asymptotic analysis of [section 12.5](#) is organized around the coordinate (298), where W_{-1} denotes the lower real branch of Lambert’s W -function [11]. That section, and the elementary expansions derived from it in [section 12.9](#), use three facts about W_{-1} without proving any of them: that the branch exists on $(-e^{-1}, 0)$, that it is characterized by being the solution below -1 , and that it has the two-term expansion recorded below. We supply them here, since they are the foundational input for (300) and for every subsequent estimate in the Lambert variable.

Proposition 11.22 (Existence, defining equation, and uniqueness). *For $z \in (-e^{-1}, 0)$, let $W_{-1}(z)$ denote the totalized lower branch `lowerLambertW`. Then*

$$W_{-1}(z) < -1, \quad W_{-1}(z) e^{W_{-1}(z)} = z, \quad (277)$$

and $W_{-1}(z)$ is the unique real $w < -1$ with $we^w = z$.

Formal proof. These are respectively `lowerLambertW_lt_neg_one`, `lowerLambertW_mul_exp`, and `lowerLambertW_unique` in `LowerLambertW.lean`. $\square$

On its natural domain, W_{-1} is continuous and strictly decreasing, and its image is exactly $(-\infty, -1)$. At every $z \in (-e^{-1}, 0)$, it is differentiable with

$$\frac{d}{dz} W_{-1}(z) = \frac{W_{-1}(z)}{z(1 + W_{-1}(z))} < 0. \quad (278)$$

These four assertions are `lowerLambertW_continuousOn`, `lowerLambertW_strictAntiOn`, `lowerLambertW_image`, `deriv_lowerLambertW`, and `deriv_lowerLambertW_neg`; the underlying derivative witness is `lowerLambertW_hasDerivAt`, all in `LowerLambertW.lean`.

Corollary 11.23 (Solution of the saddle equation). *Let $x > 0$ with $Lx < e^{-1}$, where $L = \log 2$ as in (283), and put $\lambda = -W_{-1}(-Lx)/L$ as in (299). Then $\lambda 2^{-\lambda} = x$, which is (300).*

Formal proof. The saddle equation is `paperLambertN_eq9` in `LowerLambertW.lean`. $\square$

Proposition 11.24 (Two-term expansion). *As $\epsilon \downarrow 0$,*

$$W_{-1}(-\epsilon) - (\log \epsilon - \log |\log \epsilon|) \longrightarrow 0. \quad (279)$$

Formal proof. This is `tendsto_lowerLambertW_expansion` in `LowerLambertW.lean`. $\square$

11.9 A binomial–logarithm formula for the Thue–Morse bit

The zero-one Thue–Morse bit is $\tau(n) = \text{wt}_2(n) \bmod 2$, so that $\varepsilon_n = 1 - 2\tau(n)$. Its primitive-recursive evaluator is `tmBitPR`; the theorems `tmBitPR_of_pos` and `tmBitPR_eq_thueMorseBit` prove its agreement with the binary-weight definition. The bit also admits a closed formula in which the binary structure is hidden inside a single row of Pascal’s triangle. Set

$$\mathcal{X}(n) = \sum_{k=0}^n (-1)^{\binom{n}{k}} \in \mathbb{Z}. \quad (280)$$

Lemma 11.25 (Glaisher count). *The number of odd entries in the n th row of Pascal’s triangle is $2^{\text{wt}_2(n)}$.*

Proof. Let $\nu(n)$ denote that number, so that $\nu(n)$ is the number of nonzero coefficients of $(1+z)^n$ in $\mathbb{F}_2[z]$. There $(1+z)^2 = 1+z^2$, hence

$$(1+z)^{2n} = (1+z^2)^n, \quad (1+z)^{2n+1} = (1+z^2)^n(1+z).$$

The substitution $z \mapsto z^2$ does not change the number of nonzero coefficients, so $\nu(2n) = \nu(n)$; and multiplying a polynomial in z^2 by $1 + z$ produces two summands with disjoint supports, one even and one odd, so $\nu(2n + 1) = 2\nu(n)$. Since $\text{wt}_2(2n) = \text{wt}_2(n)$ and $\text{wt}_2(2n + 1) = \text{wt}_2(n) + 1$ by (238), and $\nu(0) = 1 = 2^{\text{wt}_2(0)}$, induction on the binary representation gives $\nu(n) = 2^{\text{wt}_2(n)}$. $\square$

Proposition 11.26 (The logarithm is exact). *For every $n \geq 0$,*

$$n + 1 - \mathcal{X}(n) = 2^{\text{wt}_2(n)+1}. \quad (281)$$

In particular $n + 1 - \mathcal{X}(n)$ is a positive power of two, and $\log_2(n + 1 - \mathcal{X}(n)) = \text{wt}_2(n) + 1$ is an integer.

Proof. Each summand of (280) is -1 at an odd entry and $+1$ at an even entry, and the row has $n + 1$ entries. Hence $\mathcal{X}(n) = (n + 1) - 2\nu(n)$ with ν as in lemma 11.25, so $n + 1 - \mathcal{X}(n) = 2\nu(n) = 2^{\text{wt}_2(n)+1}$. $\square$

Theorem 11.27 (Binomial–logarithm formula). *For every $n \geq 0$,*

$$\tau(n) = \frac{1 + (-1)^{\log_2\left(n+1-\sum_{k=0}^n (-1)^{\binom{n}{k}}\right)}}{2}, \quad \varepsilon_n = 1 - 2\tau(n). \quad (282)$$

Proof. By proposition 11.26 the exponent equals $\text{wt}_2(n) + 1$. If $\text{wt}_2(n)$ is even then $\tau(n) = 0$ and $(-1)^{\text{wt}_2(n)+1} = -1$, so the right side is 0; if $\text{wt}_2(n)$ is odd then $\tau(n) = 1$ and $(-1)^{\text{wt}_2(n)+1} = 1$, so the right side is 1. The second identity is $\varepsilon_n = (-1)^{\text{wt}_2(n)} = 1 - 2(\text{wt}_2(n) \bmod 2)$. $\square$

Remark 11.28. The order of the two steps matters. As printed, (282) raises -1 to a logarithm, an operation that is meaningful only because the argument of that logarithm has first been proved to be exactly a power of two. The formal development therefore proves (281) before it interprets the outer expression at all, evaluates the logarithm on the natural numbers, and reads the final division in $\mathbb{Q}$. A floating-point evaluation of the same display, or an interpretation of the logarithm as a real-valued function whose output is then rounded, is an approximation to a formula rather than the formula.

12 Endpoint asymptotics: the exact formal hierarchy

The behavior of $F(x)$ as $x \downarrow 0$ is governed by three nested structures: an exact dyadic product on the negative Laplace axis, a genuinely nonconstant periodic correction on logarithmic scale, and a lower-Lambert saddle coordinate. This section states the hierarchy in the form proved in Lean. Longer heuristic derivations, numerical experiments, and proposed refinements without exact theorem counterparts are collected in docs/non-formalized-research-frontiers/non-formalized-research-frontiers.tex.

Throughout, put

$$L = \log 2. \quad (283)$$

12.1 The negative-Laplace product and its exact decomposition

For $s > 0$, the entire generating function G from (120) satisfies

$$G(-s) = \prod_{j=1}^{\infty} \frac{1 - \exp(-s/2^j)}{s/2^j}. \quad (284)$$

This is the real specialization of `complexGeneratingFunction_neg_eq_tprod` in `NegativeLaplaceVertical.lean`. If

$$\kappa(u) = \log\left(\frac{1 - \exp(-u)}{u}\right), \quad \Lambda(s) = \log G(-s) = \sum_{j=1}^{\infty} \kappa(s/2^j),$$

then the series is absolutely summable for every $s > 0$; the precise declarations are `summable_negativeLaplaceTerm`, `exp_negativeLaplaceLog_eq_generatingFunction_neg`, and `negativeLaplaceLog_eq_log_laplaceMoment`.

Let P denote `negativeLaplacePeriodicCorrection` and let E denote `negativeLaplaceTailError`. The exact decomposition is

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s) \quad (s > 0). \quad (285)$$

Moreover,

$$0 \leq E(s) \leq \frac{\exp(-s)}{(1 - \exp(-s))^2}, \quad E(s) \leq 4 \exp(-s) \quad (s \geq L). \quad (286)$$

These are exactly `negativeLaplaceLog_exact_periodic_decomposition`, `negativeLaplaceTailError_nonneg`, `abs_negativeLaplaceTailError_le`, and `abs_negativeLaplaceTailError_le_four_exp` in `NegativeLaplace.lean`. The correction P is C^∞ and one-periodic by `contDiff_infty_negativeLaplacePeriodicCorrection` and `negativeLaplacePeriodicCorrection_add_one`.

Write

$$\bar{P} = \int_0^1 P(u) \, du, \quad \Psi(t) = P(t) - \bar{P}. \quad (287)$$

Then Ψ is C^∞ , one-periodic, and has mean zero:

$$\Psi(t+1) = \Psi(t), \quad \int_0^1 \Psi(t) \, dt = 0.$$

The formal counterparts are `contDiff_infty_negativeLaplacePsi`, `negativeLaplacePsi_add_one`, and `integral_negativeLaplacePsi_zero`.

The first logarithmic derivative has a correspondingly sharp full-ray expansion. As $s \rightarrow +\infty$,

$$\Lambda'(s) = \frac{-\log s/L + 1/2 + \Psi'(\log_2 s)/L}{s} + \mathcal{O}(s^{-2}). \quad (288)$$

Here Λ' is the formal `negativeLaplaceLogFirst`, identified with the derivative of Λ by `negativeLaplaceLog_hasDerivAt`. The exact $\mathcal{O}(s^{-2})$ statement is `negativeLaplaceLogFirst_sub_periodic_main_isBigO_inv_sq_real` in `LaplacePeriodicSecondOrder.lean`. This promotes the proved derivative estimate without importing the longer unformalized contour and Mellin derivations from the frontier volume.

12.2 The Gamma–zeta Fourier series

For $k \in \mathbb{Z}$, use the Fourier convention

$$\widehat{\Psi}(k) = \int_0^1 \Psi(t) \exp(-2\pi i k t) \, dt, \quad \chi_k = \frac{2\pi i k}{L}. \quad (289)$$

The zero mode vanishes, and every nonzero mode has the closed form

$$\widehat{\Psi}(0) = 0, \quad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L} \quad (k \neq 0). \quad (290)$$

Every coefficient with $k \neq 0$ is nonzero, so Ψ is genuinely nonconstant. The coefficient sequence is absolutely summable and reconstructs Ψ pointwise:

$$\Psi(t) = \sum_{k \in \mathbb{Z}} \widehat{\Psi}(k) \exp(2\pi i k t). \quad (291)$$

These assertions are `negativeLaplacePsiFourierCoeff_zero`, `negativeLaplacePsiFourierCoeff_eq_neg_gamma_zeta`, `negativeLaplacePsiFourierCoeff_ne_zero`, `negativeLaplacePsi_not_constant`, `summable_negativeLaplacePsiGammaZetaFourierCoeff`, and `hasSum_negativeLaplacePsi_gammaZeta_fourierSeries` in `PeriodicFourier.lean`. No numerical truncation or stronger coefficient-decay claim is needed here.

12.3 The evaluated constant

Let γ be Euler's constant and let γ_1 denote the formally defined first Stieltjes constant `firstStieltjesConstant`. The regularized Gamma-zeta combination

$$A_0 = \frac{\gamma^2}{2} + \gamma_1 - \frac{\pi^2}{12}$$

is `gammaZetaConstant`; its equivalent common-denominator form is proved by `gammaZetaConstant_eq_div_twelve`. The split, convergent finite-part Bose integral is evaluated as this same constant by `boseFinitePartIntegral_eq_gammaZetaConstant`.

The formal finite part is independent of where the convergent integral is split. More precisely, put

$$\kappa_0(x) = \frac{1}{x} \log\left(\frac{1 - \exp(-x)}{x}\right), \quad \kappa_\infty(x) = \frac{1}{x} \log(1 - \exp(-x)).$$

For every $c > 0$, both integrals below are convergent and

$$\boxed{\int_{(0,c]} \kappa_0(x) dx + \int_{(c,\infty)} \kappa_\infty(x) dx + \frac{(\log c)^2}{2} = A_0.} \quad (292)$$

If $0 < a \leq b$, changing the splitting point is governed by the exact transition identity

$$\int_{(a,b]} (\kappa_\infty(x) - \kappa_0(x)) dx = \frac{(\log b)^2 - (\log a)^2}{2}. \quad (293)$$

Here the two kernels are `boseFinitePartSmallKernel` and `boseFinitePartLargeKernel`; their integrability is `integrableOn_boseFinitePartSmallKernel_Ioc` and `integrableOn_boseFinitePartLargeKernel_Ioi_of_pos`. Equations (292) and (293) are respectively `boseFinitePartIntegral_eq_gammaZetaConstant_of_pos` and `integral_boseFinitePartLargeKernel_sub_small_Ioc` in `BoseFinitePartIntegral.lean`. Thus the constant is represented by ordinary convergent real integrals, without assigning a value to a divergent integral.

In turn, `negativeLaplacePeriodicMean_eq` gives

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}.$$

After the Gaussian normalization, the sharp Fabius constant is

$$C_F = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi. \quad (294)$$

This is the exact identity `fabiusSharpAsymptoticConstant_eq` in `FabiusSharpConstant.lean`.

12.4 A fully effective dyadic bound

The dyadic specialization admits a proved numerical threshold and error constant. For $n \geq 1$, define

$$D_n = -\frac{L}{2}n^2 - n \log n + \left(1 + \frac{L}{2}\right)n - \frac{(\log n)^2}{2L} + C_F + \Psi(\log_2 n) - \frac{n}{2}(\Lambda''(n) + \Lambda'(n)^2). \quad (295)$$

Here the final term is exactly `dyadicEndpointSecondOrder`; the derivative identifications are `negativeLaplaceLog_hasDerivAt` and `negativeLaplaceLogFirst_hasDerivAt`. Then, for every $n \geq 224043$,

$$\left| \log F(2^{-n}) - D_n \right| \leq \frac{2512945}{12n}. \quad (296)$$

This is `abs_log_fabius_dyadic_sub_explicitCumulantMain_le` in `FabiusDyadicSharpCumulant.lean`. The threshold is a sufficient certified threshold, not an optimality claim. Its proof uses the pointwise bounds `dyadicEndpointSecondOrder_sq_le`, `dyadicHigherLaplaceMoments_le`, and `abs_dyadicEndpointLaplaceLogError_add_secondOrder_le_unconditional` in `LaplaceMomentBounds.lean`, with respective constants $256/n$, $104448/n$, and $209408/n$ on their stated ranges.

12.5 The lower-Lambert saddle and exact reduction

On the endpoint-inclusive lower-branch domain

$$\mathcal{D}_L = \left(0, \frac{e^{-1}}{L}\right], \quad (297)$$

let

$$W = W_{-1}(-Lx), \quad (298)$$

and define

$$\lambda = \lambda(x) = -\frac{W}{L}, \quad r = r(x) = 2^\lambda. \quad (299)$$

The global Lean definitions are `fabiusLambertPhase` and `fabiusLambertRadius`; here we use them on $\mathcal{D}_L$. For every $x \in \mathcal{D}_L$, their exact identities are

$$\lambda 2^{-\lambda} = x, \quad \frac{\lambda}{r} = x, \quad rx = \lambda, \quad \log r = L\lambda. \quad (300)$$

The endpoint-inclusive quotient and product identities are `fabiusLambertPhase_div_radiu s_of_le` and `fabiusLambertRadius_mul_argument_of_le`; the first equality in (300) is the quotient identity rewritten using $r = 2^\lambda$. The logarithmic identity in fact holds for every real x , not only on $\mathcal{D}_L$; this is `log_fabiusLambertRadius`. The corresponding strict-interior wrappers are `fabiusLambertPhase_div_radius` and `fabiusLambertRadius_mul_argument`. All five declarations are in `FabiusLambertSaddle.lean`.

On the same closed-upper domain, the phase has the sharp lower bound and endpoint characterization

$$\lambda(x) \geq \frac{1}{L}, \quad \lambda(x) = \frac{1}{L} \iff x = \frac{e^{-1}}{L}. \quad (301)$$

In particular,

$$\lambda\left(\frac{e^{-1}}{L}\right) = \frac{1}{L}. \quad (302)$$

Moreover, λ is strictly decreasing on $\mathcal{D}_L$, and its exact image is

$$\lambda(\mathcal{D}_L) = \left[\frac{1}{L}, \infty \right). \quad (303)$$

The bound, equality characterization, endpoint value, strict antitonicity, and image are respectively `fabiusLambertPhase_ge_inv_log_two`, `fabiusLambertPhase_eq_inv_log_two_iff`, `fabiusLambertPhase_branchPoint`, `fabiusLambertPhase_strictAntiOn_Ioc`, and `fabiusLambertPhase_image_Ioc` in `FabiusLambertSaddle.lean`.

Equivalently, under the hypotheses $x > 0$ and $Lx \leq e^{-1}$, the same lower bound and the strict positivity $0 < \lambda(x)$ are `one_div_log_two_le_fabiusLambertPhase` and `fabiusLambertPhase_pos_of_le`. Thus the algebraic saddle identities extend to the finite branch point even though the smooth formulas below do not.

The smooth API is deliberately restricted to the open interior $0 < x < e^{-1}/L$. There λ is continuous and strictly decreasing, its image is $(1/L, \infty)$, and it is differentiable with

$$\lambda'(x) = \frac{\lambda(x)}{x(1 - L\lambda(x))} < 0. \quad (304)$$

These are `fabiusLambertPhase_continuousOn`, `fabiusLambertPhase_strictAntiOn`, `fabiusLambertPhase_image`, `deriv_fabiusLambertPhase`, and `deriv_fabiusLambertPhase_neg` in `FabiusLambertSaddle.lean`; the derivative witness is `fabiusLambertPhase_hasDerivAt`, and pointwise continuity is `fabiusLambertPhase_continuousAt`. At the small-argument endpoint $x = 0$, the exact phase and its reciprocal have the limits

$$\lambda(x) \longrightarrow +\infty, \quad \lambda(x)^{-1} \longrightarrow 0 \quad (x \downarrow 0). \quad (305)$$

These are `tendsto_fabiusLambertPhase_nhdsGT_zero_atTop` and `tendsto_inv_fabiusLambertPhase_nhdsGT_zero` in `FabiusLambertRates.lean`.

The corrected main term is

$$\mathcal{M}(x) = -\frac{L}{2}\lambda(x)^2 + \left(1 + \frac{L}{2}\right)\lambda(x) - \frac{1}{2}\log \lambda(x) + C_F + \Psi(\lambda(x)). \quad (306)$$

This is definition `fabiusSharpLambertMain`. The exact saddle action itself is endpoint-inclusive: whenever $x > 0$ and $Lx \leq e^{-1}$,

$$r(x)x + \Lambda(r(x)) - \frac{1}{2}\log(2\pi\lambda(x)) = \mathcal{M}(x) + E(r(x)). \quad (307)$$

This is `fabiusLambertSaddleAction_eq_of_le` in `FabiusSharpLambertMain.lean`. If $\mathcal{R}(x)$ denotes the normalized saddle ratio `fabiusSaddleRatio` at $r(x)$ and $\lambda(x)$, then Lean proves the exact error separation

$$\log F(x) - \mathcal{M}(x) = \log \mathcal{R}(x) + E(r(x)). \quad (308)$$

The declaration is `log_fabius_sub_sharpLambertMain_eq_log_ratio_add_tail` in `FabiusSharpExactReduction.lean`.

The forward product tail in this separation is negligible at every algebraic order of the exact phase. Along $x = 2^{-t}$, for each $N \in \mathbb{N}$,

$$E(r(2^{-t})) = o(\lambda(2^{-t})^{-N}) \quad (t \rightarrow +\infty). \quad (309)$$

This is `negativeLaplaceTailError_dyadicLambert_isLittleO_inv_pow` in `FabiusSharpLambertTransfer.lean`. At $N = 0$ it says directly that the tail tends to zero; larger N show that it contributes to none of the inverse-phase coefficients.

The analytic input to that reduction is an exact Bromwich formula. For every $r > 0$ and $x \in \mathbb{R}$, as an identity in $\mathbb{C}$,

$$F(x) = \exp(rx) \int_{\mathbb{R}} \exp(2\pi i \xi x) \frac{G(-(r + 2\pi i \xi))}{r + 2\pi i \xi} d\xi. \quad (310)$$

This is `fabius_bromwich`; the dimensionless, Gaussian-normalized version used by the saddle machinery is `fabius_bromwich_scaled`. The supporting Laplace-transform identity is `generatingFunction_neg_div_eq_laplace`. All three occur in `LaplaceTransform.lean` and `FabiusBromwichInput.lean` with the positivity and integrability hypotheses shown there.

12.6 The compact online formula and its missing periodic factor

A January 2022 comment in a Mathematics Stack Exchange discussion asked whether the preceding calculation implied a particularly compact lower-Lambert formula for the endpoint behavior [20]. With the notation of (298), its logarithmic main term is

$$\mathcal{W}_{\text{MSE}}(x) = -\frac{7L}{12} - \frac{1}{2} \log(\pi x) + \frac{1}{L} \left(\gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12} - W - \frac{W^2}{2} \right). \quad (311)$$

This is the human-readable form of definition `fabiusWikipediaLambertMain` in `FabiusWikipediaMain.lean`. Written in ordinary mathematical notation, the exponential of this logarithmic encoding is the expression printed in the discussion:

$$A_{\text{MSE}}(x) = \frac{1}{2^{7/12} \sqrt{\pi x}} \exp \left[\frac{1}{L} \left(\gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12} - W - \frac{W^2}{2} \right) \right]. \quad (312)$$

For $x > 0$, this factor is strictly positive. Lean names it `fabiusWikipediaLambertFactor`; `exp_fabiusWikipediaLambertMain_eq_WikipediaLambertFactor` identifies it with $\exp(\mathcal{W}_{\text{MSE}}(x))$, and `fabiusWikipediaLambertFactor_pos` proves its positivity. All three declarations occur in `FabiusWikipediaMain.lean`. The proposed symbol $F(x) \sim A_{\text{MSE}}(x)$ means that the ratio should tend to 1 as $x \downarrow 0$.

That ratio-one claim is false. The answer on the cited page first derives a Gamma–zeta periodic term, then explicitly omits it from the subsequent approximation while noting that it must be retained for more precise asymptotics. The exact formal decomposition shows what is lost: on the positive lower-Lambert domain,

$$\mathcal{M}(x) = \mathcal{W}_{\text{MSE}}(x) + \Psi(\lambda(x)). \quad (313)$$

Here `fabiusCorrectedWikipediaMain_eq_WikipediaLambertMain_add` is the all-real identity between the two named compact definitions, while `fabiusCorrectedWikipediaMain_eq_sharpLambertMain` identifies the corrected one with $\mathcal{M}$ on the stated domain. Consequently the proved ratio-one asymptotic is

$$\boxed{F(x) \sim \frac{1}{2^{7/12} \sqrt{\pi x}} \exp \left[\frac{1}{L} \left(\gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12} - W - \frac{W^2}{2} \right) \right] \times \exp(\Psi(-W/L))} \quad (x \downarrow 0). \quad (314)$$

This is the direct theorem `fabius_isEquivalent_WikipediaLambertFactor_mul_periodic` in `FabiusSharpAsymptotic.lean`; the underlying exact positive-side factorization is `exp_fabiusCorrectedWikipediaMain_eq_WikipediaLambertFactor_mul_periodic` in `FabiusWikipediaMain.lean`. The plus sign in (313) is essential: the missing multiplier is $\exp(\Psi(\lambda(x)))$, with $\lambda(x) = -W/L$.

The logical bridge from logarithmic error to ratio-one equivalence is itself formalized. Along any filter, if f is eventually positive, then

$$f \sim \exp(g) \iff \log f - g \longrightarrow 0. \quad (315)$$

This is `isEquivalent_exp_iff_tendsto_log_sub`. In the present case Lean proves both

$$\log F(x) - \mathcal{W}_{\text{MSE}}(x) \not\longrightarrow 0 \quad (x \downarrow 0) \quad (316)$$

and that the same residual is not $\mathcal{O}((-\log x)^{-1})$. These are respectively `log_fabius_sub_WikipediaLambertMain_not_tendsto_zero` and `log_fabius_sub_WikipediaLambertMain_not_isBigO`. Applying (315) gives the direct theorem

$$F \not\sim A_{\text{MSE}}, \quad (317)$$

namely `fabius_not_isEquivalent_exp_WikipediaLambertMain`; its direct literal-factor form is `fabius_not_isEquivalent_WikipediaLambertFactor`. The criterion and all four obstruction declarations in this paragraph occur in `FabiusSharpAsymptotic.lean`. Thus the nonconvergence of the sampled periodic correction, rather than its numerical size, is decisive for ratio-one equivalence.

Theorem 12.1 (Exact cluster interval for the online quotient). *For $x > 0$, put*

$$R_{\text{MSE}}(x) := \frac{F(x)}{A_{\text{MSE}}(x)}, \quad m_{\Psi} := \min_{0 \leq t \leq 1} \Psi(t), \quad M_{\Psi} := \max_{0 \leq t \leq 1} \Psi(t).$$

Then $m_{\Psi} < M_{\Psi}$, and the quotient has the sharper additive asymptotic model

$$R_{\text{MSE}}(x) - \exp(\Psi(\lambda(x))) \longrightarrow 0 \quad (x \downarrow 0). \quad (318)$$

Write $\text{Cl}_{0+}(R_{\text{MSE}})$ for the set of all $z \in \mathbb{R}$ for which there is a sequence $x_n \rightarrow 0$, eventually positive, such that $R_{\text{MSE}}(x_n) \rightarrow z$. Its complete cluster set is

$$\text{Cl}_{0+}(R_{\text{MSE}}) = [\exp(m_{\Psi}), \exp(M_{\Psi})]. \quad (319)$$

In particular,

$$\liminf_{x \downarrow 0} R_{\text{MSE}}(x) = \exp(m_{\Psi}) < \exp(M_{\Psi}) = \limsup_{x \downarrow 0} R_{\text{MSE}}(x). \quad (320)$$

Consequently no real constant repairs the omitted factor: in Mathlib’s filter-theoretic asymptotic-equivalence relation,

$$\forall c \in \mathbb{R}, \quad F \not\sim c A_{\text{MSE}} \quad (x \downarrow 0). \quad (321)$$

For $c \neq 0$, the last assertion is the usual failure of $F(x)/(c A_{\text{MSE}}(x)) \rightarrow 1$. The formal declaration also includes $c = 0$, without interpreting a quotient by zero.

The online discussion supplies A_{MSE} ; the exact cluster interval is a consequence of the corrected formal asymptotic. The endpoint descriptions $\exp(m_{\Psi})$ and $\exp(M_{\Psi})$ are exact variational formulas in terms of the explicit periodic correction. No finite elementary evaluation, numerical certification, sign claim for the extrema, or uniqueness claim for their phases is asserted here.

The phase-sampling mechanism is reusable. Every value of a one-periodic map p is a cluster value of $p(\lambda(x))$ as $x \downarrow 0$; if p is continuous and its codomain is Hausdorff, the complete cluster set is exactly $p(\mathbb{R})$. These statements are `mapClusterPt_periodic_comp_fabiusLambertPhase` and `mapClusterPt_periodic_comp_fabiusLambertPhase_iff` in `FabiusWikipediaObstruction.lean`.

For the specialization above, R_{MSE} is `fabiusWikipediaLambertRatio`, while m_Ψ and M_Ψ are `negativeLaplacePsiPeriodMin` and `negativeLaplacePsiPeriodMax`. The exact global range, pointwise extremal bounds, and endpoint order are respectively `range_negativeLaplacePsi_eq_Icc_periodMin_periodMax`, `negativeLaplacePsiPeriodMin_le_value`, `negativeLaplacePsi_value_le_periodMax`, and `negativeLaplacePsiPeriodMin_le_periodMax`; attainment and strict separation are `exists_mem_Icc_negativeLaplacePsi_eq_periodMin`, `exists_mem_Icc_negativeLaplacePsi_eq_periodMax`, and `negativeLaplacePsiPeriodMin_lt_periodMax`. Exponentiation of the exact range is `range_exp_negativeLaplacePsi_eq_Icc_periodExtrema`.

The ratio-one comparison with the sampled periodic model is `fabiusWikipediaLambertRatio_isEquivalent_periodic`, and the stronger additive statement (318) is `tendsto_fabiusWikipediaLambertRatio_sub_periodic`. The interval identity (319) is `mapClusterPt_fabiusWikipediaLambertRatio_nhdsGT_zero_iff`; the two endpoint limits are `liminf_fabiusWikipediaLambertRatio_nhdsGT_zero_eq` and `limsup_fabiusWikipediaLambertRatio_nhdsGT_zero_eq`; and the universal obstruction is `fabius_not_isEquivalent_const_mul_WikipediaLambertFactor`. These declarations all occur in `FabijsSharpAsymptotic.lean`.

12.7 The first two periodic saddle corrections

Define

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \quad (322)$$

This is `fabiusFirstSaddleCorrection`. It is continuous, one-periodic, and bounded by `continuous_fabiusFirstSaddleCorrection`, `fabiusFirstSaddleCorrection_periodic`, and `isBounded_range_fabiusFirstSaddleCorrection`. The first corrected expansion is

$$\boxed{\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda(x))}{\lambda(x)} + \mathcal{O}(\lambda(x)^{-2})} \quad (x \downarrow 0). \quad (323)$$

It is the $N = 2$ specialization of `log_fabius_sub_sharpLambertExpansion_isBigO`, together with `fabiusSharpLambertExpansion_two`.

The second coefficient has a compact closed form in the bounded exponent jets. For each m , write $J_m(u)$ for the jet `negativeLaplaceBoundedExponentJet(m, u)`, and define

$$A_2(u) = \frac{1}{24} (8J_0^3 + 12J_0^2J_1 + 12J_0^2 + 48J_0J_1 + 12J_0J_2 + 6J_1^2 + 24J_1 + 20J_2 + 3J_3)(u). \quad (324)$$

This is exactly `fabiusSecondSaddleCorrection`; it is continuous, one-periodic, and bounded by `continuous_fabiusSecondSaddleCorrection`, `fabiusSecondSaddleCorrection_periodic`, and `isBounded_range_fabiusSecondSaddleCorrection`. The formally proved second-correction theorem is

$$\boxed{\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda(x))}{\lambda(x)} + \frac{A_2(\lambda(x))}{\lambda(x)^2} + \mathcal{O}(\lambda(x)^{-3})} \quad (x \downarrow 0). \quad (325)$$

Its exact declarations are `fabiusSharpLambertExpansion_three` and `log_fabius_sub_twoSaddleCorrections_isBigO` in `FabijsSecondSaddleExpansion.lean`; the companion theorem `log_fabius_sub_twoSaddleCorrections_isBigO_negLog` gives the literal remainder $\mathcal{O}((-\log x)^{-3})$.

12.8 The recursive all-orders theorem

For $j \in \mathbb{N}$, let $a_j(u)$ denote `fabiusSaddleLogCoefficient`(j, u). Every a_j is continuous, one-periodic, and bounded, by `continuous_fabiusSaddleLogCoefficient`, `fabiusSaddleLogCoefficient_periodic`, and `isBounded_range_fabiusSaddleLogCoefficient`. The initial identifications are

$$a_0 = 0, \quad a_1 = A_1, \quad a_2 = A_2,$$

proved by `fabiusSaddleLogCoefficient_zero`, `fabiusSaddleLogCoefficient_one_eq_firstSaddleCorrection`, and `fabiusSaddleLogCoefficient_two_eq_secondSaddleCorrection`.

For every $N \in \mathbb{N}$, the complete exact-phase expansion is

$$\log F(x) = \mathcal{M}(x) + \sum_{j=0}^{N-1} \frac{a_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N}) \quad (x \downarrow 0). \quad (326)$$

The precise formal statement is `log_fabius_sub_sharpLambertMain_hasAsymptoticExpansion`; its finite-remainder forms are `log_fabius_sub_sharpLambertExpansion_isBigO` and `log_fabius_sub_sharpLambertExpansion_isBigO_negLog` in `FabiusFullAsymptoticExpansion.lean`.

The coefficients are constructive rather than existential. The formal development identifies the jet weights as coefficients of

$$\prod_{k=0}^{n-1} (X - (k+1)),$$

proves closed forms for the periodic and bounded exponent jets, and gives recursive exponent-polynomial identities. The exact declarations are `negativeLaplaceJetPolynomial_eq_prod`, `negativeLaplaceJetStirling_eq_coeff`, `negativeLaplacePeriodicJet_eq_closedForm`, `negativeLaplaceBoundedExponentJet_eq_closedForm`, `negativeLaplaceExponentCoefficient_succ_eq`, and `negativeLaplaceExponentPolynomial_succ_eq`. This coefficient-of-a-product description is the proved statement; no identification with an external Stirling-number API is asserted here.

The algebraic conversion between relative-mass and logarithmic coefficients is normalized at every index rather than only asymptotically. If `ExpCoeff` and `LogCoeff` denote the formal recurrences `expCoeff` and `logCoeff`, then

$$\begin{aligned} \text{ExpCoeff}(\text{LogCoeff}(a))_n &= \begin{cases} 1, & n = 0, \\ a_n, & n > 0, \end{cases} \\ \text{LogCoeff}(\text{ExpCoeff}(E))_n &= \begin{cases} 0, & n = 0, \\ E_n, & n > 0. \end{cases} \end{aligned} \quad (327)$$

These are `expCoeff_logCoeff_eq_ite` and `logCoeff_expCoeff_eq_ite`; the corresponding identities of formal power series are `expSeries_logCoeff_eq_massSeries_normalize_zero` and `logSeries_expCoeff_eq_exponentSeries_normalize_zero` in `SaddleLogExpansionAlgebra.lean` and `SaddleLogExpansionPowerSeries.lean`.

12.9 Recursive elementary re-expansion of the phase

The exact Lambert phase can itself be expanded to arbitrary order. With

$$T(x) = -\log_2 x,$$

write $\mathcal{L}_N(x)$ for `fabijsLambertLogarithmicApproximation(N, x)`. This approximation is built from the recursively defined polynomials `dyadicLambertDisplacementPolynomial`. For every $N \in \mathbb{N}$,

$$\lambda(x) - \mathcal{L}_N(x) = \mathcal{O}(T(x)^{-N}) \quad (x \downarrow 0). \quad (328)$$

This is `fabijsLambertPhase_sub_logarithmicApproximation_isBigO`; the equivalent dyadic-scale form is `dyadicLambertAllOrderRemainder_isBigO`. Explicit coefficient polynomials are proved through index two, while higher orders are represented by the formal recursion.

The periodic term cannot simply be discarded from a sharp estimate. If $\mathcal{M}_{\text{Wiki}}$ denotes the uncorrected elementary expression `fabijsWikipediaElementaryMain`, then Lean proves that $x \mapsto \log F(x) - \mathcal{M}_{\text{Wiki}}(x)$ is not $\mathcal{O}((-\log x)^{-1})$ as $x \downarrow 0$. This is `log_fabijs_sub_wikipediaElementaryMain_not_isBigO` in `FabijsSharpAsymptotic.lean`. The obstruction is therefore a theorem, not a numerical diagnosis.

12.10 Three endpoint consequences already formalized

On the dyadic logarithmic scale, let $g(t) = -\log F(2^{-t})$. Then

$$\lim_{t \rightarrow \infty} \frac{g(t)}{t^2} = \frac{L}{2}. \quad (329)$$

This is `fabijsLogProfile_normalized_tendsto` in `FabijsLogSquaredAsymptotic.lean`. On the original small-argument scale, for every $c > 0$,

$$\exp(-c/x) = o(F(x)) \quad (x \downarrow 0). \quad (330)$$

This is `exp_neg_div_isLittleO_fabijs` in `FabijsDecayComparison.lean`. At the opposite endpoint, the exact global symmetry already proved in (15) gives only the identity

$$1 - F(1 - x) = F(x). \quad (331)$$

No additional right-endpoint asymptotic-transfer theorem is asserted here.

13 The Fourier–Legendre expansion of up

Let P_n be the Legendre polynomial normalized by $P_n(1) = 1$, and define

$$u_n = \frac{4n+1}{2} \int_{-1}^1 \text{up}(x) P_{2n}(x) \, dx. \quad (332)$$

Odd Legendre coefficients vanish by `rvachevFullLegendreCoefficient_odd_eq_zero`. The even coefficients have the exact finite formula

$$\begin{aligned} u_n &= 4^{-n} (4n+1) \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} \\ &\quad \times (2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \end{aligned} \quad (333)$$

This is `rvachevLegendreCoefficient_eq_fabijs_sum` in `FabijsLegendreCoefficients.lean`.

The even series

$$\text{up}(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x) \quad (-1 \leq x \leq 1) \quad (334)$$

converges absolutely and uniformly in the formal development. The exact declarations are `summable_abs_rvachevLegendreCoefficient`, `hasSum_rvachevLegendreSeries`, and `hasSum_rvachevLegendreSeries_uniform` in `FabiusLegendreSeries.lean`. Its N -th polynomial partial sum minimizes squared $L^2([-1, 1])$ error among all polynomials of degree at most $2N + 1$, and every distinct competitor has strictly larger error. These are `rvachevLegendrePartialSum_least_squares` and `rvachevLegendrePartialSum_strict_least_squares` in `FabiusLegendreLeastSquares.lean`. On $[0, 2]$, the signed global extension has the corresponding translated Legendre series `globalFabius_eq_tsum_translatedLegendre_formula`.

14 Additional proved summation identities

On the support interval $-1 \leq t \leq 1$, Poisson summation yields the exact cosine series

$$\text{up}(t) = \frac{1}{2} + \sum_{k=0}^{\infty} \widehat{\text{up}}\left(\frac{2k+1}{2}\right) \cos((2k+1)\pi t). \quad (335)$$

This is `rvachev_cosine_series`. The two-term support specialization

$$\text{up}(t) + \text{up}(t-1) = 1 \quad (0 \leq t \leq 1) \quad (336)$$

is `rvachev_add_shift_eq_one`. More general scaled Poisson support identities are sharp for either orientation of the lattice: whenever $1/2 \leq |a|$,

$$1 + 2 \text{up}(a) = \sum_{m \in \mathbb{Z}} |a|^{-1} \widehat{\text{up}}(m/a), \quad (337)$$

$$|a| + 2|a| \text{up}(a) = \sum_{m \in \mathbb{Z}} \widehat{\text{up}}(m/a). \quad (338)$$

These are `rvachev_poisson_support_specialization_unscaled_of_one_half_le_abs` and `rvachev_poisson_support_specialization_of_one_half_le_abs` in `PoissonSummation.lean`; the oriented frequency m/a is retained while the covolume is $|a|$. They supplement the partition formulas (87)–(88); no unformalized derivative partition, probabilistic convolution interpretation, or moment statistic is inferred.

15 Effective computability

The centered spline construction gives a primitive-recursive rational approximation procedure. In the exact coding used by Lean, for every dyadic input code c and precision p ,

$$|F(c.\text{value}(p+3)) - \text{Approx}(c, p).\text{value}(p)| \leq 5 \cdot 2^{-(p+3)}. \quad (339)$$

This is `fabiusSplineApproxPR_error`; the procedure itself is proved primitive recursive by `fabiusSplineApproxPR_primrec`. Together with the sharp Lipschitz bound it proves that every bounded Fabius solution is sequentially computable and effectively uniformly continuous, hence a computable real function in the formal Grzegorzczuk-style definition. The declarations are `fabiusReal_sequentiallyComputable`, `fabiusReal_effectivelyUniformContinuous`, and `fabiusReal_isComputableRealFunction` in `FabiusComputableSpline.lean` and `FabiusComputability.lean`. The same effective continuity holds for every signed global extension: `extendedFabius_effectivelyUniformContinuous`. For the canonical extension $\mathcal{F} = \text{globalFabius}$, the explicit modulus is $d(n) = 2n$: whenever $n > 0$ and $|x - y| < 1/(2n)$,

$$|\mathcal{F}(x) - \mathcal{F}(y)| < \frac{1}{n}. \quad (340)$$

This is `globalFabius_effectively_uniformly_continuous`, together with the definition `fabiusEffectiveUniformModulus`, in `FabiusComputability.lean`.

The same finite spline code also gives a signed-global algorithm. Instead of clamping the positive numerator to the unit grid, it retains the unrestricted centered spline and rounds the signed rational numerator pair. Negative dyadic inputs collapse to the zero grid point, which is exact because $\mathcal{F}(x) = 0$ for $x \leq 0$. The resulting primitive-recursive procedure satisfies

$$|\mathcal{F}(c.\text{value}(p+3)) - \text{Approx}_{\text{glob}}(c, p).\text{value}(p)| \leq 5 \cdot 2^{-(p+3)}. \quad (341)$$

These are `extendedFabiusSplineApproxPR_primrec` and `extendedFabiusSplineApproxPR_error`. Hence, for every F satisfying `IsFabiusF`, its signed extension `extendedFabiusF` is a computable real function, by `extendedFabius_isComputableRealFunction`; the canonical specialization is `globalFabius_isComputableRealFunction`. This is a primitive-recursive correctness statement, not a practical running-time estimate: the unrestricted positive grid numerator sets the length of the finite fold.

The quantitative analytic estimate is stable under moving the evaluation point:

$$|S_p(x) - \mathcal{F}(y)| \leq 2^{-p} + 2|x - y|. \quad (342)$$

This is `abs_fabiusUniformSpline_sub_extendedFabius_le_add`. In particular, if $x_p \rightarrow x$, then $S_p(x_p) \rightarrow \mathcal{F}(x)$, as packaged by `fabiusUniformSpline_tendsto_extendedFabius_of_tendsto`.

16 Formalization boundary and audit conventions

This exposition follows a fail-closed rule. A mathematical assertion remains in the main document only when an exact proved declaration can be identified in the Lean development. A nearby theorem, an executable computation, a numerical check, or an argument that appears straightforward is not treated as a substitute. When the formal theorem has stricter hypotheses, narrower domains, or different endpoint conventions, the prose uses those exact conditions.

That rule explains several choices made above:

- The finite probability approximation uses atomic convolution measures and the exact weak-convergence API, not an unformalized continuous box-convolution density.
- Step cells use the formal half-endpoint indicator, including its point value at a shared boundary.
- The Fourier chapter states the exact complex zero classification and canonical product, and the exact positive-integer zero order in its proved real punctured-limit form; it does not promote that real limit to an unstated complex local-multiplicity theorem.
- Flatness uses natural powers, and the analytic-locus claims use precisely the domains proved in `NowhereAnalytic.lean`.
- The q -calculus is restricted to the half-base identities actually present in Lean.
- Endpoint asymptotics retain exact phases and the proved remainder scopes; numerical fits and unproved coefficient formulas are absent.

The source drafts have been discharged into two destinations. Proved material was rewritten into the theorem-backed narrative here. The remaining research material is preserved under

`Analysis/FabiusFunction/docs/non-formalized-research-frontiers/`.

In particular, the `inverse/saddle`, `repeated-integration`, `small-argument`, generic q -calculus, and other deferred claims remain available there with an explicit warning that they are not part of the formal corpus.

17 Conclusion

The formal development links three global presentations without confusing their domains. The bounded F supplies the probability and arithmetic viewpoints; the compactly supported u supplies refinement, Fourier, spline, and Legendre viewpoints; and the signed $\mathcal{F}$ supplies the global dilation and binary-translation formulas. Exact bridges connect them to finite atomic laws, weak convergence, centered splines, moments, reciprocal-power values, dyadic algorithms, half-base q -binomial identities, prefix sums, endpoint Laplace analysis, and computable approximation.

The most important methodological conclusion is equally concrete: the primary document is now a map of proved mathematics. Research claims can move back from the frontier only after a matching Lean theorem exists, with the same hypotheses, domain, normalization, endpoint convention, and conclusion.

A Lean module map

Every retained result above points to one or more exact declaration names. The principal source groups are:

Topic	Modules
Definitions, existence, uniqueness	<code>Basic.lean</code> , <code>Existence.lean</code> , <code>OriginalCharacterization.lean</code> , <code>OriginalUniqueness.lean</code>
Shape and regularity	<code>Differential.lean</code> , <code>Monotonicity.lean</code> , <code>Convexity.lean</code> , <code>Regularity.lean</code>
Probability and finite approximations	<code>ProbabilityRepresentation.lean</code> , <code>EarlyApproximations.lean</code> , <code>EarlyMeasureBridge.lean</code> , <code>WeakConvergence.lean</code> , <code>ProbabilityLaplaceMoments.lean</code> , <code>LaplaceMoments.lean</code> , <code>StepMeasureBridge.lean</code> , <code>StepApproximationLimit.lean</code> , <code>FabiusUniformSpline.lean</code> , <code>FabiusComputability.lean</code>
Fourier analysis	<code>FourierProduct.lean</code> , <code>FourierAnalytic.lean</code> , <code>OriginalPaperSupplement.lean</code> , <code>PoissonSummation.lean</code> , <code>RenormalizationIdentity.lean</code> , <code>SincProductShells.lean</code> , <code>SincCanonicalProduct.lean</code> , <code>PhiZeroOrder.lean</code> , <code>ThueMorseLobeSign.lean</code> , <code>RvachevProductContinuity.lean</code> , <code>PeakRayEnvelope.lean</code> , <code>GlobalDecayEnvelope.lean</code> , <code>RvachevFourierDecay.lean</code> , <code>VarianceBookkeeping.lean</code> , <code>BirkhoffVariance.lean</code> , <code>BirkhoffProductBridge.lean</code>
Global derivatives and flatness	<code>GlobalExtension.lean</code> , <code>GlobalBounds.lean</code> , <code>BoundedDerivatives.lean</code> , <code>EffectiveFlatness.lean</code> , <code>SharpFlatness.lean</code> , <code>FabiusFlatness.lean</code>
Analyticity, inverse, elementary obstruction	<code>NowhereAnalytic.lean</code> , <code>FabiusInverse.lean</code> , <code>ElementaryFunction.lean</code> , <code>NotElementary.lean</code> , <code>InverseBranch.lean</code> , <code>InverseNotElementary.lean</code>
Moments and exact arithmetic	<code>Arithmetic.lean</code> , <code>AnalyticMoments.lean</code> , <code>MomentPowerSeries.lean</code> , <code>UnitLaplaceMomentBounds.lean</code> , <code>NormalizedEvenMoments.lean</code> , <code>BernoulliRecurrences.lean</code> , <code>MomentCumulantAlgebra.lean</code> , <code>ExactInversePower.lean</code> , <code>Parity.lean</code> , <code>TwoAdic.lean</code>

Topic	Modules
Dyadic and q -binomial formulas	DyadicClosedForm.lean, DyadicAnalytic.lean, GlobalDyadic.lean, DyadicCorrectness.lean, HalfQBinomial.lean, FabiusDyadicQBinomialScalar.lean, FabiusQBinomialFormula.lean, FabiusRawQBinomialScalar.lean, FabiusBinaryReductionSeries.lean, FabiusParityPowerSeries.lean, FabiusGlobalQBinomialSeries.lean, FabiusDiscreteLimitToeplitz.lean, FabiusDiscreteLimitIntegration.lean
Thue–Morse algebra, prefixes, disc series, and Prouhet identities	ThueMorsePrefix.lean, ThueMorseGenerating.lean, ThueMorseBlockAlgebra.lean, ThueMorseInfiniteProduct.lean, ThueMorseDiscSeries.lean, ThueMorseNaturalBoundary.lean, ThueMorseExponential.lean, ThueMorseApproximation.lean, ThueMorseSparseProuhet.lean, PaperKFoldThueMorse.lean
Endpoint asymptotics	NegativeLaplace.lean, NegativeLaplaceVertical.lean, PeriodicCorrection.lean, PeriodicFourier.lean, MellinBose.lean, MellinFinitePart.lean, BoseFinitePartIntegral.lean, LowerLambertW.lean, FabiusLambertPhase.lean, FabiusLambertSaddle.lean, FabiusSharpLambertMain.lean, FabiusWikipediaMain.lean, FabiusDecayComparison.lean, FabiusSharpAsymptotic.lean, FabiusFullAsymptoticExpansion.lean, FabiusSharpLambertTransfer.lean, LaplaceMomentBounds.lean, FabiusDyadicSharpCumulant.lean, SaddleLogExpansionAlgebra.lean, SaddleLogExpansionPowerSeries.lean, FabiusSecondSaddleCorrection.lean, FabiusSecondSaddleExpansion.lean
Legendre expansion and computability	FabiusLegendreCoefficients.lean, FabiusLegendreSeries.lean, FabiusLegendreLeastSquares.lean, FabiusTranslatedLegendreSeries.lean, FabiusComputableSpline.lean

B Notation crosswalk

Article	Lean	Meaning
F	fabiusRealfabius	bounded canonical Fabius function
up	rvachevUpfabius	compactly supported fold
$\mathcal{F}$	extendedFabiusfabius	signed global extension
I	fabiusInvfabiusfabius_spec	totalized inverse
c_n	momentn	even moment
d_n	halfMomentn	half moment
ε_n	thueMorseSignn	signed Thue–Morse value
$\text{wt}_2(n)$	binaryWeightn	binary digit sum

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